WEEKLY ROLLUP • PhD RESEARCH WATCH

Particulate Matter
Week in Review

Week ending 31 October 2026 | 25 October 2026 to 31 October 2026 | 135 records pooled from 7 daily issues

135
RECORDS POOLED
(135 UNIQUE DOIs)

37%
MEASUREMENT SIDE
(50 RECORDS)

10
SUBTOPICS
ACTIVE

29
POOLED EFFECT
ESTIMATES (29 WITH A
CI)

23
ABSTRACT-FREE
(17% OF WEEK)

The week in one paragraph

Volume up to 135, the health side swung from cardiorenal components to the airway and the developing brain, and fire smoke became the week's common exposure — prenatal smoke and autism in 893,212 Medicaid pairs, smoke plus power shutoffs in California, smoke days in Head Start classrooms, a lead-laden urban firestorm, and a preregistered null for smoke and depression. On the measurement side, humidity came back to the sensor: a heated PurpleAir sample path, Kelvin-suppressed growth of ultrafine particles, and organic-aerosol hygroscopicity by source.

Scale. 135 records over seven issues on daily totals of **3**, 24, **30**, 22, 22, 24 and 10. Against recent rollups: 135 after 119, 123, 87, 125, 162 and 148. **All ten subtopics are populated** for the sixth week running.
The measurement side slipped to 37 % (50 records): *Exposure assessment & modelling* 31 → **34**, *Sensing* 15 → **16**; the share fell only because the health side grew faster. Share series 33, 42, 44, 39, 39, **37 %**.
Respiratory led the health side's growth: 11 → **17**; *Neuro / mental health* doubled 4 → **8**; *Mechanistic toxicology* 10 → 14 and *Occupational & indoor* 11 → 14. *Cardiovascular & metabolic* halved, 12 → **6**.
Estimates: 29, every one with an interval (sixth complete week at 100 %), one per 4.7 records. 21 on ratio scales (OR 12, HR 5, RR 3, IRR 1) and 8 on difference or percentage scales.
Abstract gap 23 of 135 (17 %), down from 21 %. **Tier mix 16 A, 54 B, 65 C** — tier A at **12 %**, back from last week's 18 % high; ten EUPHA conference abstracts inflate tier C.

Subtopic trend and inflection

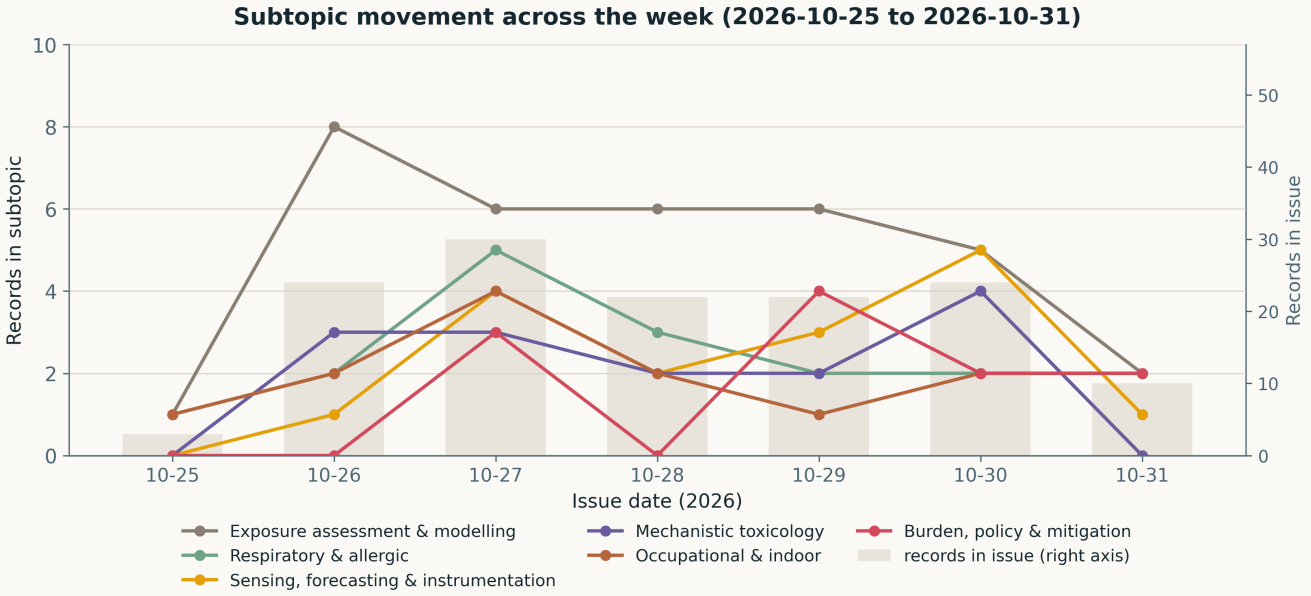


Fig. Per-issue subtopic counts across the week from `state/metrics.csv`, seven issues. Daily totals 3, 24, 30, 22, 22, 24 and 10. The scheduled runs for 23–29 October did not complete; 25–29 October were built on the **30 October recovery run**, and 30 and 31 October with this rollup on the **1 November run**. The 3-record Sunday (25 October) is the week's floor, as Sundays usually are.

Expanding. *Respiratory* 11 → **17** is the largest gain and is cohort- and event-driven: COPSAC lung-function trajectories to age 18 (FEV_1 −2.70 %, sRaw +7.30 % per IQR at age 6), the California PSPS–smoke interaction (joint OR 1.16), Louisiana paediatric asthma admissions (OR 1.079), a biomass-plus-tobacco COPD series, and a cluster of EUPHA abstracts (Polish DLNM admissions, Dublin spores, La Palma volcanic ash). *Neuro* 4 → **8** brings two tier-A records (Medicaid prenatal smoke and ASD; the All of Us depression null) and a purifier-controlled crossover on cognition.

Contracting. *Cardiovascular & metabolic* 12 → **6** — the stroke time series from Guiyang and the 123-city heart-disease cohort are what is left. *Burden* 14 → 11.

The inflection worth naming. Last week the effect was carried by $PM_{2.5}$ components; this week it is carried by the source — fire. Nine records put a fire-derived exposure on a health or exposure endpoint: prenatal smoke $PM_{2.5}$ and ASD (HR **1.30**, 1.07–1.59, per 0.75 $\mu g\,m^{-3}$) but not ADHD (1.03, 0.91–1.17); PSPS outages compounding smoke for respiratory encounters; FIRMS fire counts and Head Start sick absence (OR 1.106); fire $PM_{2.5}$ and eGFR ($-0.83\,mL\,min^{-1}\,1.73\,m^{-2}$); a Brazilian fire-density null for circulatory admissions; the Eaton firestorm plume enriched >50-fold in lead over wildland smoke; Idaho's fire-season $PM_{2.5}$ rising 2.54 $\mu g\,m^{-3}$ per decade while the winter wood-smoke mode declines; and the preregistered All of Us analysis in which smoke $PM_{2.5}$ showed **no** association with incident depression. Every one of these exposures is a model or a satellite product (ECHO smoke, FIRMS, CTM, kriged AQS); the composition contrast the Eaton record measured is the exception, and it is the one that matters most for toxicity. **The test for next week**: a smoke-specific estimate built on measured network data, or a smoke-vs-non-smoke $PM_{2.5}$ contrast with composition attached.

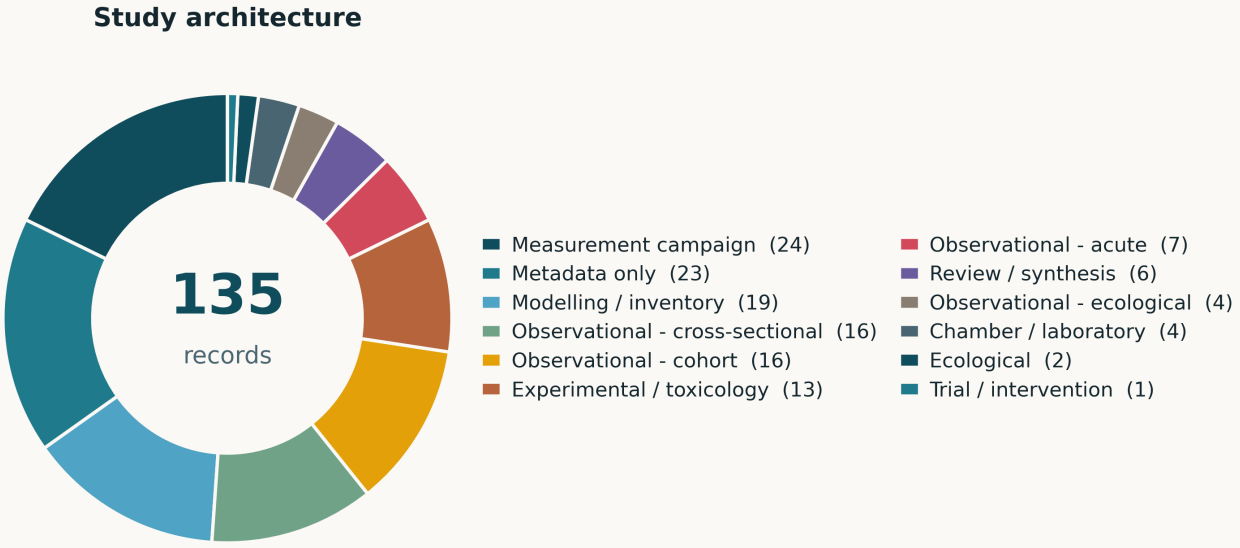


Fig. Study architecture over the pooled week. *Measurement campaign* 24 and *Metadata only* 23 lead, then *Modelling / inventory* 19, cross-sectional and cohort designs 16 each and experimental / toxicology 13. **Interventional evidence**: a purifier vs sham-filter randomised crossover (cognition, HRV), the OK-AIR cluster-randomised HEPA/UVGI classroom trial (secondary analysis), and a difference-in-differences evaluation of China's climate-adaptive city pilot; only OK-AIR carries the *Trial / intervention* label.

Pooled analytics

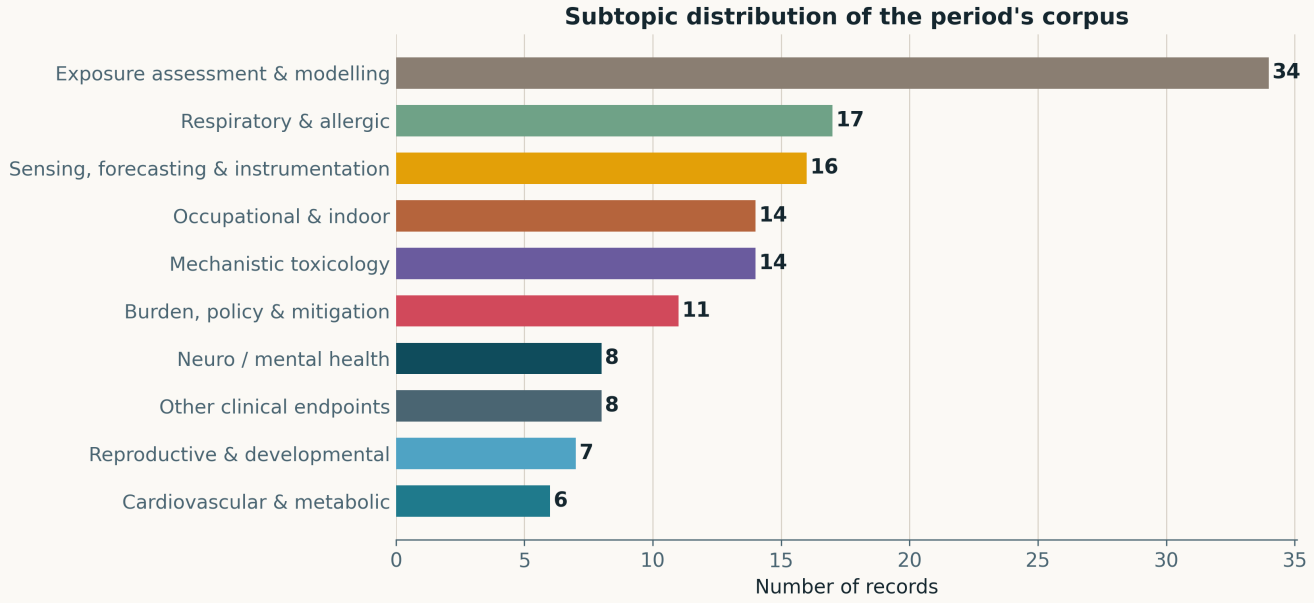


Fig. Pooled subtopic distribution, all ten bands populated: *Exposure* 34 and *Sensing* 16 put the measurement side at **50 of 135 (37 %)**. On the health side *Respiratory* 17 leads, then *Mechanistic toxicology* and *Occupational & indoor* 14 each and *Burden* 11; *Cardiovascular* is smallest at 6. The canonicalised endpoint chart (f2b) is omitted: at this volume it resolves into >60 single-record categories and is unreadable; respiratory and neurodevelopmental endpoints lead, and end-of-induction MRD in childhood leukaemia, transient tachypnoea of the newborn and infant antibiotic prescribing are new endpoint classes for the watch.

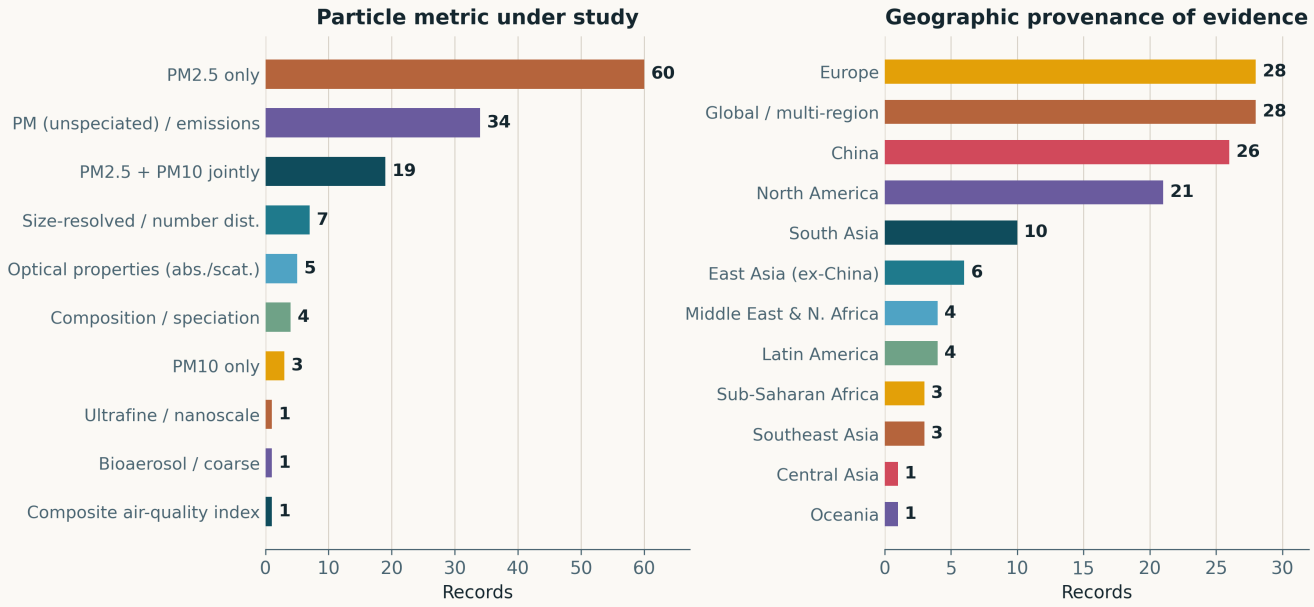


Fig. Left — particle metric; $PM_{2.5}$ alone dominates, with BC/eBC (Oakland inversion, Nepal on-road) and number-concentration work (ironing, UFP kidney, hygroscopic deposition) the secondary metrics. Right — geography; Europe and *Global / multi-region* tie at 28 (the EUPHA abstracts lift Europe), then China 26 and North America 21; South Asia 10 is a series high.

Life-course windows addressed this period (records may span several stages)

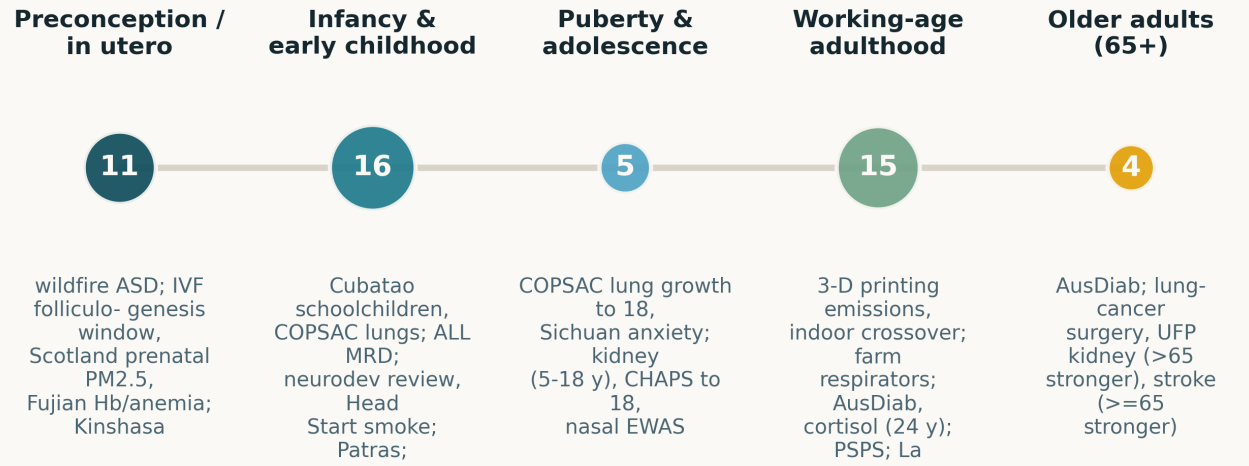


Fig. Life-course windows across the pooled week; counts non-exclusive. **Childhood leads** for the first time (COPSAC, REDIAL, Head Start, Louisiana asthma, infant antibiotics), ahead of working age; preconception / in utero is carried by Medicaid smoke, Taiwan TTN, Chengdu IVF and Indian stillbirth.

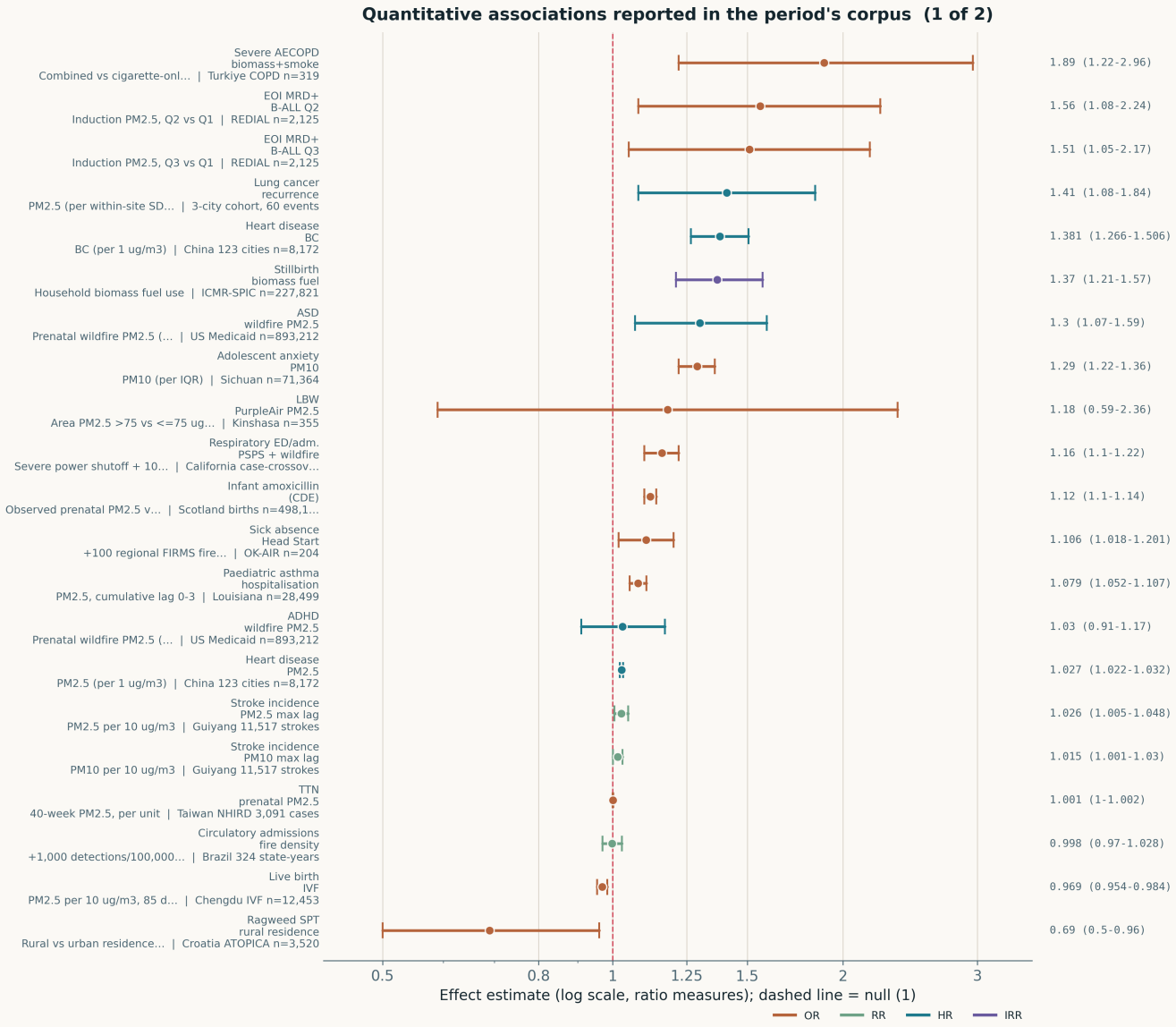


Fig. The **21** ratio-scale estimates (OR 12, HR 5, RR 3, IRR 1), every one with a 95 % CI. **Not mutually comparable and not meta-analysed**: increments include per 1, per 10 $\mu g\,m^{-3}$, per IQR, per within-site SD, quartile and binary contrasts and fire-count units, so no pooled estimate or I^2 is reported. **Heterogeneity is informative inside families**: the Medicaid prenatal-smoke pair diverges by outcome (ASD 1.30 vs ADHD 1.03); REDIAL is non-monotone (Q2 1.56, Q3 1.51, Q4 null); in the 123-city cohort the per-μg BC HR (1.381) is 12 times the $PM_{2.5}$ HR (1.027) in log terms, as expected for a minor, toxic mass fraction. **Three intervals cross unity**: Kinshasa PurpleAir LBW 1.18 (0.59–2.36), Medicaid ADHD 1.03 (0.91–1.17) and Brazilian fire density 0.998 (0.97–1.028). Below-null points are not protective effects: IVF live birth (0.969) is a harm direction, and ragweed sensitisation (0.69) contrasts rural with urban residence. Taiwan TTN (1.001 per $\mu g\,m^{-3}$ over 40 weeks) is precise and negligible. Nested families: REDIAL 2, Medicaid 2, China 123 cities 2, Guiyang 2.

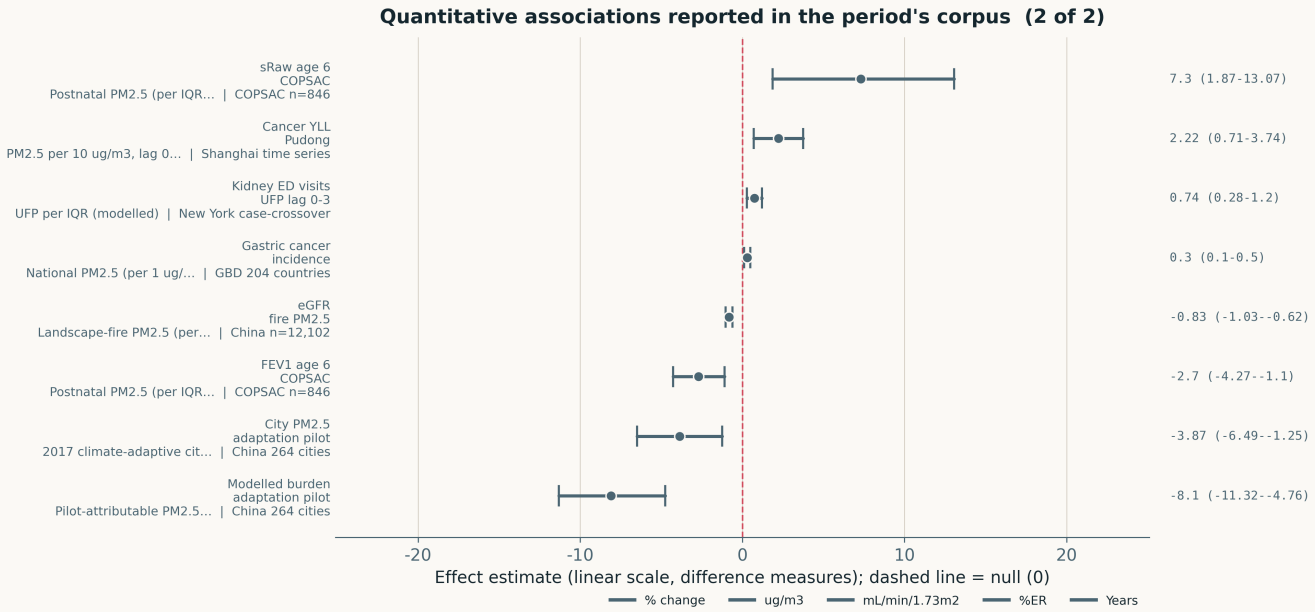


Fig. The 8 difference- and percentage-scale estimates. COPSAC per IQR (2.2 $\mu g\,m^{-3}$) postnatal $PM_{2.5}$: FEV_1 at age 6 −2.70 % (−4.27 to −1.10) and sRaw +7.30 % (1.87–13.07). Landscape-fire $PM_{2.5}$ and eGFR −0.83 mL min^{-1} 1.73 m^{-2} (−1.03 to −0.62). Modelled UFP and kidney ED visits +0.74 % (0.28–1.20) per IQR. China's climate-adaptive city pilot: $PM_{2.5}$ −3.87 $\mu g\,m^{-3}$ (−6.49 to −1.25) and a GEMM-derived burden −8.1 % — the second is an accounting transform of the first, not an independent estimate. GBD gastric cancer (0.3 % per $\mu g\,m^{-3}$) and Pudong cancer YLL (2.22 years) are ecological. Different units on one axis: read each row alone.

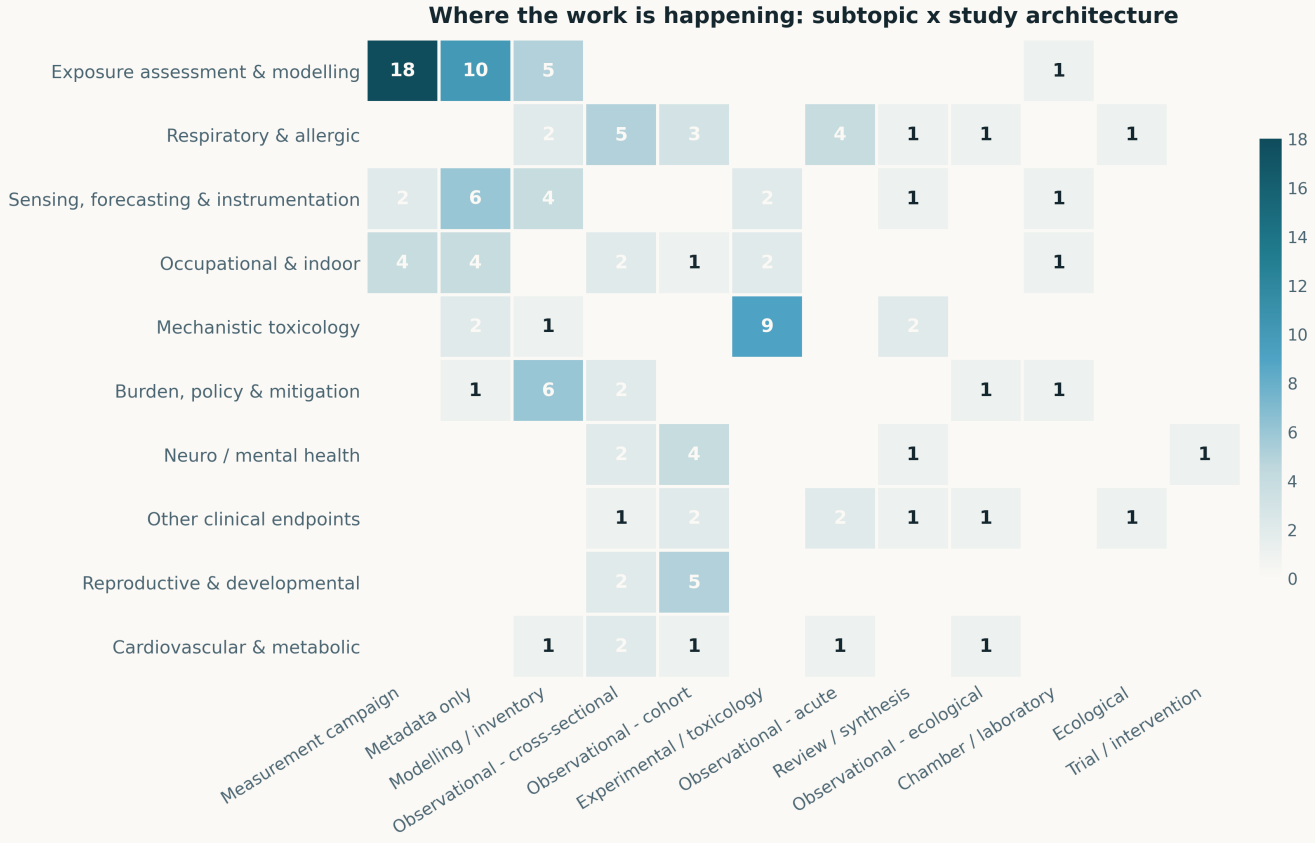


Fig. Subtopic × architecture, pooled. The densest cells are *Exposure* × *Measurement campaign* and *Mechanistic toxicology* × experimental; respiratory work spreads across cohort, acute and cross-sectional designs; sensing is split between instrument evaluation, method papers and ML prediction.

Venues, trial watch, and what the pipeline did

Venues. *Eur J Public Health* leads at **10** — all conference abstracts from the EUPHA 2026 supplement, a batch deposit rather than a trend — then *Toxics* 6, and *Atmos Chem Phys*, *Atmosphere*, *Front Public Health* and Research Square at 5. **Preprints supplied 7** (Research Square 5, medRxiv 1, Preprints.org 1). *Atmos Chem Phys* at 5 is the PubMed-invisible instrumentation literature arriving through the Crossref-ISSN leg, as designed.

Recurring groups and most-cited work. One genuine repeat-group cluster: the **K-HEALTHinAIR** consortium supplied two records (four-country low-cost-sensor monitoring of homes and schools; a Spanish indoor $PM_{2.5}$–COPD burden estimate). Otherwise surname frequency is a naming artefact. Citation counts for records days old are uninformative; the largest datasets — 893,212 Medicaid mother–child pairs, 498,123 Scottish births, 269,945 kidney ED visits, 227,821 Indian pregnancies, 179,218 All of Us adults — each supplied one record.

Trial watch. ClinicalTrials.gov `LastUpdatePostDate` 23–30 October: 4 hits, **1 new** — **NCT07536178** (UBC, *Lungs on Fire*), a controlled woodsmoke-exposure chamber study with FeNO and a CANTAB cognitive battery as co-primary endpoints; MARKOPOLO NCT07111208, APCIQ-BP NCT05903950 and the Taif COPD survey NCT07886346 were record updates. 30 October–1 November: **0 hits** (the weekend days are a structural registry zero). `trials.json` holds **20 trials**. The chamber study is the registry's answer to this week's smoke literature: it is the only design in the set that can separate smoke from co-exposures. The week's other interventional evidence came from the literature (purifier crossover, OK-AIR).

What the harvesting legs did. Europe PMC re-surfaced PubMed records on later days throughout — 18 of the 31 October candidates were 30 October records and went to that issue; the recovery run caught ∼40 cross-day duplicates for 23–29 October. **Crossref-by-ISSN** 22–66/day. **PubMed connector** appended 4–10 records/day. **OpenAlex** HTTP 429 throughout. **arXiv** no in-window entry. **Consensus** 10 hits, **0 new** (both unmatched titles resolved by DOI to shipped records). **Scholar Gateway** not called.

Open gaps

- **Smoke health effects are estimated on modelled smoke.** ECHO, FIRMS and CTM fields carry every smoke estimate this week; the one record using a PurpleAir exposure (Kinshasa low birthweight, n=355, OR 1.18, 0.59–2.36) is too small to inform. A corrected low-cost network is the obvious smoke-exposure instrument and is not yet being used as one.
- **Humidity is being fixed in hardware and still ignored in exposure.** Heating the PMS5003 sample path moved readings toward regulatory conditions, and Kelvin-suppressed growth changes UFP deposition by up to 25 %; no health record propagates RH-correction uncertainty into its estimate.
- **Ultrafine particles: modelled in the epidemiology, measured only indoors.** The New York kidney signal (0.74 % per IQR) rests on CTM number concentration; the only measured PNC in the week is from ironing.
- **Composition faded as fast as it rose.** Last week's ion-resolved cohorts had no successor; measured speciation (Accra 146-site ED-XRF, Czech dual-height ACSM) stayed on the exposure side and did not reach an outcome.
- **The abstract gap eased to 17 %**, still concentrated in day-0 Elsevier and Taylor & Francis deposits.
- **Pipeline defects carried forward**: scheduled runs for 23–29 October did not complete (recovered on 30 October); late Europe PMC re-surfacing makes per-day dedup order-dependent; OpenAlex dead; `.git` lock debris on the delete-restricted mount.

Research direction. The field is moving from *how much* $PM_{2.5}$ to *which* $PM_{2.5}$, and is answering the question first by source (fire, waste burning, freight) rather than by measured chemistry. That favours networks that can label source in real time — absorption/BC channels, size ratios, co-located gas signatures — over networks that only report mass.

Paper-level digest — the week by subtopic

Every record that appeared in a daily issue keeps that issue's critical summary verbatim; nothing here is re-written or re-assessed. Records surfaced without an abstract are shown as such. Ordered by subtopic, then by the date of the issue that first carried the record.

Sensing, forecasting & instrumentation — 16 records

Carried from the daily issues — full summaries (16 of 16 records in this subtopic)

Ai optical trapping Raman single carbonaceous aerosol smoke 2026 Sensors (Basel)

In-Situ Single-Particle Sensing of Carbonaceous Aerosols in Freshly Generated Smoke Using Optical Trapping-Raman Spectroscopy

Optical trapping-Raman spectroscopy captured smoke particles from nylon rope and plastic fibre ignited by the trapping laser itself, and analysed them airborne within seconds. Particles of several micrometres (~10 pg) gave full Raman fingerprints; band analysis separated two particle types, and spatially resolved spectra showed organic carbon and soot segregated within single particles. Limitation: freshly generated, supermicron, laser-ignited particles from two polymer fuels in an open chamber; ambient carbonaceous PM is mostly accumulation-mode, aged and far lighter, and the trapping laser may heat and alter what it measures. No throughput or detection-limit figures. Relevance: low-moderate - a reference technique for particle chemistry, not a field sensor.

doi: 10.3390/s26206527

Bei UPLC-MS/MS 74 antibiotics in $PM_{2.5}$ Beijing 2026 J Chromatogr A

Optimization and application of a sensitive UPLC-MS/MS method for the quantitative determination of antibiotics in $PM_{2.5}$

Method for 74 antibiotics in $PM_{2.5}$: T3 column, formic-acid mobile phase, Oasis HLB extraction with Na2EDTA. Recoveries 69.7-131.7%, RSD 1.0-18.1% at three spike levels, quantification limits down to 0.37 pg/m3. On 29 Beijing 24-h samples, 22 compounds exceeded detection limits (max 121.38 pg/m3); roxithromycin was in every sample and oxytetracycline had the highest mean (14.48 pg/m3). Limitation: 29 samples from one city and season give occurrence, not exposure; recoveries near 70% for some classes bias low, and the link to resistance spread is asserted, not tested. Relevance: low for sensing - a filter-bound chemical fraction no optical sensor sees, but co-located filter sampling at sensor nodes could support it.

doi: 10.1016/j.chroma.2026.467564

Kara Pythagorean fuzzy $PM_{2.5}$ monitoring station placement Trabzon 2026 J Air Waste Manag Assoc

A Pythagorean Fuzzy decision making framework for $PM_{2.5}$ monitoring station placement in urban areas: A case study of Trabzon, Turkiye

Modified Delphi to set eight criteria, PF-SWARA to weight them and PF-CODAS to rank five candidate sites in Trabzon. $PM_{2.5}$ level weighted highest, then population density and distance to existing stations; a shopping-mall site ranked first. Limitation: the criterion weights are expert judgment; there is no objective function (coverage, representativeness, variance reduction) and no test that the chosen site improves exposure estimation. Weighting '$PM_{2.5}$ concentration' highest is circular when the network is being sited to measure it, and five alternatives make the ranking near-trivial. Relevance: medium - a contrast case for data-driven siting (kriging variance, information gain) that LCS deployments use.

doi: 10.1080/10962247.2026.2754941

Manchanda hyperlocal BC sensing Bayesian inverse modeling West Oakland 2026 Sci Adv

Hyperlocal sensing and inverse modeling reveal community impacts of urban air pollutant emissions

Dense mobile and fixed-site BC measurements were assimilated in a Bayesian inversion to produce hourly 150 x 150 m emission maps for West Oakland, a freight-corridor community. The posterior raised total BC emissions by ~33% over the bottom-up inventory and quadrupled the share of population-weighted exposure attributed to neighbourhood and port-related sources (5% to 20%), concentrated in low-income blocks. Sensitivity runs with ~10 fixed monitors (~1 site/km2) plus time-averaged mobile maps recovered similar results. Limitation: the posterior is only as good as the transport kernel and prior error covariance; unpermitted sources are inferred, not verified on the ground, and one community limits transfer. Relevance: high - a direct recipe for turning a sparse low-cost BC/PM network into source-resolved emission corrections.

doi: 10.1126/sciadv.aeb7647

Mercado-Velazquez low-cost sensors three-tier London calibration transferability 2026 Sensors (Basel)

How Accurate Are Low-Cost Air Quality Sensors? A 10-Month Study of Three-Tier Comparison with Three Calibration Methods and Cross-Site Transferability

Ten-month co-location of low-cost (MOS, optical), mid-cost electrochemical and reference instruments at two London sites. Calibration by manufacturer curve, T/RH linear regression and ML. Field-calibrated $PM_{2.5}$ explained over half of reference variance at both sites, and linear calibration matched ML. $PM_{2.5}$ calibrations trained April-October 2025 held through winter, whereas NO_2 and CO did not; transferred between sites, $PM_{2.5}$ reached R2 >=0.56 and gases R2 <=0.28. The mid-cost NO_2 and low-cost O_3 sensors failed under all methods. Limitation: two sites 1.6 km apart in one city share aerosol type and humidity regime, so 'transferability' is tested at its easiest. Relevance: high - supports simple T/RH regression for $PM_{2.5}$ nodes and mandatory co-location for gas channels.

doi: 10.3390/s26206441

Ouimette PMS5003 PurpleAir internal heating hygroscopic mass reduction 2026 Preprints.org (preprint)

Air Flow, Heat Transfer, and Hygroscopic Aerosol Mass Reduction in the Plantower PMS5003 Sensor and PurpleAir Monitor

Lab and field work on temperature in the particle detection zone of the PMS5003, standalone and inside a PurpleAir. The PurpleAir housing warms the detection zone more than a bare PMS5003, lowering sample RH; a wraparound USB heater pushed sample RH down to regulatory-monitor conditions. In Santa Rosa and Winston-Salem (against a T640x), external heating lowered reported $PM_{2.5}$ at high ambient RH, and the unheated PurpleAir already read below a standalone PMS. CFD characterised airflow, residence time and temperature. Limitation: two sites, no quantified bias reduction or growth-factor fit in the abstract; preprint. Relevance: high - EPA-style RH corrections assume ambient RH, but the sensor sees a warmed, drier sample; corrections built on bare-PMS data may not transfer to housed units.

doi: 10.20944/preprints202610.2325.v1

Schmidt AOD gap filling iterative singular spectrum analysis AERONET Cuiaba 2026 Environ Monit Assess

Gap filling in long-term AOD time series using Iterative Singular Spectrum Analysis at a tropical AERONET site

Applies iterative singular spectrum analysis (ISSA) to fill a 2005-2024 daily AOD record from the Cuiaba-Miranda AERONET station, where ~47% of days are missing, mostly in the cloudy wet season. Climatological (day-of-year mean) initialisation beat global-mean initialisation, giving lower error and better temporal consistency, and the cross-validation design materially changed the optimal reconstruction parameters. No error magnitudes are given in the abstract. Limitation: gaps are not missing at random - they coincide with cloud and wet-season aerosol regimes - so a climatology-anchored fill will bias toward dry-season behaviour exactly where data are absent. Relevance: low-medium - the same informative-missingness problem affects sensor networks with humidity-triggered dropouts and satellite-surface fusion.

doi: 10.1007/s10661-026-16067-6

Spinosa volcanic Saharan dust biomass burning aerosol advections Naples 2026 Atmos Res

Multiple advections of volcanic, Saharan dust and biomass burning aerosols over Naples

METADATA ONLY. Title and authors only (Spinosa, Manzo, Amoruso, Sannino et al.). No abstract was retrievable at harvest. The title describes overlapping advections of volcanic, Saharan dust and biomass-burning aerosol over Naples; the author group operates lidar there, so a profiling case study is likely but is not confirmed. Instruments, dates and surface PM impacts are not available; nothing is inferred. Relevance: medium - mixed dust/smoke/volcanic layers are the cases where optical low-cost sensors misreport mass most, because refractive index and size change simultaneously.

doi: 10.1016/j.atmosres.2026.109425

Trivanovic periodic technical inspection particle number devices gasoline exhaust 2026 J Aerosol Sci

Assessment of existing PN-PTI devices to measure gasoline exhaust

METADATA ONLY. Title and authors only (Trivanovic, Melas, Auderset, Hammer et al.). No abstract was retrievable at harvest. The title indicates an evaluation of existing particle-number periodic-technical-inspection (PN-PTI) instruments - developed and type-approved for diesel vehicles with particle filters - when applied to gasoline exhaust, whose size distribution sits lower and nearer the counters' 23 nm or 10 nm cut. Instruments, test vehicles and error magnitudes are not available; nothing is inferred. Relevance: high - PN-PTI counters are the first mass-produced, low-cost, legally metrological particle counters, and their size-cut behaviour is the same question for community UFP sensing.

doi: 10.1016/j.jaerosci.2026.106901

Vannini biochar passive biomonitor potentially toxic elements lichens 2026 Atmos Environ

Wood-derived biochar as a potential sustainable alternative to lichens for monitoring airborne potentially toxic elements

METADATA ONLY. Title and authors only (Vannini, Sidoli, Pagano, Magnani). No abstract was retrievable at harvest. The title proposes wood-derived biochar as a standardisable passive accumulator of airborne potentially toxic elements in place of transplanted lichens. Exposure design, elements, comparison data and agreement are not available; nothing is inferred. Relevance: medium - a cheap passive element sampler could add composition to dense optical-mass networks, provided uptake can be calibrated to air concentration.

doi: 10.1016/j.atmosenv.2026.122457

Waghmare machine learning explainable AI $PM_{2.5}$ prediction Nagpur 2026 MethodsX

Machine learning models with explainable AI for $PM_{2.5}$ prediction for Urban air quality assessment

Six regressors (decision tree, extra trees, random forest, KNN, XGBoost, SVR) trained on ~4,000 observations of $PM_{2.5}$, NO_2, SO_2, NH_3, O_3 and meteorology in Nagpur, 70/30 split with 10-fold CV. A single decision tree scored best: test R2 0.90, RMSE 7.5-8.9 $\mu g\,m^{-3}$, MAE 5.4-6.6 $\mu g\,m^{-3}$; SHAP ranked NO_2, O_3, RH and NH_3 top. Limitation: random splits of autocorrelated time series leak information, so skill is inflated; co-pollutants measured at the same station make this a nowcast, not a forecast, and a lone tree beating ensembles is a warning sign. Relevance: low-medium - illustrates the validation discipline (blocked or spatial CV) that sensor-network ML calibration needs.

doi: 10.1016/j.mex.2026.104217

Wang $PM_{2.5}$-bound PAH association rules machine learning forecasting Nagoya 2026 Environ Monit Assess

Characterizing statistical association patterns and local forecasting of $PM_{2.5}$-bound polycyclic aromatic hydrocarbon concentrations in Nagoya, Japan

Hourly $PM_{2.5}$-bound PAHs in Nagoya with association-rule mining and ML under time-blocked cross-validation. Low wind speed was the most consistent condition for PAH accumulation; BC and NOx peaked at rush hours. XGBoost and RF predicted high-molecular-weight PAHs best (R2 0.81-0.93, RMSE 0.02-0.11); SVR and MLP were poor. BaP dominated toxicity; lifetime cancer risks were below the USEPA negligible threshold. Limitation: a 10-day observation period, so the models learn one synoptic episode, and the authors themselves restrict the rules to site-specific description. Relevance: moderate - a BC-plus-meteorology proxy for PAHs is the kind of virtual sensor a network could run, if trained on far longer records.

doi: 10.1007/s10661-026-16083-6

Washeem machine learning condensation bioaerosol sampler collection volume 2026 J Aerosol Sci

Machine-Learning-Guided Prediction and Temperature Optimization for Collection Volume Control in a Condensation-Based Bioaerosol Sampler

METADATA ONLY. Title and authors only (Washeem, Li, Jansen, Shirkhani et al.). No abstract was retrievable at harvest. The title describes using machine learning to predict and control the liquid collection volume of a condensation-growth bioaerosol sampler by optimising its temperature set points. Instrument, training data and performance are not available; nothing is inferred. Relevance: low - sampler control engineering for bioaerosol, adjacent to but outside mass-concentration sensing.

doi: 10.1016/j.jaerosci.2026.106907

Wu source-dependent organic aerosol hygroscopicity eastern China biomass burning 2026 Atmos Chem Phys

Source-dependent organic aerosol hygroscopicity in eastern China: observational evidence for conditional enhancement under biomass-burning influence

Winter field campaign at a rural eastern-China site retrieving organic-aerosol hygroscopicity from humidified nephelometry and ZSR closure. kappa_OA ranged from ~0 to 0.49 (mean 0.11 +/- 0.11) and correlated only weakly with O:C; instead it rose with biomass-burning influence, and ageing enhanced kappa_OA only when biomass burning dominated. Highly oxidised OA under weak biomass influence stayed weakly hygroscopic. Limitation: kappa_OA is a residual after subtracting inorganic contributions, so errors in the inorganic closure load onto the organic term; one site, one season. Relevance: high - humidity correction of optical PM sensors usually assumes a fixed kappa; this shows kappa shifts with source, so smoke-affected periods need a different growth correction.

doi: 10.5194/acp-26-15333-2026

Zhao spin-trapping EPR radicals atmospheric fine particulate matter cavity quality factor 2026 Aerosol Sci Technol

Optimized spin-trapping method for radicals detection induced by atmospheric fine particulate matter: Critical roles of temporal evolution and cavity quality factor

METADATA ONLY. Title and authors only (Zhao, Chen, Ding, Zhang et al.); no abstract retrievable from Crossref, Europe PMC or Semantic Scholar at harvest. The title describes optimising EPR spin-trapping for radicals generated by $PM_{2.5}$, identifying the time course of adduct formation and the cavity quality factor as critical. Sample type, n and performance metrics cannot be stated. Limitation: unverifiable beyond the title; spin-trap adducts decay on minute timescales, so inter-lab oxidative-potential comparisons hinge on exactly these timing choices. Relevance: low - offline oxidative-potential metrology.

doi: 10.1080/02786826.2026.2750406

zum Berge multicopter vibration optical particle counter accuracy 2026 Aerosol Sci Technol

Influence of multicopter-induced vibrations on the measurement accuracy of optical particle counter

METADATA ONLY. Title and authors only (zum Berge, Platis, Bange). No abstract was retrievable from Crossref or Semantic Scholar at harvest. The title describes quantifying how airframe vibration from a multicopter degrades the accuracy of an optical particle counter carried aloft - relevant to the growing use of drone-borne OPCs (often the same Alphasense/Plantower-class sensors used in ground networks) for vertical PM profiling. Design, sensor model, vibration spectrum and error magnitudes are not available; nothing is inferred. Relevance: high - vibration is an uncharacterised error term for mobile and UAV-mounted low-cost OPCs; retry the abstract first.

doi: 10.1080/02786826.2026.2747765

Exposure assessment & modelling — 34 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (34 OF 34 RECORDS IN THIS SUBTOPIC)

Accra $PM_{2.5}$ elemental composition PMF source shift e-waste burning 2026 Environ Health Perspect

Shifting Dominant Air Pollution Sources in Sub-Saharan Africa's Changing Urban Environment: Analysis of Elemental Composition and Source Contributions of Ambient Fine Particulate Matter ($PM_{2.5}$) Pollution in Accra, Ghana

Citywide campaign April 2019-June 2020: >600 weekly gravimetric $PM_{2.5}$ filters from 146 sites, ED-XRF and PMF, compared with four legacy sites from 2007-2008. Harmattan crustal dust was 52-68% of elemental mass. Outside Harmattan, 4-6 factors: biomass burning 11-32%, traffic/road dust 9-36%, solid and e-waste combustion 0-35%. Over a decade the biomass contribution fell by ~16% (5.2-25.8) while waste burning rose by 23 points (15.9-30.0). Limitation: ED-XRF elements miss the carbonaceous fraction, so the shares are of elemental mass, not $PM_{2.5}$ mass, and the decadal comparison rests on four sites. Relevance: high - a 146-site gravimetric network is the speciation reference a West African LCS network needs; e-waste burning is a source optical sensors cannot distinguish.

doi: 10.1021/ehp.6c00323

Aktypis nucleation-mode particles marine atmosphere downwind urban 2026 Atmos Environ

Frequent Appearance of Nucleation-Mode Particles in the Marine Atmosphere Downwind of an Urban Area

METADATA ONLY. Title and authors only (Aktypis, Vasilakopoulou, Matrali, Kaltsonoudis et al.). No abstract was retrievable at harvest. The title reports frequent nucleation-mode particles in marine air downwind of an urban area, implying urban plumes seed new-particle formation or carry fresh emissions over water. Site, instrumentation and event statistics are not available; nothing is inferred. Relevance: low - sub-25 nm particles are invisible to optical sensors and carry negligible mass.

doi: 10.1016/j.atmosenv.2026.122458

Aloraini negative control PSCF CWT trajectory dust attribution Riyadh 2026 Atmosphere

Transport Pathways Versus Dust Sources: A Negative-Control Test of Trajectory-Statistical Attribution in Riyadh, Saudi Arabia

Ten years of Riyadh METARs separated locally lifted dust (BLDU/BLSA, 179 days) from suspended dust (DU/SA, 345). ERA5 models predicted local lifting better (held-out AUC 0.914 vs 0.765). PSCF and CWT fields from 14,612 HYSPLIT trajectories were then run on the local class as a negative control: it was more enriched over a priori source regions than the suspension class (permutation p <=0.007), and wind matching erased most structure in both. Dust days also fell 85.9% in 2023-2024 without model explanation. Limitation: METAR codes are visibility-based, not PM. Relevance: high - trajectory 'source maps' from single receptors, common in LCS papers, mostly reflect transport climatology.

doi: 10.3390/atmos17111055

Alvi PMF trace elements PM_{10} Faisalabad health risk 2026 J Hazard Mater

PMF Resolved Source Identification & Health Risk Assessment of Trace-elements in Airborne Particulate Matter (PM_{10}) from South-Asian Megacity

Filter PM_{10} in Faisalabad analysed by ICP-OES: mean 517 +/- 206 $\mu g\,m^{-3}$, ~10x WHO, higher in summer than autumn or winter. Element order Fe > Al > Si > Zn > Cu > Pb > Cd. PMF resolved seven sources: metal works, vehicle exhaust and non-exhaust, industrial fossil fuel, municipal waste incineration and industrial mixed dust. Cumulative carcinogenic risk 1.22e-3 (adults) and 5.05e-4 (children). Limitation: ICP-OES detection limits for As, Cr and Cd are marginal at these loadings, PMF with ~20 elements and no carbon or ions cannot separate combustion sources robustly, and n of samples is not given. Relevance: low-moderate - coarse, crustal-dominated PM_{10} at these levels saturates optical LCS assumptions.

doi: 10.1016/j.jhazmat.2026.144032

Anjum Idaho $PM_{2.5}$ 2005-2024 wildfire vs wood-smoke regimes 2026 Research Square (preprint)

Long-Term Spatiotemporal Variability of Fine Particulate Matter ($PM_{2.5}$) Across Idaho, USA, Using Statistical and Geospatial Tools

EPA AQS daily $PM_{2.5}$ at 32 Idaho sites, 2005-2024, aggregated to 4,260 site-months; Mann-Kendall/Theil-Sen trends, PCA and IDW. Mean 7.53 $\mu g\,m^{-3}$ (site range 3.16-12.08). Bimodal climatology: an August-September wildfire peak and a November-February wood-smoke/inversion peak; July-September carries 36.6% of annual mass. Cold-season and background series declined while fire-season $PM_{2.5}$ rose 2.54 $\mu g\,m^{-3}$ per decade, holding the annual mean flat; 32.7% of site-years exceed the 9.0 $\mu g\,m^{-3}$ standard. Limitation: unreviewed preprint; IDW over 32 sites in mountainous terrain is not an exposure surface, and the 'regimes' are PCA loadings, not source apportionment. Relevance: high - two diverging seasonal regimes in inversion valleys are precisely where sparse FRM coverage fails and a calibrated PurpleAir layer (smoke-specific correction) adds most.

doi: 10.21203/rs.3.rs-11373778/v1

Baliaka Eaton urban firestorm airborne toxic metals lead Los Angeles 2026 Sci Adv

Fire at the boundary: Airborne toxic metals from the 2025 Los Angeles urban firestorm

Hourly airborne trace-element measurements during the January 2025 Eaton fire tracked the transition from structure to wildland burning via the K/Pb ratio. Relative to wildfire smoke, the urban plume was enriched >50-fold in lead, ~7-fold in halogens and ~3-fold in arsenic, attributed to vehicles, electronics and building materials. An estimated 7 million residents (range 4.3-8.0 million) experienced 24-h lead above 50 ng/m3, >80 times background. Limitation: population exposure is extrapolated from a small number of monitoring points through a dispersion estimate, and the 24-h lead threshold has no health-based standard counterpart (NAAQS is a 3-month mean). Relevance: high - optical PM sensors would register the smoke but not its metal toxicity; speciation is needed.

doi: 10.1126/sciadv.aei4455

Bei air moisture particulate matter winter Guanzhong Basin 2026 Atmos Pollut Res

Contributions of air moisture to particulate matter pollution during wintertime in the Guanzhong Basin, China

METADATA ONLY. Title and authors only (Bei, Chen, Jiang, Tie et al.). No abstract was retrievable at harvest. The title reports an analysis of how air moisture contributes to wintertime particulate pollution in the Guanzhong Basin. Design, mechanism, metrics and magnitude are not available; nothing is inferred. Relevance: potentially high - humidity-driven aerosol water uptake is the dominant error term in optical low-cost PM sensing, so a basin-scale moisture budget bears directly on correction. Retry abstract first.

doi: 10.1016/j.apr.2026.103256

Chakraborty fog intensity droplet size aqueous SOA Kanpur 2026 Environ Sci Technol

Fog Intensity and Droplet Size-Resolved Aqueous Secondary Organic Aerosol (aqSOA) Formation in Polluted Winter Fog: New Insights from a Combined Observational and Modeling Study

Two winters of size-resolved fogwater at Kanpur analysed by offline HR-AMS and PMF; three highly oxidised factors (O/C 0.95-1.19) were summed as an aqSOA proxy and scaled by liquid water content and WSOC. Across seven events aqSOA was 3.1-8.4 $\mu g\,m^{-3}$, a median ~7% of ambient SOA and 5% of OA, up to 13% of SOA in the densest fogs. Fine droplets carried 43-84% of aqSOA, linked to dissolved Fe and Cu and longer residence. Limitation: the aqSOA proxy is a PMF construct, and offline AMS of fogwater may not represent residual particle mass after evaporation. Relevance: medium - Indo-Gangetic fog episodes are when optical PM sensors fail outright from droplet interference, while secondary mass is being added.

doi: 10.1021/acs.est.6c09700

Dhital on-road black carbon PM exposure Nepal route travel mode 2026 J Air Waste Manag Assoc

On-road exposure to black carbon and particulate matter in Nepal: Effects of route attributes and travel modes

Mobile monitoring across Nepali road types, gradients and travel modes. Mean on-road eBC 45.7 +/- 32.5 $\mu g\,m^{-3}$, ~11x background; eBC made up 51% of on-road $PM_{2.5}$ and 75% of PM_1. Concentrations rose from residential streets to intercity highways. Upgraded orbital-highway segments had lower $PM_{2.5}$ and PM_{10} but unchanged eBC and PM_1, so paving cut resuspended dust rather than exhaust. High-gradient (4.5%) segments: median eBC 86.7 vs 41.5 $\mu g\,m^{-3}$ on 0.6% segments. Motorcycles gave the highest exposure, then non-AC and AC buses. Limitation: no n of runs or season control is reported, and the old-vs-upgraded comparison is not a before-after design. Relevance: high - shows optical $PM_{2.5}$ nodes miss the exhaust-dominated BC contrast that drives on-road exposure.

doi: 10.1080/10962247.2026.2754932

Fang end of uniformity spatial climatic toxicological redefinitions air quality 2026 Environ Sci Technol

The End of Uniformity: Spatial, Climatic, and Toxicological Redefinitions of Air Quality

METADATA ONLY. Title and authors only (Fang, Song, Weon). No abstract was retrievable at harvest; the format is likely a viewpoint or perspective. The title argues against uniform air-quality definitions along three axes - spatial heterogeneity, climate-driven change and toxicity beyond mass - which maps directly onto hyperlocal sensing, wildfire-era exposure and component- or oxidative-potential-based metrics. Arguments and any proposals are not available; nothing is inferred. Relevance: medium - if it advocates spatially resolved or toxicity-weighted standards, it is a policy argument for distributed networks; retry the text on a later run.

doi: 10.1021/acs.est.6c14170

Jabbari-Hichri on-road emissions light-duty vehicles particulate gaseous pollutants 2026 Atmos Pollut Res

On-road emissions of light-duty vehicles: detailed analysis of particulate and gaseous pollutants

METADATA ONLY. Title and authors only (Jabbari-Hichri, Azizi, Guiot, Boreave et al.); no abstract retrievable at harvest. The title describes detailed on-road measurement of particulate and gaseous emissions from light-duty vehicles. Vehicles, instruments and emission factors cannot be stated. Limitation: unverifiable beyond the title; on-road (PEMS) particle number measurements usually exclude volatile and sub-23 nm particles, which matters for comparing with roadside data. Relevance: low.

doi: 10.1016/j.apr.2026.103252

Julaha vertical aerosol chemical composition 230 m 4 m ACSM rural background 2026 Environ Pollut

Meteorological Effects on the Vertical Variability of Aerosol Chemical Composition at a Rural Background Site

Concurrent dual-height speciation, December 2019-May 2020: compact ToF-AMS at 230 m, ToF-ACSM at 4 m, Sunset OCEC. Every component was higher at 4 m; median 230/4 m ratios 0.55-0.82. Carbonaceous species had the steepest gradients and winter peaks; secondary inorganics were more homogeneous. Under inversions with mixing height <230 m, surface concentrations were 2-3x those aloft. Limitation: two different instruments at the two heights (AMS vs ACSM) - inter-instrument bias contributes directly to the ratio, and the abstract gives no co-location intercomparison. One site, one half-year. Relevance: high - a 2-3x surface/aloft contrast under shallow mixing is the bias any satellite-AOD or tower-based calibration of ground sensors must model.

doi: 10.1016/j.envpol.2026.129431

Kalkavouras ammonia sulfuric acid BVOC urban new particle formation Athens 2026 Environ Pollut

Assessing the combined roles of ammonia, sulfuric acid, and biogenic volatile organic compounds in urban new particle formation

Seven months (June 2023-January 2024) of particle number size distributions, NH_3 and BVOCs at the Thissio urban background site in Athens, with a proxy for H2SO4. Mean NH_3 2.3 +/- 0.7 $\mu g\,m^{-3}$. NPF occurred on 19% of days, most often in autumn (29-33%) when ventilation and condensation sink were low. Precursor levels barely differed between event and non-event days; the highest J14 came at intermediate NH_3 and H2SO4 with low BVOCs, while growth rates rose with NH_3 and isoprene/monoterpenes. Limitation: H2SO4 is a proxy, not measured, and the record lacks a full annual cycle. Relevance: medium - NPF bursts dominate urban number concentration but are invisible to mass-based optical sensors, which see nothing below ~300 nm.

doi: 10.1016/j.envpol.2026.129453

Kwon seasonal $PM_{2.5}$ secondary inorganic aerosol livestock rural Ulsan 2026 Atmosphere

Seasonal Composition of $PM_{2.5}$ and Secondary Inorganic Aerosols at a Rural Site near Livestock Facilities in Ulsan, South Korea

Weekly 24 h filters, June 2023-November 2025, at a rural site near livestock facilities. $PM_{2.5}$ peaked in spring; nitrate peaked in winter and was lowest in summer (seasonal means 3.00 and 0.26 $\mu g\,m^{-3}$). Elevated ammonium-to-sulfate ratios were frequent in spring and winter, but nitrate exceeded sulfate by mass far more often in winter, so the $PM_{2.5}$ maximum and the nitrate maximum do not coincide. Limitation: one sample a week, no gas-phase NH_3/HNO_3 matched in time and no thermodynamic modelling, so ammonia-limited vs nitrate-limited regimes cannot be assigned, as the authors state. Relevance: low-medium - semi-volatile nitrate is a known source of LCS-vs-FEM divergence.

doi: 10.3390/atmos17111042

Lian firework barium aerosols single-particle SPAMS Guangzhou 2026 Atmos Chem Phys

Evolution of firework-related barium aerosols: insights from single-particle analysis and mass concentration monitoring

Online Ba mass monitoring and SPAMS at four Guangzhou sites through firework events. Ba rose 1-3 orders of magnitude above background. The urban-core site under a complete firework ban peaked lower than a suburban site with time- and place-restricted use, despite higher population density - consistent with, but not proof of, a ban effect. Chloride-rich and nitrate-dominated Ba-containing particle classes were interpreted as fresh and aged, tracking chloride displacement and nitrate uptake. Limitation: four sites and a few events; the ban contrast is confounded by local emissions and transport, as the authors acknowledge, and SPAMS is semi-quantitative. Relevance: medium - firework episodes are a known optical-sensor artefact (hygroscopic, metal-rich particles), and this characterises their chemistry.

doi: 10.5194/acp-26-15101-2026

Liang deep learning low-visibility events fog-haze eastern China 2026 Atmos Chem Phys

Classification and quantification of low-visibility events using deep learning over eastern China

A unified CNN fuses geostationary imagery, meteorology and surface $PM_{2.5}$ to map and classify low-visibility events, separating polluted fog-haze from clean fog over land and sea. Scenario analysis: a 20% cut in $PM_{2.5}$ shrinks winter fog-haze area by 43% but population exposure to fog-haze only by 15%, because the most populated and most polluted regions coincide; >40% $PM_{2.5}$ reduction is needed there. Limitation: the scenario perturbs $PM_{2.5}$ as a model input, so it inherits the learned correlation rather than fog microphysics (hygroscopic growth, aerosol-fog feedback); classification skill metrics are not in the abstract. Relevance: medium - humidity-driven haze is the regime where optical LCS fail.

doi: 10.5194/acp-26-15003-2026

Lu heterogeneous ammonia uptake SOA WRF-Chem North China Plain 2026 Atmos Chem Phys

Impacts of the heterogeneous ammonia uptake on air quality in the North China Plain

A first-order NH_3 uptake on SOA forming NOCs was added to WRF-Chem for November 2017, with uptake coefficient gamma 1e-5 to 1e-3. Laboratory decline in reaction efficiency and measured NOC burdens support gamma=1e-5 as a conservative upper limit. At that value, NH_3 fell 0.62%, inorganic aerosol and $PM_{2.5}$ fell <0.06%, and the pathway produced 1.2 $\mu g\,m^{-3}$ of NOCs. Limitation: one month, one region, and the constraining gamma is chosen partly from the NOC burden it is meant to explain. Relevance: low - a negligible $PM_{2.5}$ mass term; useful only as a null for the chemistry.

doi: 10.5194/acp-26-15257-2026

Mandujano change-point detection trend estimation long-term PM records 2026 Environ Sci Atmos

Assessing the Effect of Change-Point Detection Techniques on Trend Estimation in Long-Term Particulate Matter Records

METADATA ONLY. Mandujano, Gonzalez, Hernandez, Guillen et al. Only the opening sentence of the abstract was retrievable from Crossref or Semantic Scholar: long-term PM trend analyses assume a single trend direction across the record, which can mask structural changes. The title states that change-point detection techniques are compared for their effect on trend estimates. Location, data set, methods and results are not available at harvest; nothing is inferred. Relevance: medium - instrument changes and station relocations are exactly the breakpoints that contaminate trends from mixed reference/LCS archives; retry the abstract.

doi: 10.1039/d6ea00110f

Netto metal distribution $PM_{2.5}$ smelting Rio de Janeiro solubility risk 2026 Atmos Pollut Res

Metal distribution in atmospheric particulate matter ($PM_{2.5}$) from urban areas with smelting activities in the Rio de Janeiro state, Brazil: Solubility, correlation and risk assessment

METADATA ONLY. Title and authors only (Netto, da Silva, Diniz et al.); no abstract retrievable at harvest. The title describes $PM_{2.5}$ metal concentrations, their soluble fractions and risk assessment in urban areas with smelting in Rio de Janeiro state. Concentrations, n and risk estimates cannot be stated. Limitation: unverifiable beyond the title; risk assessments on total metal mass overstate dose unless the bioaccessible (soluble) fraction is used, which the title suggests is addressed. Relevance: low - filter chemistry.

doi: 10.1016/j.apr.2026.103250

Pani Southeast Asian biomass burning smoke aerosol light extinction apportionment high-altitude western North Pacific 2026 Atmos Res

Impact of peninsular Southeast Asian biomass-burning smoke on chemical apportionment of aerosol light extinction at a high-altitude remote environment in the western North Pacific

METADATA ONLY. Title and authors only (Pani, Lee, Lin); no abstract retrievable at harvest. The title describes apportioning aerosol light extinction by chemical component at a high-altitude remote site in the western North Pacific under transported Southeast Asian biomass-burning smoke. Concentrations and apportionment fractions cannot be stated. Limitation: unverifiable beyond the title; IMPROVE-type mass extinction efficiencies are fixed assumptions that aged smoke violates. Relevance: medium - mass-extinction relations are what optical PM sensors implicitly assume.

doi: 10.1016/j.atmosres.2026.109421

Park $PM_{2.5}$ composition sources COVID-19 social distancing Seoul 2026 Toxics

Restructuring of $PM_{2.5}$ Composition and Source Contributions Across COVID-19 Social Distancing Periods: A Case Study at an Urban Monitoring Site in Seoul, South Korea

Feb 2018-Mar 2024 at one Seoul site, split into pre-, during- and post-distancing periods; PMF with GAM adjustment for meteorology and season. $PM_{2.5}$ stayed lower during and after distancing. Secondary inorganic aerosol remained dominant with no period effect after adjustment; metal industry fell then partly rebounded; construction dust stayed elevated and paved-road dust rose post-distancing. Limitation: one site, and 'post-distancing' is also the period of Korea's seasonal-management PM policy and Chinese emission reductions, so the periods confound policy and transboundary change with social activity. No effect sizes in the abstract. Relevance: medium - non-exhaust road dust is a coarse-mode source optical LCS undercount.

doi: 10.3390/toxics14100886

Qu $PM_{2.5}$-bound heavy metals source apportionment Laibin winter 2026 Toxics

Source Apportionment and Health Risk Assessment of $PM_{2.5}$-Bound Heavy Metals in Wintertime Laibin, Guangxi, China

42-day winter 2025 campaign: mean $PM_{2.5}$ 47.6 $\mu g\,m^{-3}$, higher at night with lower wind and shallower boundary layer; nine metals summed 363.0 ng/m3, dominated by Zn, Cr and Ni, with severe Cd and significant Ni enrichment. PMF: non-ferrous smelting plus coal combustion 43%, industrial combustion 34%, traffic 23%; traffic share rose at night. Hazard index <1 for all groups; carcinogenic risk slightly above 1e-4 under winter scenarios, Cr dominant. Limitation: PMF on nine species and ~84 day/night samples is under-determined (the authors say 'preliminarily'), and Cr risk depends on an assumed Cr(VI) fraction. Relevance: low-medium - nocturnal boundary-layer peaks are visible to any LCS node.

doi: 10.3390/toxics14100923

Rosu multi-instrument Saharan dust event South-Eastern Europe 2026 Atmos Res

A multi-instrument analysis of a Saharan dust event in South-Eastern Europe

Metadata only: no abstract at Crossref, Europe PMC or Semantic Scholar on the issue date (Elsevier day-0 deposit). The title reports a multi-instrument analysis of a Saharan dust intrusion over South-Eastern Europe; instruments, sites, PM metrics and results are not available here and are not inferred. Limitation (structural): single-event case studies cannot establish event frequency or generalise dust contributions to PM_{10} exceedances. Relevance: medium - dust events are where optical LCS under-read coarse mass most strongly, so multi-instrument events are useful calibration benchmarks. Retry the abstract before the weekly; if $PM_{10}/PM_{2.5}$ ratios are reported, they matter for the coarse-mode LCS correction.

doi: 10.1016/j.atmosres.2026.109409

Somsunun $PM_{2.5}$-bound metals health risk haze season Chiang Mai 2026 Toxics

Health Risk Assessment of Ambient $PM_{2.5}$-Bound Metals During Haze Seasons in Urban and Rural Chiang Mai, Thailand

101 24 h samples, 15 Jan-22 Apr 2020 haze season, at urban (CMU), agricultural rural (Mae Chaem) and roadside rural (Chom Thong) sites. Median $PM_{2.5}$ 50.3, 42.1 and 64.9 $\mu g\,m^{-3}$. Despite lower mass, Mae Chaem had HI 2.10 and TCR 2.37e-4 vs CMU 1.34 and 1.18e-4; HI >1 and TCR >1e-4 at all sites, highest at Chom Thong (2.69; 3.63e-4), driven by Cr. Limitation: risk is dominated by total Cr with an assumed Cr(VI) share, and spring 2020 overlaps COVID-19 restrictions. Relevance: medium - a direct case that equal $PM_{2.5}$ mass from a sensor does not mean equal hazard across land uses.

doi: 10.3390/toxics14100869

Szalata waste fire toxic emissions Gaussian plume environmental risk 2026 Toxics

Environmental Risk Assessment of Toxic Emissions into the Air Resulting from Waste Fires

Receptor-oriented screening: Gaussian plume (Pasquill classes) for a 14,000 kg municipal-waste fire burning 3.5 h, compared against Polish, US EPA, OSHA and NDS limits. Under the worst case analysed (class F, 1 m/s southerly) the residential receptor 700 m away stayed below reference values. Limitation: emission factors for mixed-waste fires are highly uncertain, and a steady-state Gaussian plume ignores fire-driven plume rise, lofting and near-field downwash, and comparing to occupational limits is not a public-health criterion. No measurement validates the field. Relevance: medium - rapid-deployment LCS around waste fires are the only practical check on such screening models.

doi: 10.3390/toxics14100892

Szczepanik Warsaw wintertime air quality improvement 2016-2025 2026 Aerosol Air Qual Res

Improvement in wintertime air quality in urban and suburban areas of Warsaw within 2016-2025

METADATA ONLY. Title and authors only (Szczepanik, Deptula, Klapinski, Rakowska). No abstract was retrievable at harvest. The title reports improved wintertime air quality across urban and suburban Warsaw over 2016-2025, the period of Poland's anti-smog resolutions and boiler-replacement programmes. Pollutants, data sources and magnitudes are not available; nothing is inferred. Relevance: medium - suburban solid-fuel heating is where Polish LCS networks (Airly and others) are densest; the paper may use them.

doi: 10.1007/s44408-026-00173-8

UK Air Quality Cell PM_{10} plume back trajectory tyre fire Mexborough vulnerability 2026 Eur J Public Health (conference abstract)

Back trajectory modelled PM_{10} plumes: population vulnerability profile from an uncontrolled fire

The UK Air Quality Cell monitored PM_{10} during an uncontrolled tyre fire; 22 hourly ADMS-STAR back-trajectory plumes were spatially averaged and overlaid on Output Areas. 24,952 residents fell in 'Unhealthy for Sensitive Groups' or worse; 7,856 (31.5%) in 'Hazardous' or beyond. Exposed areas had above-median deprivation, prior ill health and lower education. The modelling suggested field monitoring under-captured peaks. Limitation: AQI bands from a 24 h index were applied to hourly plume concentrations, and exposure is residential, ignoring sheltering and evacuation; no health outcomes. Conference abstract. Relevance: high - rapid-deploy sensor arrays around industrial fires would close the peak-capture gap the authors identify.

doi: 10.1093/eurpub/ckag169.809

Upaul Khulna $PM_{2.5}$ meteorology HYSPLIT random forest STL 2026 Research Square (preprint)

Meteorological and HYSPLIT-Derived Transport Influences on $PM_{2.5}$ Exposure in a Khulna City, Bangladesh

One year (2024) of hourly $PM_{2.5}$ at a single Khulna station, analysed with MLR, Random Forest, STL and K-means on HYSPLIT trajectories. Winter mean ~121 $\mu g\,m^{-3}$ vs ~15 in the monsoon. Temperature and pressure were top RF predictors (OOB R2 0.64); temperature and RH were negative in MLR. Winter air arrived on short north-westerly continental paths, monsoon air on long southerly maritime paths; STL residuals flagged pre-monsoon spikes (~138 $\mu g\,m^{-3}$). Limitation: one station, one year, and reanalysis meteorology at ~50 km; season is collinear with every predictor, and the 'variance explained' shares are not additive. Preprint. Relevance: low-moderate - a coastal South Asian city with almost no monitoring, where LCS would add the spatial dimension.

doi: 10.21203/rs.3.rs-11078552/v1

Vikas wavelet background $PM_{2.5}$ India MERRA-2 2026 Environ Monit Assess

Estimating background $PM_{2.5}$ concentrations using a wavelet-based convolution framework

Wavelet coherence of low-frequency components across neighbouring MERRA-2 cells yields a spatiotemporal $PM_{2.5}$ background map for India, capturing the IGP winter build-up and pre-monsoon dust in the northwest. Against a GMM-HMM background at four cities, MAD 2.2 (Kolkata) to 10.8 (Delhi) $\mu g\,m^{-3}$. Against 5th-percentile PCHIP backgrounds across the Delhi network, MAE 21.95 and RMSE 29.45 $\mu g\,m^{-3}$, improving to 14.03/17.65 at the Delhi background site. Limitation: validation is against other statistical background definitions, not tracer or model source attribution, and MERRA-2 $PM_{2.5}$ is a reconstructed aerosol product with known bias over South Asia. Relevance: high - background/increment separation is a standard post-processing step for dense LCS networks.

doi: 10.1007/s10661-026-16082-7

Wang urban morphology street-scale $PM_{2.5}$ PM_{10} change points 2026 Atmos Pollut Res

Non-linear associations, model-supported change points, and seasonal variations of urban morphology with street-scale $PM_{2.5}$ and PM_{10} concentrations

METADATA ONLY. Metadata-only record: Crossref deposit with no abstract on the day of entry, and no abstract recoverable from Europe PMC or Semantic Scholar. The title reports non-linear associations and model-supported change points between urban morphology indicators and street-scale $PM_{2.5}$ and PM_{10}, with seasonal variation. Design, city, sample size, morphology metrics and effect sizes cannot be stated from the record and are not inferred here. Limitation: unassessable until the abstract or full text is available; threshold-type 'change points' in morphology-PM regressions are prone to overfitting when sites are few. Relevance: medium - street-scale morphology thresholds bear directly on how a dense sensor network should be sited across canyon geometries.

doi: 10.1016/j.apr.2026.103242

Wei $PM_{2.5}$ ozone nonlinear coupling Puyang oilfield city GAM 2015-2025 2026 Atmosphere

Long-Term Trends and Nonlinear $PM_{2.5}$-O_3 Coupling in a Typical Oilfield City of the "2 + 26" Region

Daily $PM_{2.5}$ and MDA8 O_3 in Puyang 2015-2025 with nested GAMs adding meteorology, NO_2, humidity and season interactions. Raw correlation reversed by concentration band: r 0.271 below 35 $\mu g\,m^{-3}$, -0.208 at 35-75 and -0.313 above 75. After controlling for meteorology and NO_2, the conditional partial effect of $PM_{2.5}$ on O_3 increased over most of the range, so the negative raw correlation is largely meteorological. The low-band positive coupling strengthened over time (0.190 to 0.306). Limitation: one city, observational correlations; GAM partial effects do not identify chemical mechanisms (aerosol radiative or heterogeneous HO2 uptake). Relevance: low.

doi: 10.3390/atmos17111057

Yingkar $PM_{2.5}$-bound PAHs Central Asia urban trends sources risk 2026 Atmos Environ

Environment-to-Health Linkages of $PM_{2.5}$-Bound Polycyclic Aromatic Hydrocarbons in Urban Cities of Central Asia: Trends, Sources, and Risk Assessment

METADATA ONLY. Title and authors only (Yingkar, Omarova, Sovetova, Alibekov et al.). No abstract was retrievable from Crossref or Semantic Scholar at harvest. The title describes trends, source apportionment and health-risk assessment of $PM_{2.5}$-bound PAHs across urban Central Asian cities - a region with sparse speciation data and heavy coal heating. Cities, sampling design, concentrations and risk estimates are not available; nothing is inferred. Relevance: medium - coal-heating PAH peaks in Almaty-type basins are where low-cost PM networks already operate without any chemical reference; retry the abstract on a later run.

doi: 10.1016/j.atmosenv.2026.122447

Yousaf nanoscale curvature hygroscopic growth respiratory deposition ultrafine 2026 Environ Pollut

Nanoscale Curvature Controls Hygroscopic Growth and Respiratory Deposition of Atmospheric Nanoparticles

HTDMA measurements of 15-100 nm NaCl particles fed into the MPPD lung-deposition model. The Kelvin effect raised deliquescence RH to 77-81% for 15-36 nm particles and suppressed growth; at 80-90% RH, growth factors were 1.95-2.25 (15 nm) and 1.99-2.36 (100 nm). Ignoring hygroscopic growth overestimated total deposition by up to 25%, with an 18% tracheobronchial difference for particles <=50 nm, and shifted larger UFPs deeper into the lung. Limitation: pure NaCl is a best-case hygroscopic aerosol; ambient UFP is organic-rich with far lower kappa, so the correction is an upper bound. Relevance: medium - the same size-dependent water uptake that biases optical sensors in humid air also changes delivered dose, so dry-size number counts understate the problem.

doi: 10.1016/j.envpol.2026.129429

Yu Paris urban plume aerosol composition optical properties airborne ground 2026 Atmos Chem Phys

Urban-to-downwind variations in aerosol composition and optical properties in the Paris urban plume from coordinated airborne and ground-based observations

Coordinated ground and airborne measurements in Paris and its downwind plume, ~2-6 h transport. The plume modestly raised 80-200 nm particle number and moderately enhanced PM_1; organics dominated PM_1 throughout, with more oxidised organics downwind. Optical properties changed sharply: higher SSA at 450 and 630 nm downwind; refractive index moved from ~1.35-1.40 + 0.03-0.08i near source to ~1.45-1.50 + 0.015-0.02i downwind; AAE rose. Limitation: limited flights under specific meteorology; plume age is estimated, not tracked. Relevance: high - optical LCS use a single assumed refractive index; a 2-6 h change of this size shifts OPC mass conversion between urban and downwind nodes.

doi: 10.5194/acp-26-15055-2026

Cardiovascular & metabolic — 6 records

Carried from the daily issues — full summaries (6 of 6 records in this subtopic)

Andreotti active fire detections circulatory hospital admissions Brazil ecological panel 2026 Cureus

Active Fire Detections and Hospital Admissions for Circulatory Diseases in Brazil: An Ecological Panel Study (2013-2024)

27 Brazilian federative units x 12 years (324 unit-years), 13.8 million circulatory admissions and 2.3 million fire detections. Correlations were null (rho -0.03 to 0.01). In negative-binomial models with state fixed effects, year effects, temperature, poverty and primary-care coverage, the rate ratio per additional 1,000 detections/100,000 km2/year was 0.998 (0.970-1.028), null in every subset and with clustered SEs. Limitation: annual state-level fire counts are far removed from short-term smoke $PM_{2.5}$ at residence; smoke from Amazon fires is exported to other states, so the exposure is misplaced as well as aggregated. The authors correctly state this is not evidence of no effect. Relevance: low - illustrates why fire counts cannot substitute for measured PM.

doi: 10.7759/cureus.118676

Cao $PM_{2.5}$ chemical components heart disease longitudinal China 123 cities 2026 BMC Cardiovasc Disord

Association between $PM_{2.5}$ and its chemical components exposure and heart disease risk: a longitudinal study from China

8,172 participants from 123 Chinese cities; Cox models with three adjustment levels plus WQS mixtures. Per 1 $\mu g\,m^{-3}$ $PM_{2.5}$: HR 1.027 (1.022-1.032); BC had the largest per-unit estimate, HR 1.381 (1.266-1.506). All components showed non-linear exposure-response; southern residents and overweight participants had stronger associations; PAF 2.63%. Limitation: per-$\mu g\,m^{-3}$ component HRs are not comparable because component ranges differ by an order of magnitude, and components are highly collinear, so 'BC largest' is a scaling artefact unless expressed per IQR. Heart disease is likely self-reported (cohort not named in the abstract). Relevance: medium - again points to absorption (BC) as the health-relevant channel.

doi: 10.1186/s12872-026-06395-3

Cerin air pollution road environment cardiometabolic walking AusDiab 2026 Toxics

Associations of Ambient Air Pollution and Road-Related Environmental Characteristics with Cardiometabolic Health: The Role of Walking

4,141 middle-aged and older adults; GAMMs with FDR control within test families. $PM_{2.5}$ associated positively with diastolic and non-linearly with systolic BP; road density with lower HDL and higher triglycerides. Higher $PM_{2.5}$ and NO_2 were associated with more walking for transport, setting up opposing pathways via physical activity. Coefficients are not reported in the abstract. Limitation: cross-sectional, so walking cannot be ordered between exposure and outcome, as the authors state; Australian $PM_{2.5}$ contrasts are narrow (single-digit $\mu g\,m^{-3}$) and track urbanity, which drives both walkability and pollution. Relevance: low-medium - co-located PM and mobility data would let LCS studies separate these pathways.

doi: 10.3390/toxics14100920

Musa outdoor air pollution cardiovascular health school children Dar es Salaam 2026 BMC Public Health

Effects of outdoor air pollution on the cardiovascular health of school-aged children residing near polluted areas and the efficacy of mitigation measures in Dar es Salaam, Tanzania

Community cross-sectional study, December 2024-July 2025, of 653 school-aged children living in areas classed as highly polluted by proximity to industrial zones, major roads and waste-burning sites. Single oscillometric BP reading; 19.9% had elevated BP. In log-binomial models, prior respiratory disease carried ARR 8.81 (6.12-12.67) for cardiovascular symptoms. Awareness of mitigation policy was mixed. Limitation: no pollutant was measured and there is no unexposed comparison group, so the title's causal framing is not supported; one-visit BP overstates prevalence in children. Relevance: high as a gap marker - urban East Africa has almost no PM monitoring, and this design shows what health studies look like without it.

doi: 10.1186/s12889-026-28148-9

Zhang short-term air pollution stroke incidence Guiyang karst city 2026 Front Public Health

Time-series analysis of the impact of short-term exposure to atmospheric pollutants on stroke risk among residents in Guiyang City, 2016-2021

Guiyang 2016-2021, 11,517 incident strokes (5.25/day), quasi-Poisson GAM with single-day and cumulative lags, controlling trend, weather, day-of-week and holidays. Per 10 $\mu g\,m^{-3}$ at the maximal lag: $PM_{2.5}$ RR 1.026 (1.005-1.048), PM_{10} 1.015 (1.001-1.030); SO_2 1.075, NO_2 1.057, CO 1.346 per mg/m3. Women and adults >=65 carried more of the risk; under-65s responded only to NO_2 and PM_{10}. Limitation: single-pollutant models with the lag chosen at the maximum inflate the PM coefficients, co-pollutant collinearity is unaddressed, and the stroke series does not split ischaemic from haemorrhagic. Relevance: moderate - a low-concentration karst basin is where fixed-site averaging hides terrain-driven gradients that a dense low-cost network would resolve.

doi: 10.3389/fpubh.2026.1777772

Zhao interpretable machine learning multi-window $PM_{2.5}$ hypertension CHNS Beijing 2026 Toxics

An Interpretable Machine Learning Framework Integrating Multi-Window Environmental Exposure for Hypertension Risk Assessment

Five classifiers (GLM, Lasso, tree, RF, XGBoost) with SHAP and partial-dependence layers on 2,523 participant-visits from the CHNS Beijing subsample; person-level and year-based splits; BP excluded from predictors. Test AUC 0.69-0.77, Brier 0.17-0.23. The 60-day $PM_{2.5}$ window was the top exposure feature and added ~0.02-0.03 AUC (p=0.03-0.05). Adjusted window-specific analyses showed inverse $PM_{2.5}$-hypertension associations, stronger for longer windows. Limitation: a ~0.02 AUC gain at borderline p with inverse associations indicates that the $PM_{2.5}$ windows act as calendar/season proxies, not causal exposure; feature importance is not effect estimation, and the CHNS visit spacing is irregular. Relevance: low.

doi: 10.3390/toxics14100871

Neuro / mental health — 8 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (8 OF 8 RECORDS IN THIS SUBTOPIC)

Al-Atram early-life air pollution child neurodevelopment critical review 2026 J Xenobiot

Early Life Ambient Air Pollution and Child Neurodevelopment: Critical Review of Cohort Evidence

Narrative critical review (purposive selection, PubMed through 30 Aug 2026) weighing null findings, exposure assessment, confounding and causal interpretation. European and US multicohort studies show small associations with selected outcomes alongside substantial nulls; Canadian and Korean cohorts show pollutant- and age-specific heterogeneity. Birth weight and prematurity require total vs conditional effect distinctions; placental and imaging data support plausibility but not mediation. The authors conclude the evidence does not make pollution a determinative cause of ASD or ADHD. Limitation: no protocol, duplicate screening or risk-of-bias scoring, so selection is the authors' own, and it predates today's Medicaid wildfire-ASD cohort. Relevance: low.

doi: 10.3390/jox16060191

Holm CHAPS Fresno neurobehavior BDNF children air pollution 2026 Front Public Health

Neurobehavioral assessments and biomarkers in school children who are highly exposed to air pollution

Children's Health and Air Pollution Study, n=193 aged 7-18. Third-grade test scores correlated with NIH Toolbox total and crystallized composites (r 0.55-0.60) and with BDNF (r 0.35). First-trimester traffic pollutants were associated with aggressive and rule-breaking behaviour; first-year exposure to all pollutants with lower BDNF, window around 5-6 months. Limitation: 193 children, many pollutants, windows and outcomes, and no effect sizes or multiple-testing control in the abstract - the window findings are exploratory. Pollutants are highly collinear in Fresno, so 'all pollutants' cannot attribute the BDNF signal to PM or EC. Relevance: medium - EC and PAH contrasts again point to absorption channels rather than mass.

doi: 10.3389/fpubh.2026.1970714

Li built environment sleep duration air pollution mediation Yinzhou cohort 2026 BMC Public Health

Association between built environment and sleep duration mediated by air pollution: a cohort study in China

Yinzhou cohort, 18,891 adults, baseline 2015-2017 and follow-up 2021-2023; quantile g-computation for a built-environment mixture and SEM for mediation. Per quartile of the mixture: +0.057 h/year sleep (0.039-0.075). Per IQR, PM_{10}, $PM_{2.5}$ and NO_2 were associated with -0.027, -0.048 and -0.071 h sleep. $PM_{2.5}$ mediated up to 21.5% of the green-space effect; NO_2 up to 45.3% of the walkability effect. Limitation: self-reported sleep, and SEM mediation needs no unmeasured mediator-outcome confounding - noise and light at night co-vary with traffic pollution and are only partly modelled. Relevance: low-moderate - mediation of this kind needs better spatial exposure, not more covariates.

doi: 10.1186/s12889-026-28717-y

Papatheodorou prenatal wildfire $PM_{2.5}$ ASD ADHD Medicaid 2026 JAMA Pediatr

Prenatal Exposure to Wildfire $PM_{2.5}$ and Risk of ASD and ADHD in Children Enrolled in Medicaid

Nationwide Medicaid (MAX) cohort of 893,212 mother-child pairs with stable ZIP code in pregnancy; 8,357 ASD and 22,302 ADHD diagnoses (cumulative incidence to age 8: 1.9% and 10.4%). Cox models with distributed-lag nonlinear terms. Per IQR (0.75 $\mu g\,m^{-3}$) of cumulative smoke $PM_{2.5}$: ASD HR 1.30 (1.07-1.59); ADHD HR 1.03 (0.91-1.17). The ASD window sat in mid to late gestation. Limitation: a 30% hazard increase per 0.75 $\mu g\,m^{-3}$ is very steep for a ZIP-level smoke fraction; smoke years and regions coincide with secular and regional shifts in ASD claims coding, and claims-based diagnosis in Medicaid depends on service access. Relevance: high - smoke-specific PM at sub-$\mu g\,m^{-3}$ contrasts is exactly where dense sensor networks add resolution over 1 km model fields.

doi: 10.1001/jamapediatrics.2026.5045

Santoso air pollution noise greenness salivary cortisol BAMSE 2026 J Expo Sci Environ Epidemiol

Air pollution, road traffic noise, and greenness in relation to salivary cortisol in young adults

BAMSE birth cohort, n=1,148 at age 24; linear regression of salivary cortisol on prior-year modelled pollutants, noise and NDVI. No overall associations. In post hoc male subgroups, PM_{10} was associated with higher morning cortisol (beta 0.058, 0.003-0.112) as were NOx and noise, with greenness inverse; females showed opposite trends. Limitation: the only signals are exploratory sex-by-season subgroup contrasts after a null main analysis, with lower CIs at ~0; single-day cortisol is highly variable. Treat as null. Relevance: low - low-contrast Stockholm exposures modelled at the address.

doi: 10.1038/s41370-026-00995-z

Wang air pollution adolescent anxiety depression Sichuan cross-sectional 2026 BMC Public Health

Air pollution and mental health in adolescents: a cross-sectional study in Sichuan, China

71,364 school students aged 10-19, surveyed over two winters (2022-23, 2023-24). Per IQR, PM_{10} anxiety OR 1.29 (1.22-1.36) and O_3 1.24 (1.19-1.29), with similar depression associations; a weighted combined score rose stepwise. $PM_{2.5}$ and SO_2 were inversely associated. Limitation: exposure was assigned from the IP address of the online survey, i.e. a point-in-time location of the device, and the PM_{10}-positive/$PM_{2.5}$-inverse split from highly correlated pollutants is the signature of collinearity and spatial confounding rather than a fraction-specific effect. Relevance: low - coarse exposure assignment.

doi: 10.1186/s12889-026-28370-5

Yang heat anomaly wildfire smoke $PM_{2.5}$ incident depression All of Us 2026 J Affect Disord

Heat anomaly, wildfire smoke, social-material vulnerability, and incident depression in a national EHR-linked cohort: A preregistered secondary analysis

Preregistered Cox analysis in All of Us: 179,218 adults free of recorded depression, 5,484 incident cases over a mean 371 days. Exposures were the 30-day pre-index PRISM maximum-temperature anomaly, Stanford ECHO smoke $PM_{2.5}$ and a social-material vulnerability (SMV) composite. Heat anomaly: HR 1.06 (1.03-1.10) and 1.08 (1.04-1.11) per SD; SMV: HR 1.36 (1.33-1.40) per SD. Mean smoke $PM_{2.5}$ showed no association in the primary model; secondary smoke metrics were heterogeneous by metric and follow-up window. Limitation: exposure is assigned at the 3-digit ZIP level and frozen at baseline, so smoke-episode contrast is heavily smoothed and a null is weak evidence of no effect; EHR ascertainment depends on care access. Relevance: high - this is the misclassification problem that residence-scale sensor-corrected smoke $PM_{2.5}$ exists to fix.

doi: 10.1016/j.jad.2026.122663

Zhou short-term indoor $PM_{2.5}$ cognition HRV randomized crossover 2026 iScience

Short-term indoor fine particulate matter exposure impedes cognitive performance and autonomic nervous system in adults

Randomised single-blind crossover in 90 working-age adults, two stable chamber conditions set by an air purifier with real or sham filter: 1.70 vs 31.22 $\mu g\,m^{-3}$ $PM_{2.5}$ from candle burning. The high condition lowered heart rate variability and attention and memory scores, but not coordination or reasoning; reduced HRV partly mediated some cognitive declines. Effect sizes with CIs are not given in the abstract. Limitation: candle smoke carries VOCs, CO, odour and visible haze, so blinding and the PM-specific attribution are both weak; multiple cognitive domains invite selective significance. Relevance: medium - controlled indoor steps of this size are what consumer PM monitors can verify in occupied rooms.

doi: 10.1016/j.isci.2026.117904

Respiratory & allergic — 17 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (17 OF 17 RECORDS IN THIS SUBTOPIC)

Dublin air pollution fungal spores respiratory admissions time series 2026 Eur J Public Health (conference abstract)

Short-term effects of air pollution and aeroallergens on respiratory admissions

Dublin daily HIPE admissions with NO_2, $PM_{2.5}$, PM_1, fungal spores and meteorology; negative-binomial time series, lags 0-7. NO_2 was consistently associated with asthma, COPD and all respiratory admissions (~3-5% per unit, p <0.01); particulate effects were 'outcome-specific' and not quantified in the abstract; spores showed no independent short-term effect. Limitation: per-unit NO_2 effects of 3-5% are implausibly large per ppb or $\mu g\,m^{-3}$ and suggest an unstated scale; study years and PM results are missing; conference abstract. Relevance: low-moderate - Dublin's PM_1 series is unusual, and spore counts pair naturally with optical sensor coarse channels.

doi: 10.1093/eurpub/ckag169.1252

ISVOLCAN Tajogaite volcano La Palma lung function obstructive pattern 2026 Eur J Public Health (conference abstract)

Lung function after Tajogaite volcano eruption in La Palma: ISVOLCAN study

Random population sample of 604 adults (mean age 50; 62% women) with spirometry a median 424 days after the 85-day eruption. Obstruction prevalence 28.8%. Western (more exposed) residence: OR 1.64 (1.01-2.67); 1.74 (1.03-2.93) without prior respiratory disease. Intermediate SO_2 exposure OR 1.98 (1.18-3.34). Limitation: a fixed FEV1/FVC <70% cut-off over-calls obstruction in older adults, no pre-eruption baseline exists, and residence area is a coarse proxy that cannot separate SO_2 from volcanic PM; non-monotone SO_2 categories. Conference abstract. Relevance: moderate - volcanic events are a case for rapidly deployed PM and SO_2 sensor networks.

doi: 10.1093/eurpub/ckag169.069

Kobza respiratory ambulatory care air quality Upper Silesia 2026 Int J Occup Med Environ Health

The assessment of respiratory diseases in ambulatory care attributable to air quality in the Upper Silesian agglomeration

Five-year ecological analysis correlating daily mean concentrations from about 50 monitoring stations with National Health Fund outpatient incidence for seven respiratory ICD codes across eight regions of the Upper Silesian voivodeship. The strongest signal was for SO_2, not PM: tau 0.99 and rho 0.9 in regions 7 and 8, rho 0.78 in regions 1 and 4. Limitation: rank correlations on aggregated series without control for season, influenza, temperature or autocorrelation - seasonality alone would generate these coefficients, so the results carry no causal or PM-specific information. Relevance: low - included because $PM_{2.5}/PM_{10}$ were analysed; the coal-heating region is a classic case where denser residential PM sensing would sharpen exposure contrast.

doi: 10.13075/ijomeh.1896.02875

Kusumaningtiar school children $PM_{2.5}$ PM_{10} health risk systematic review 2026 Research Square (preprint)

Environmental Health Risks in School Children Due to Air Pollution Exposure: Systematic Literature Review

PRISMA review (PROSPERO CRD420261282406) of 12 studies published 2015-2025 on school $PM_{2.5}$ and PM_{10}, combining environmental health risk assessment, epidemiology, MPPD dosimetry and STELLA modelling. Several studies reported risk quotients >1; one Bandung study found inverse $PM_{2.5}$-lung function associations; dosimetry indicated higher airway deposition in children. Limitation: 12 heterogeneous studies with mixed methods and no quantitative synthesis; RQ>1 depends on default intake assumptions rather than measured outcomes. Preprint. Relevance: medium - underlines the need for standardised school-level monitoring that low-cost networks could provide.

doi: 10.21203/rs.3.rs-11094408/v1

Li spatiotemporal causal machine learning $PM_{2.5}$ asthma Colorado counties 2026 GeoHealth

A Spatiotemporal Causal Machine Learning Framework for Quantifying the Impact of $PM_{2.5}$ on Asthma at the Community Level

Colorado county-level monthly asthma hospitalisations and ED visits, 2011-2019, modelled with a spatiotemporal representation-learning network plus a causal-inference layer that estimates counterfactual outcomes under hypothetical $PM_{2.5}$ reductions. Estimated effects were heterogeneous, larger in communities with compounded environmental and social stressors, with wider uncertainty in data-sparse counties. No effect size per $\mu g\,m^{-3}$ is given in the abstract. Limitation: county-month aggregation with monthly $PM_{2.5}$ is ecological; 'causal' here rests on unconfoundedness given learned covariates, which a neural model cannot verify, and sparse rural counties are exactly where exposure error peaks. Relevance: moderate - the model's uncertainty is concentrated where monitors are absent, an explicit argument for sensor densification in rural Colorado.

doi: 10.1029/2025gh001735

Lovric ragweed sensitisation PM_{10} pollen model comparison Croatia ATOPICA 2026 Children (Basel)

Outdoor Environmental Risk Factors for Paediatric Ragweed Sensitisation: A Formal Model Comparison

Re-analysis of 3,520 Croatian children from ATOPICA within EDIAQI; logistic models compared by AIC, adjusted for age and parental atopy. Sensitisation prevalence 10.8%. PM_{10} and categorical pollen area fit equally well (delta AIC 1.7) because both encode the same inland-coastal contrast; NO_2 added nothing. Transitional woodland-shrub aOR 0.93 (0.90-0.97) survived Bonferroni; rural residence aOR 0.69 (0.50-0.96). Limitation: the authors state it directly - area-level exposure cannot separate particulate pollution from pollen, so a PM contribution can be neither confirmed nor excluded. Relevance: moderate - an explicit case where only sub-regional, co-located PM and pollen measurement could break the collinearity.

doi: 10.3390/children13101366

Luglio air pollution pediatric asthma hospitalizations Louisiana vulnerable groups 2026 Environ Epidemiol

Associations between air pollution and pediatric asthma hospitalizations across vulnerable groups in Louisiana

28,499 asthma hospitalizations in children in Louisiana 2010-2019, time-stratified case-crossover with DLNM to lag 3. Cumulative lag 0-3: $PM_{2.5}$ OR 1.079 (1.052-1.107), O_3 1.082 (1.050-1.116), NO_2 1.071 (1.034-1.111), with increments as defined in the paper. In multipollutant models O_3 and NO_2 attenuated while $PM_{2.5}$ persisted. Effect modification by race and Medicaid status was not significant, although lag patterns differed (lags 1-2 Black children, lag 3 White children). Limitation: exposure assigned from area-level estimates in a state with sparse monitoring outside the industrial corridor, so differential misclassification by race/rurality could mask modification; NO_2 covers only 2010-2016. Relevance: medium - higher-burden groups sit where reference coverage is thinnest.

doi: 10.1097/ee9.0000000000000544

Oliveira air pollution HRV lung function schoolchildren Cubatao 2026 Epidemiol Serv Saude

Effect of air pollution on heart rate variability and pulmonary function in schoolchildren: a cross-sectional study, Cubatao, Brazil, 2017.

Convenience sample of 77 schoolchildren in the Cubatao industrial zone, with pollutant levels from the state agency's monitors. Associations reported in the abstract are for gases: SO_2 and HFnu -0.68 (-1.27 to -0.11) daily, -1.29 (-2.21 to -0.38) over three days; O_3 and HFnu -0.32 (-0.62 to -0.02); NO_2 with 0.23% lower FEV1 and 0.30% lower FEF25-75 (no CI given). No PM association is reported in the abstract. Limitation: 77 children, one fixed-site monitor and a single visit leave essentially no exposure contrast across subjects - the estimates rest on day-to-day variation within a cross-sectional design, with multiple pollutants and lags tested. Relevance: low - illustrates the exposure-assignment ceiling of a single regulatory monitor.

doi: 10.1590/s2237-96222026v35e20250368.en

Pedersen early-life air pollution lung function COPSAC 2026 BMJ Open Respir Res

Impact of air pollution exposure in early life on lung function development in the COPSAC cohorts

Two Copenhagen birth cohorts (n=846) with repeated spirometry and plethysmography from age 3 to 18 years; linear mixed models. Per IQR (2.2 $\mu g\,m^{-3}$) of postnatal $PM_{2.5}$: FEV1 at age 6 -2.70% (-4.27 to -1.10) and specific airway resistance +7.30% (1.87-13.07). The deficit and the resistance increase persisted in growth trajectories through childhood, stronger in males. Limitation: COPSAC is an asthma-enriched cohort (COPSAC2000 enrolled children of asthmatic mothers), so effect sizes may not transfer to the general population; a 2.2 $\mu g\,m^{-3}$ IQR in Copenhagen is small and partly a proxy for urban residence. Relevance: medium - effects at very low contrast argue for address-level precision rather than municipal means.

doi: 10.1136/bmjresp-2025-003297

Perez-Garcia nasal EWAS respiratory biomarkers Mexican American CHAMACOS 2026 medRxiv

Nasal Epigenome-Wide Association Study of Respiratory and Allergic Biomarkers in Mexican American Young Adults

Preprint. Nasal EWAS (EPICv2, 804,220 CpGs) in 354 CHAMACOS young adults aged 22, replicated in 540 Project Viva adolescents aged 13. Forty-nine CpGs were associated with FeNO and 557 DMRs with respiratory outcomes (FDR<0.05); 26 CpGs and 103 DMRs replicated (e.g. LRRFIP1, EPX, IL13). Ten FeNO-related CpGs were associated with prenatal and/or past-year $PM_{2.5}$ or O_3 at nominal p<0.05 only. Limitation: the air-pollution step is a secondary look-up without multiple-testing control, so the $PM_{2.5}$ link is hypothesis-generating; cross-sectional nasal sampling cannot order exposure and methylation. Relevance: low-medium - an agricultural-valley cohort where PM sources are local and poorly served by regulatory monitors.

doi: 10.64898/2026.10.27.26366640

Poland air pollution COPD asthma emergency admissions DLNM heating season 2026 Eur J Public Health (conference abstract)

Air pollution and respiratory admissions in Poland: a nationwide DLNM analysis of COPD and asthma

National registry: 625,597 COPD and 153,563 asthma emergency admissions (32,893 paediatric), DLNM lag 0-10, stratified by heating season. Associations appeared only in the heating season: COPD RR per IQR $PM_{2.5}$ 1.044, PM_{10} 1.039, SO_2 1.055; lead 1.072, benzo[a]anthracene 1.057. Adult asthma stronger ($PM_{2.5}$ 1.100). In children, PM was not significant; arsenic (1.124) and ozone (1.115) were. NO_2 null throughout. No CIs in the abstract. Limitation: component and pollutant RRs come from single-pollutant models on collinear coal-combustion tracers, so 'lead' and 'PAH' effects cannot be separated from $PM_{2.5}$; conference abstract. Relevance: moderate - heating-season coal signatures are where speciation-capable nodes add value over mass.

doi: 10.1093/eurpub/ckag169.807

Shi air pollution pulmonary tuberculosis incidence China GTWR 2026 Front Public Health

Analysis of influencing factors of atmospheric pollution on pulmonary tuberculosis incidence in China from 2014 to 2020

Provincial panel 2014-2020 (31 units), spatial autocorrelation, space-time scanning and geographically and temporally weighted regression (R2 0.889), with 1-5-year moving-average exposure windows. High-burden clusters persisted in Xinjiang, Xizang and Qinghai (RR 3.56). $PM_{2.5}$ coefficients ran 'east-negative, west-positive'; SO_2, NO_2 and CO were jointly positive in the north-east; O_3 mostly negative. Limitation: 217 province-years and locally varying coefficients amount to a flexible fit of an ecological surface; the sign flip of $PM_{2.5}$ across China shows confounding by development and TB control, not toxicity. Relevance: low.

doi: 10.3389/fpubh.2026.1951817

Wang air pollutants sleep apnea polysomnography architecture Shenzhen 2026 Environ Res

Associations of multiple ambient air pollutants and sleep apnea, sleep architecture and nocturnal oxygenation in adult patients with or without obstructive sleep apnea (OSA)

Retrospective polysomnography study of 4,049 adults in Shenzhen. Over a 0-180 day lag window, $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO and diurnal temperature range were associated with greater OSA severity. $PM_{2.5}$ and PM_{10} were associated with 1.71% and 1.32% of total sleep time more N1 and 1.51% and 1.32% less N3 sleep (increments as reported). O_3 and DTR showed the clearest associations with oxygenation; $PM_{2.5}$ had a rising exposure-response. Limitation: the lag window was chosen post hoc, which inflates false-positive risk across many pollutant-outcome pairs; a clinic-referred sample means referral itself may track season and pollution. CIs are not given in the abstract. Relevance: low - city-level exposure.

doi: 10.1016/j.envres.2026.126044

Wang climate air pollutants influenza 48 countries convergent cross mapping 2026 Glob Health Res Policy

Delayed and nonlinear associations and predictive causal signals between climate factors, air pollutants, and influenza epidemics in 48 countries

48 countries, 2022-2024; FluNet weekly positivity against national climate and pollutant series in hemisphere-specific Bayesian distributed-lag models, then convergent cross mapping (CCM) for 'predictive causal' drivers. Northern Hemisphere peak OR 1.74 (1.34-2.26) at weekly cumulative SO_2 3100 $\mu g\,m^{-3}$, lag 1.6 weeks; Southern peak for organic carbon OR 1.72 (0.65-4.58), null. CCM ranked dust $PM_{2.5}$ the leading pollutant driver (19 countries). Limitation: national-level exposure against surveillance positivity is fully ecological; three post-pandemic seasons cannot separate pollution from NPI relaxation, and CCM detects shared dynamics, not causation. Relevance: low - surveillance-scale; dust $PM_{2.5}$ is a size fraction low-cost optical sensors undercount.

doi: 10.1016/j.ghrp.2026.04.002

Wilner public safety power shutoff wildfire $PM_{2.5}$ healthcare California 2026 PLoS Med

Co-exposure to planned power outages and wildfire $PM_{2.5}$ and acute healthcare utilization in California: A time-stratified case-crossover study

Statewide California encounters for adults 20+, time-stratified case-crossover, temperature-adjusted. Severe PSPS events alone: same-day respiratory encounters OR 1.06 (1.02-1.11), stronger for COPD. Joint PSPS and +10 $\mu g\,m^{-3}$ wildfire $PM_{2.5}$ vs neither: OR 1.16 (1.10-1.22), with multiplicative interaction. Cardiovascular and psychiatric encounters null. Limitation: PSPS is circuit-level but assigned by ZIP, and outage plausibly changes the exposure itself (no power for filtration or HVAC, more time outdoors or relocation), so the interaction mixes effect modification with exposure modification. Relevance: high - outages disable mains-powered sensors and purifiers precisely during smoke; battery-backed nodes are needed to measure this window.

doi: 10.1371/journal.pmed.1005270

Yuksel COPD cigarette biomass combined exposure exacerbations Turkiye 2026 Tob Induc Dis

Clinical outcomes of COPD associated with cigarette smoke, biomass smoke, and combined exposure: A retrospective observational study

319 stable COPD patients: smoking-associated 159, biomass-associated 80, combined 80. Combined exposure presented younger (59.2 vs 61.4 and 63.5 y), with lower FEV1 (48% vs 55-56% predicted), more chronic respiratory failure (26.2% vs 17.0% and 11.2%) and, adjusted, severe exacerbations AOR 1.89 (1.22-2.96) vs smoking alone. Limitation: exposure is a retrospective chart classification with no dose for biomass, and the equal group sizes (80/80) suggest selection rather than consecutive enrolment. Pack-years differ by group and are not shown to be balanced, so 'combined' may simply mean heavier total exposure. Relevance: low - no measured particle exposure.

doi: 10.18332/tid/231161

Zhou preoperative short-term $PM_{2.5}$ postoperative pulmonary complications lung cancer surgery 2026 Am J Respir Crit Care Med

Preoperative short-term $PM_{2.5}$ Exposure and Postoperative Pulmonary Complications After Lung Cancer Surgery

Metadata only: the record (AJRCCM research letter) carried no abstract at PubMed, Crossref or Semantic Scholar on the issue date. The title reports an association of short-term $PM_{2.5}$ exposure before surgery with postoperative pulmonary complications after lung cancer resection; design, n and effect size are not available here and are not inferred. Limitation (structural): preoperative exposure windows are short, so day-of-week and seasonal surgical scheduling and hospital load are the confounders any such analysis has to defeat. Relevance: medium - a perioperative window is one where short-horizon PM forecasts could change clinical scheduling. Retry the abstract before the weekly.

doi: 10.1093/ajrccm/aamag575

Reproductive & developmental — 7 records

Carried from the daily issues — full summaries (7 of 7 records in this subtopic)

Ai air pollution IVF outcomes Chengdu retrospective cohort 2026 Int J Fertil Steril

Association between Ambient Air Pollution Exposure and Reproductive Outcomes following In Vitro Fertilization: A Retrospective Cohort Study

12,453 women with fresh embryo transfer at one Chengdu centre, 2019-2022; exposure windows defined around oocyte retrieval and transfer; GLMs adjusted for clinical, sociodemographic and paternal factors. Exposure in the 85 days before oocyte retrieval (folliculogenesis window) was associated with lower odds of biochemical pregnancy, clinical pregnancy and live birth; live birth OR 0.969 (0.954-0.984) per 10 $\mu g\,m^{-3}$ $PM_{2.5}$. Limitation: four pollutants over several overlapping windows from 4 years of gridded data raise multiplicity, and in a single-city clinic cohort the exposure contrast is mostly temporal, confounded by season and protocol changes (including COVID-era disruption). Relevance: low.

doi: 10.22074/ijfs.2026.2072188.1959

Gaylord DNA methylation signature prenatal air pollution ECHO cohort 2026 Environ Epigenet

A methodology to identify a DNA methylation-based signature of prenatal air pollution exposure in the ECHO Cohort

Refines a pipeline that predicts high versus low in-utero air-pollution exposure from cord-blood DNA methylation (450K/EPIC) across multiple ECHO pregnancy cohorts. Exposures were ML-modelled at geocoded addresses; training and validation sets were matched on exposure level, site and array. Validation AUCs: $PM_{2.5}$ 0.64-0.68, NO_2 0.62-0.95, O_3 0.50-0.75, varying with region, exposure window and the high/low threshold. Limitation: the target is itself a model output, so the epigenetic classifier inherits exposure-model error, and $PM_{2.5}$ discrimination is modest; dichotomisation discards dose information. Relevance: medium - a biomarker of cumulative exposure is only as good as the exposure surface it is trained on, which is where denser monitoring would raise the ceiling.

doi: 10.1093/eep/dvag034

ICMR-SPIC stillbirth risk factors biomass fuel India pooled cohort 2026 Lancet Reg Health Southeast Asia

Prevalence and risk factors for stillbirths in India: secondary analysis of ICMR-Stillbirth Pooled India Cohort (ICMR-SPIC)

Ten prospective pregnancy cohorts 1994-2023 harmonised, 227,821 pregnancies, 3,935 stillbirths; weighted rate 14.8/1000 (13.4-16.3). In mixed-effects modified Poisson models, biomass fuel use aIRR 1.37 (1.21-1.57) with the largest population attributable fraction of any factor, 13.95% (9.11-18.79), ahead of nulliparity and hypertensive disorders. Limitation: biomass use is a binary household marker strongly tied to rural residence, poverty and access to obstetric care; mutual adjustment for education does not remove that, so the PAF is an upper bound, not a smoke effect. Cohorts span three decades with different ascertainment. Relevance: low - the household-air-pollution burden case, without a particle metric.

doi: 10.1016/j.lansea.2026.100841

Lin prenatal air pollution transient tachypnea newborn Taiwan 2026 Front Pediatr

Prenatal air pollution exposure and the risk of transient tachypnea of the newborn

Nationwide matched design on the Taiwan NHIRD linked to the Maternal and Child Health Database, 2004-2020: 3,091 TTN cases in first-born infants and 12,364 controls (1:4 on maternal age, sex, birth date). Exposure over the 40 weeks before birth from the nearest of 71 stations to the residential postal code. Adjusted ORs: CO 1.806 (1.312-2.487), NOx 1.010 (1.005-1.015), $PM_{2.5}$ 1.001 (1.000-1.002), the last per unit concentration with a CI touching the null. Effects concentrated in the Taipei basin. Limitation: nearest-station assignment at postal-code scale, no multipollutant model reported, and the $PM_{2.5}$ signal is marginal while CO dominates, pointing to traffic as a mixture rather than PM. Relevance: medium - traffic-gradient exposure in a dense basin is where street-level sensor networks change exposure contrast.

doi: 10.3389/fped.2026.1941575

Scotland prenatal $PM_{2.5}$ infant amoxicillin birthweight mediation 2026 Eur J Public Health (conference abstract)

Prenatal $PM_{2.5}$ and Infant Antibiotics: Birth-Weight Mediation via Causal Analysis

All Scottish singleton births 2009-2018, n=498,123, linked registries; g-computation mediation comparing observed $PM_{2.5}$ with a counterfactual of 5 $\mu g\,m^{-3}$. Controlled direct effect OR 1.12 (1.10-1.14); interventional direct OR 1.12 (1.11-1.14); indirect via birthweight 0.99 (0.99-1.01). Limitation: exposure IQR is only 6-8 $\mu g\,m^{-3}$, so a 12% excess over a ~2 $\mu g\,m^{-3}$ counterfactual shift is implausibly large and points to residual area-level confounding (deprivation, prescribing culture); amoxicillin dispensing is a weak infection proxy. Preliminary conference abstract. Relevance: low-moderate - fine-scale exposure contrast is too narrow for postcode annual means.

doi: 10.1093/eurpub/ckag169.955

Valuvunina $PM_{2.5}$ PurpleAir quarry low birth weight Kinshasa DRC 2026 J Toxicol

Atmospheric Pollution by Fine Particulate Matter ($PM_{2.5}$) and Low Birth Weight: A Cross-Sectional Study in Binza Ozone Health Zone, Democratic Republic of the Congo, 2022

355 mothers delivering in 2021 at six facilities near the Carrigres quarry. LBW prevalence 12.96% (9.7-16.9). PurpleAir PA-II-SD sensors, corrected with a Kinshasa BAM-derived equation, gave area $PM_{2.5}$ 54.6-102.5 $\mu g\,m^{-3}$. Residence <500 m from the quarry: aOR 4.12 (1.98-8.57); area $PM_{2.5}$ >75 $\mu g\,m^{-3}$: aOR 1.18 (0.59-2.36). Limitation: $PM_{2.5}$ was measured for 48-72 h in 2022 and assigned to births in 2021, i.e. after the pregnancies, at the health-area level, so the null $PM_{2.5}$ result says nothing about gestational exposure; quarry dust is coarse and poorly captured by Plantower optics. Relevance: high - a direct example of low-cost sensors used as the sole exposure metric, and of the temporal mismatch that undermines it.

doi: 10.1155/jt/7953451

Zhu prenatal PM_1 $PM_{2.5}$ PM_{10} childhood hemoglobin anemia Fujian birth cohort 2026 Ecotoxicol Environ Saf

Association of prenatal PM_1, $PM_{2.5}$, and PM_{10} exposure with early childhood hemoglobin levels and anemia: Identification of susceptibility windows

Fujian Birth Cohort paediatric subcohort, 1,235 mother-child pairs. Per 1 $\mu g\,m^{-3}$ over the whole pregnancy, child hemoglobin fell by 1.02 (0.53-1.52) for PM_1, 0.87 (0.46-1.28) for $PM_{2.5}$ and 0.37 (0.19-0.56) for PM_{10} (units as reported). First-trimester anemia ORs per unit were 1.23, 1.16 and 1.09; DLMs placed the hemoglobin window in early-mid pregnancy and the anemia window in the first trimester. Limitation: per-microgram effects of this size imply implausibly large decrements across the cohort's range; PM_1, $PM_{2.5}$ and PM_{10} are near-collinear satellite products, so the size-fraction ranking is not interpretable. Postnatal exposure and diet are not stated. Relevance: low - satellite-derived exposures only.

doi: 10.1016/j.ecoenv.2026.120980

Mechanistic toxicology — 14 records

Carried from the daily issues — full summaries (14 of 14 records in this subtopic)

Awasthi air pollution oxidative stress inflammation molecular mechanisms review 2026 J Xenobiot

Molecular Mechanisms Linking Air Pollution-Induced Oxidative Stress to Inflammation and Human Disease

Broad review of ROS/RNS generation by PM, NOx, O_3, SO_2 and VOCs and downstream Nrf2, NF-kB and MAPK signalling, mitochondrial dysfunction, ER stress and epigenetic reprogramming, mapped to respiratory, cardiovascular, neurological, metabolic and carcinogenic outcomes; also covers biomarkers, susceptible windows and antioxidant-based interventions. No quantitative synthesis. Limitation: pollutants with very different chemistry are pooled under one oxidative-stress narrative, and in vitro dose relevance is not appraised; nothing here is new relative to the established mechanistic literature. Relevance: low - background for oxidative-potential (OP) metrics, which are the measurable bridge to sensing.

doi: 10.3390/jox16060185

Chen $PM_{2.5}$ mitochondria-associated membranes mouse liver 2026 Am J Physiol Cell Physiol

Exposure to airborne $PM_{2.5}$ suppresses mitochondrial associated membranes and MAM-associated metabolic regulators in mouse liver

Mice inhaling environmentally relevant $PM_{2.5}$ showed fewer mitochondria-associated membranes and fewer MAM-forming mitochondria in liver (TEM). In hepatocytes, $PM_{2.5}$ suppressed basal and glucagon-induced MAM formation (BiFC sensor); macrophage particle uptake was blocked by antioxidants or calcium chelation. Mechanism: repression of CREBH/PPARalpha programs for the mitochondrial unfolded protein response and fatty-acid oxidation. Limitation: in vitro particle doses and the inhaled dose to liver are not reconciled, and the translocation route (systemic particles vs lung-derived mediators) is assumed rather than shown. Sample sizes are not in the abstract. Relevance: low - mechanism for PM-metabolic associations.

doi: 10.1152/ajpcell.00103.2026

Chen $PM_{2.5}$ mixtures Parkinson disease INPP5D SHIP1 PI3K-AKT multi-omics 2026 J Xenobiot

Neurotoxicity of $PM_{2.5}$-Bound Multi-Pollutant Mixtures in Parkinson's Disease: Exploring the INPP5D/SHIP1-PI3K-AKT Axis via an Integrative Multi-Omics Study

Two-sample MR on public GWAS suggested a positive genetic association of $PM_{2.5}$ with PD; network toxicology intersected $PM_{2.5}$-component targets, PD risk genes and differential transcripts (195 genes), and LASSO/RF picked INPP5D (SHIP1) as hub. Docking and SH-SY5Y experiments with SRM 1648a showed SHIP1 up-regulation, PI3K-AKT suppression, inflammation and apoptosis, rescued by INPP5D knockdown. Limitation: MR for an environmental exposure rests on GWAS of residential $PM_{2.5}$, whose instruments mostly encode where people live (pleiotropy); docking affinities are not evidence. MR estimates are not reported in the abstract, nor is the SRM dose. Relevance: low.

doi: 10.3390/jox16060184

Chen pirfenidone fine particulate matter EMT mitochondrial fission autophagy 2026 Respir Res

Pirfenidone reduces fine particulate matter-induced epithelial to mesenchymal transition in lung epithelial cells through regulating mitochondrial fission and autophagy

A549 cells exposed to 100 ug/mL fine PM lost E-cadherin and gained vimentin, alpha-SMA, TGF-beta1, Snail and Slug, turned spindle-shaped and migrated more; PM increased mitochondrial fission and LC3B/p62 accumulation, i.e. impaired autophagic flux. Intratracheal PM in mice reproduced EMT markers, inflammatory infiltration and alveolar wall thickening. Pirfenidone reversed these effects in vitro and reduced collagen deposition in vivo. Limitation: 100 ug/mL on a cell monolayer and bolus instillation are far above realistic deposited doses, and A549 is a carcinoma line; the drug result says nothing about population exposure. Relevance: low - mechanism for PM-associated fibrosis; no measurement content.

doi: 10.1186/s12931-026-03804-1

Do $PM_{2.5}$ superior cervical ganglion calcium channels satellite glia 2026 Int J Mol Sci

$PM_{2.5}$ Exposure Disrupts Sympathetic Neurotransmission by Impairing Voltage-Gated Ca2+ Channels and Remodeling Satellite Glia in the Superior Cervical Ganglion

Rats received intracerebroventricular $PM_{2.5}$; isolated superior cervical ganglia were also exposed directly. ICV $PM_{2.5}$ raised hypothalamic TNF-alpha and IL-6. Direct exposure cut medium norepinephrine without changing tissue content; depolarisation-evoked Ca2+ responses fell after either route, and patch clamp showed lower VGCC current density without Cav alpha1 mRNA change. GFAP and Kir4.1 rose (satellite glia). Limitation: intracerebroventricular injection bypasses the lung and does not model inhalation dose or translocation; particle source and dose relevance are not given in the abstract. Relevance: low.

doi: 10.3390/ijms27209184

Donohoe ultrafine amorphous silica epithelial stress pro-fibrotic 2026 Part Fibre Toxicol

Ultrafine amorphous silica induces chronic epithelial cell stress and pro-fibrotic reprogramming in a particle size- and exposure duration-dependent manner

METADATA ONLY. Title and authors only (Donohoe, Selvamani, Dharwal, O'Neil et al.). No abstract was retrievable at harvest. The title reports that ultrafine amorphous silica causes chronic epithelial cell stress and pro-fibrotic reprogramming that depends on particle size and exposure duration - relevant because amorphous silica is usually treated as low-hazard relative to crystalline silica. Model system, doses and endpoints are not available; nothing is inferred. Relevance: low - toxicology of a particle class not resolved by mass sensors.

doi: 10.1186/s12989-026-00710-8

Ganse diesel exhaust particles cardiac metabolomics glycolytic shift rats 2026 Biomolecules

Diesel Exhaust Particles Exposure Disrupts Cardiac Bioenergetics and Induces a Shift Towards Glycolytic Metabolism in Rats

Sprague-Dawley rats instilled with DEP or saline three times weekly for one or three weeks. At one week, 17 of 22 myocardial metabolites changed: succinate, acetate, formate, 3-hydroxybutyrate, ATP and ADP fell; glucose and lactate rose, consistent with reduced oxidative metabolism and a glycolytic shift, plus nucleotide degradation. Most changes resolved by three weeks; plasma changes were modest. Limitation: intratracheal bolus dosing of a single reference DEP does not reproduce inhaled dose rate, group sizes are not stated, and transient changes that resolve under continued exposure question chronic relevance. Relevance: low.

doi: 10.3390/biom16101501

Gautam diesel exhaust miR-133b-3p CTGF Hippo ECM remodeling mouse lung 2026 Environ Mol Mutagen

Diesel Exhaust-Induced miR-133b-3p Downregulation and CTGF-Linked Hippo Pathway Inactivation Promote ECM Remodeling and Inflammation in Mouse Lungs

C57BL/6 mice received diesel exhaust 30 min/day, 5 days/week for 8 weeks versus filtered air. Microarrays showed miR-133b-3p down more than 2-fold; CTGF was confirmed as a direct target in A549 cells (mimic reduced CTGF mRNA p=0.0054, protein p=0.0001). In vivo, exhaust raised CTGF, lowered P-YAP/YAP (Hippo inactivation), increased BALF IL-18, IL-1beta and GM-CSF, raised MMP-3/7/12 and lowered LOXL1/4, consistent with ECM remodelling. Limitation: exhaust dose (PM mass, NO_2) is not reported in the abstract, so particle and gas effects cannot be separated; the CTGF link is correlative in vivo. Relevance: low - traffic-exhaust mechanism; supports particle number or black carbon over mass as the near-road metric.

doi: 10.1002/em.70088

He FOXM1 lncRNA COPD-R biomass $PM_{2.5}$ pulmonary inflammation rat 2026 Ecotoxicol Environ Saf

FOXM1/LncRNA COPD-R regulates pulmonary inflammation in chronic obstructive pulmonary disease induced by biomass-related $PM_{2.5}$

Rats exposed to biomass-fuel smoke for seven months developed COPD-like disease; lung RNA-seq found 312 differentially expressed lncRNAs (294 up). lncRNA COPD-R rose from 0.15 to 1.05 (relative expression, p<0.01). In rat L2 alveolar cells, biomass-related $PM_{2.5}$ increased COPD-R; knockdown reduced PM-induced CCL2, IL-17F, MMP-9 and TIMP-1, and ChIP/luciferase assays placed FOXM1 upstream as its transcriptional activator, with rescue experiments consistent with a FOXM1-COPD-R axis. Limitation: whole-smoke exposure in vivo but $PM_{2.5}$ only in vitro, dose not reported in the abstract, and a rat lncRNA whose human orthologue is not established. Relevance: low - mechanistic support for household biomass PM, where the exposure gap is measurement in homes.

doi: 10.1016/j.ecoenv.2026.120952

Liu Streptococcus D19T fermentation supernatant asthmatic mice $PM_{2.5}$ 2026 Environ Int

Protective effect and mechanism of Streptococcus D19T fermentation supernatant in asthmatic mice exposed to $PM_{2.5}$

METADATA ONLY. Title and authors only (Liu, Sun, Wang, Jin et al.); no abstract retrievable at harvest. The title describes a probiotic-derived fermentation supernatant protecting asthmatic mice against $PM_{2.5}$-aggravated airway disease. Dose, model and outcomes cannot be stated. Limitation: unverifiable beyond the title; $PM_{2.5}$ aggravation models usually use instilled boluses far above ambient deposition. Relevance: low.

doi: 10.1016/j.envint.2026.110584

Liu cerium dioxide scavenging environmentally persistent free radicals $PM_{2.5}$ filtration review 2026 Int J Mol Sci

Cerium Dioxide-Based Scavenging of Environmentally Persistent Free Radicals in Fine Particulate Matter: A Prospective Review

A hypothesis-stating review, explicitly not a validated result. EPFRs persist on $PM_{2.5}$ for days to months and drive hydroxyl-radical generation by Fenton-like cycling after deposition. The authors propose migrating CeO2 Ce3+/Ce4+ redox cycling, an established antioxidant mechanism, into ambient-temperature air filters to quench EPFRs on captured particles, and argue that high-temperature catalytic uses of CeO2 are not a precedent. They list the open questions: direct quenching vs ROS interception, humidity, airflow, contact time, and CeO2 fate. Limitation: no data; the only cited support is gas-phase ROS scavenging in cigarette smoke. Relevance: low - filter chemistry, not measurement.

doi: 10.3390/ijms27209232

Nored lithium-ion battery fire soot PM pulmonary toxicity mice fluoride 2026 Environ Res

Pulmonary Toxicity of Particulate-Phase Combustion Byproducts from Lithium-Ion Battery Fires

Soot (10 fire tests) and smoke PM (2 tests) from lithium-ion battery thermal runaway were size-filtered to <10 um and given to mice (100 ug). Soot had more carbon (37% vs 18%), PM more anions (12% vs 6%), inorganic elements similar (23-28%); particulate fluoride reached ~5% of mass. Only soot altered ventilatory timing, peak expiratory flow and relaxation time; both raised lung inflammation to levels comparable with other combustion sources. Fluoride, Al, Fe, Ni and Li were associated with toxicity. Limitation: a single 100 ug bolus (aspiration/instillation) is not inhalation dosimetry, two smoke tests is a thin basis, and component-toxicity associations across so few samples are exploratory. Relevance: low - toxicology; flags fluoride-bearing PM as a new indoor/urban fire hazard.

doi: 10.1016/j.envres.2026.126023

Zakari PM_1 microbial profiles lung gut microbiome mice 2026 J Hazard Mater

Associations of particulate matter-associated microbial profiles with lung and intestinal microbiomes in healthy mice

Mice were housed under ambient PM_1, HEPA-filtered air or clean air; PM_1 was characterised for ions, metals, carbon fractions, oxidative potential and its own 16S microbiome (Proteobacteria-dominated). Lung diversity barely changed while the gut microbiome shifted: PM_1 reduced Actinobacteriota and Bacteroidota and raised Proteobacteria versus both control arms. Eleven families and six genera overlapped among PM_1, lung and gut communities in the PM_1 group. Limitation: group sizes and exposure concentrations are not given in the abstract, gases co-vary with PM_1, and taxon overlap does not show transfer from air to gut. Relevance: low - airborne microbial load is a PM component no optical sensor resolves.

doi: 10.1016/j.jhazmat.2026.143978

Zhao coal dust macrophage extracellular vesicles thyroid cancer miR-205-5p 2026 Front Immunol

Coal dust-stimulated macrophage-derived extracellular vesicles promote thyroid cancer cell phenotypes involving miR-205-5p, CAMK1G, and NF-kB-associated signaling

THP-1 macrophages exposed to characterised coal dust released extracellular vesicles enriched in miR-205-5p; these increased TPC-1 thyroid cancer cell viability, clonogenicity, migration and invasion and reduced apoptosis. Effects ran through CAMK1G and NF-kB (p65, IkBa phosphorylation) and were attenuated by miR-205-5p inhibition or CAMK1G knockdown; coal-dust-exposed mice and humanised xenograft mice were used in vivo. Limitation: the premise - coal dust promoting thyroid cancer - has no epidemiological anchor, and macrophage dosing in vitro is far above realistic lung burdens. Relevance: low - occupational mechanism without an exposure metric.

doi: 10.3389/fimmu.2026.1856770

Occupational & indoor — 14 records

Carried from the daily issues — full summaries (14 of 14 records in this subtopic)

Arif Agra footwear cluster ventilation IAQ tiers 2026 Research Square (preprint)

Ventilation, indoor air quality and thermal comfort in India's leather footwear manufacturing cluster: a tier-resolved field study

Agra footwear cluster, three enterprise tiers (formal factory to cottage unit), 378 worker-sessions over four seasons and 147 days, breathing-zone measurements paired with ASHRAE 55 ballots. All pollutants rose linearly from large to cottage scale as air velocity fell from 0.80 to 0.16 m/s; velocity explained 79% of between-tier pollutant variance. No session anywhere met the WHO $PM_{2.5}$ guideline; 91.3% exceeded the ACGIH formaldehyde TLV. Limitation: unreviewed preprint; PM instrument and calibration are not described in the abstract, and session means against an annual guideline overstate comparability. Relevance: moderate - informal-sector workplaces with no monitoring are an obvious deployment niche for wearable or wall-mounted low-cost PM nodes.

doi: 10.21203/rs.3.rs-10780874/v1

Blanco-Alegre ironing indoor ultrafine particles ventilation eBC 2026 Atmosphere

Ironing as a Source of Indoor Aerosol Particles: Effects of Ventilation and Ironing Conditions

Residential measurements with a conventional steam iron and a steam iron with a portable boiler, at low and higher ventilation. PNC rose by about two orders of magnitude during ironing, mostly 20-80 nm, to a maximum of 7.12e5 cm^{-3} (conventional iron, low ventilation). Emission rates 8.1e11-1.5e12 particles/min, highest for the boiler iron at low ventilation; modes shifted to larger sizes after ironing; eBC changed little. Limitation: one dwelling, few repetitions, and the source mechanism (water additives, fabric, soleplate heating) is not separated; no mass concentration is reported, so health relevance is by number only. Relevance: medium - an ultrafine source that optical home sensors will undercount almost entirely.

doi: 10.3390/atmos17111063

Fan wildfire smoke indoor $PM_{2.5}$ absenteeism Oklahoma Head Start OK-AIR 2026 J Expo Sci Environ Epidemiol

Wildfire smoke, indoor and outdoor $PM_{2.5}$, and sick-related absenteeism in Oklahoma Head Start early care and education centers

Cohort 1 of the OK-AIR cluster-randomised 2x2 factorial trial (HEPA, upper-room UVGI): 5 centres, 20 classrooms, 204 children aged 0-5, 2023-2024. On smoke-impacted days, the intervention period (pooled arms) was associated with 29.0% lower indoor $PM_{2.5}$ (22.8-34.7%) after adjustment for outdoor $PM_{2.5}$. Each +100 in the 7-day mean of regional FIRMS fire detections: sick-related absence OR 1.106 (1.018-1.201). Limitation: the period contrast pools all arms, so it is not a HEPA effect, and smoke days cluster in Feb-Mar, confounded by respiratory-virus season; fire counts are a regional proxy, not exposure. Relevance: high - classroom sensor plus EPA outdoor pairing is a deployable I/O design for childcare networks.

doi: 10.1038/s41370-026-00993-1

Geng grain dust simulator manikin respirator workplace protection factor 2026 J Agric Saf Health

Using a Grain Dust Simulator with a Manikin System to Evaluate Performances of Commonly Used Respirators

A grain dust simulator combined with a breathing manikin estimated SWPFs for four common respiratory protection devices at four respirable dust levels (5, 10, 15, 20 mg/m3) and two inhalation flows (30 and 85 L/min), benchmarked against real-time dust in on-farm grain bins. The platform was repeatable and separated device performance; the nuisance dust mask is not recommended for grain bins. SWPF values are not in the abstract. Limitation: manikin faces do not reproduce human fit variability or movement, so SWPF overstates real protection for poorly fitted FFRs; a static simulator lacks the burst exposures of bin entry. Relevance: low-medium - inside/outside real-time dust monitors are the field version of this test.

doi: 10.13031/jash.16485

Hunter e-cigarette aerosol early post-emission particle decay controlled conditions 2026 Sci Total Environ

Early post-emission particle decay dynamics of E-cigarette aerosols under controlled conditions

METADATA ONLY. Title and authors only (Hunter, Jacoby, Ferdousi-Rokib, Asa-Awuku et al.). No abstract was retrievable at harvest. The title describes the first seconds-to-minutes decay of e-cigarette aerosol under controlled conditions - evaporation of volatile droplets that shrinks particles rapidly after exhalation or emission. Instruments, conditions and decay constants are not available; nothing is inferred. Relevance: medium - e-cigarette droplets produce large, short-lived spikes on indoor optical PM sensors that mass-based readings misstate because the particles are liquid and evaporating; useful for indoor network event classification.

doi: 10.1016/j.scitotenv.2026.182379

K-HEALTHinAIR multicountry indoor air homes schools low-cost sensors 2026 Eur J Public Health (conference abstract)

Multicountry evaluation of indoor air in homes and schools: results from K-HEALTHinAIR project

Low-cost sensors in homes and schools across four countries, 2025. Home $PM_{2.5}$ was elevated with smoking (93.9 $\mu g\,m^{-3}$), air conditioning (34.8), gas cooking (29.5) and Dutch district heating (19.4). School $PM_{2.5}$ was higher in Poland (22.0) than Austria (9.5). $PM_{2.5}$ exceedance reached 56% of days in Austrian wood-heated homes and 36% in Spanish smokers' homes; CO_2 exceedance was 41% in Austrian schools. Limitation: sensor calibration and RH correction are not described, so cross-country contrasts may partly be instrument and humidity artefacts; sampling of homes is unstated. Conference abstract. Relevance: high - a multi-country indoor LCS deployment whose QA is the open question.

doi: 10.1093/eurpub/ckag169.339

Kulmala determination deposition rates indoor particle dynamics 2026 Aerosol Sci Technol

Determination of deposition rates from indoor particle dynamics

METADATA ONLY. Title and authors only (Kulmala I., Siilin, Hakala, Ehder-Gahm); no abstract retrievable at harvest. The title describes deriving particle deposition rates from indoor concentration dynamics. Method, size range and validation cannot be stated. Limitation: unverifiable beyond the title; decay-based deposition estimates are confounded by air-exchange and mixing assumptions unless these are measured independently. Relevance: medium - deposition rates are an input for indoor/outdoor ratio models used to interpret paired indoor and outdoor low-cost sensors.

doi: 10.1080/02786826.2026.2754776

Martin simplified method portable air cleaner CADR low-cost sensors 2026 MethodsX

A simplified method for evaluating fine particulate matter removal performance of portable air cleaners

Methods paper for community-run CADR testing: release a short-lived nebuliser aerosol in a sealed room, log $PM_{2.5}$ decay with and without the cleaner on low-cost sensors, and compute CADR from the decay-rate difference. Residential low-cost-sensor CADRs ran ~26% below laboratory values from research-grade instruments - a conservative bias. Of ten units, a four-filter Corsi-Rosenthal box performed best among DIY builds; a fan shroud added ~11%. Across 135 models DIY units cost far less per CADR unit. Limitation: the 26% gap mixes sensor size-response with room mixing and leakage, and is not decomposed; nebuliser aerosol is not wildfire smoke. Relevance: high - a practical protocol in which the low-cost sensor is the reference, with its bias quantified.

doi: 10.1016/j.mex.2026.104216

Nahian Siegel in-situ testing in-duct air cleaners residential 2026 Build Environ

In-situ testing of in-duct air cleaners in residential buildings

METADATA ONLY. Title and authors only (Nahian, Siegel); no abstract retrievable at harvest from Crossref, Europe PMC or Semantic Scholar. The title describes in-situ testing of in-duct air cleaners (filters or electronic cleaners in forced-air systems) in occupied homes, presumably measuring particle removal under real operation. Country, number of homes, test protocol and removal metrics cannot be stated. Limitation: unverifiable beyond the title; in-situ effectiveness depends on HVAC runtime fraction, which lab single-pass efficiency ignores. Relevance: moderate - indoor LCS networks are the practical tool for verifying such devices in homes.

doi: 10.1016/j.buildenv.2026.115494

Ng particulate matter respiratory deposition inhalation dose biomass burning homes indigenous communities 2026 Atmos Environ

Particulate Matter Exposure, Respiratory Deposition and Inhalation Dosage Assessment in Biomass Burning Homes in Rural Indigenous Communities

METADATA ONLY. Title and authors only (Ng, Monalisa, Balachandran, Hoque et al.); day-0 Elsevier deposit with no abstract at harvest. The title describes household PM measurement in biomass-burning homes in rural indigenous communities with respiratory deposition and inhaled-dose modelling. Instruments, n and concentrations cannot be stated. Limitation: unverifiable beyond the title; deposition models (e.g. MPPD) are sensitive to the assumed size distribution, which optical monitors in smoke-laden kitchens report poorly. Relevance: medium - household biomass smoke is a key use case for low-cost PM monitors.

doi: 10.1016/j.atmosenv.2026.122448

Patras primary school classroom PM airborne viruses ventilation 2026 Eur J Public Health (conference abstract)

Dual burden of PM and airborne viruses in primary schools: A pilot study in Patras, Greece

March 2026 sampling in 28 classrooms of seven primary schools. Mean $PM_{2.5}$ 37.8 and PM_{10} 109 $\mu g\,m^{-3}$; every classroom exceeded the WHO 24 h guideline, and at least one respiratory virus was detected in every room (SARS-CoV-2 most often). Frequent natural ventilation was associated with lower PM ($p < 0.033$, Cliff's delta > 0.58) and less viral RNA. $PM_{2.5}/PM_{10}$ 0.35 points to resuspension. Limitation: short occupied-hours sampling compared against a 24 h guideline overstates exceedance, and one month in one city; instrument type unstated. Conference abstract. Relevance: moderate - coarse-dominated classroom PM is where optical LCS under-read most.

doi: 10.1093/eurpub/ckag169.1262

Saborit gas cooking indoor air quality households 7 European countries 2026 Indoor Environments

Characterization of the impact of gas cooking on indoor air quality in households across 7 European countries

Metadata only: no abstract at Crossref, Europe PMC or Semantic Scholar on the issue date. The title reports household measurements of gas-cooking impacts on indoor air quality across seven European countries; which pollutants (NO_2, $PM_{2.5}$, ultrafine particles) were measured, the number of homes and the instrumentation are not available here and are not inferred. Limitation (structural): cooking PM is dominated by the food and method rather than the fuel, so attributing particle increments to gas requires an electric comparison arm. Relevance: medium - multi-country household campaigns of this type typically rely on low-cost PM and NO_2 sensors. Retry the abstract before the weekly.

doi: 10.1016/j.indenv.2026.100202

Stefaniak recycled plastic bag HDPE filament 3-D printing emissions 2026 J Chem Health Saf

Recycling Plastic Bags into Filament: Consideration of Emissions during Melting, Extrusion, and 3-D Printing.

Waste HDPE plastic bags from South Africa and the US were blended with virgin HDPE, extruded into filament and 3-D printed, with particle- and gas-phase emissions measured through each stage. Filament extrusion emitted mostly 1.2-5.2 nm particles; printing emitted mostly 5.6-560 nm particles, with particle yields (no./g processed) 2-3 orders of magnitude higher than extrusion. Printing also released acetone (1.4-39.4 ug/g printed) and formaldehyde (9.9-16.1 ug/g). Mechanical properties matched virgin HDPE. Limitation: yields are per mass processed in a controlled enclosure, not personal breathing-zone concentrations, and no uncertainty is attached to the order-of-magnitude contrast; feedstock contamination of bag plastic was not characterised. Relevance: low-medium - nearly all the number emission sits below the ~300 nm optical cut of consumer PM sensors, so maker-space monitoring needs a CPC or diffusion charger, not a PMS-class node.

doi: 10.1021/acs.chas.5c00138

Zargarian spent coffee grounds MOF filtration membrane $PM_{2.5}$ bioaerosol photothermal 2026 Small

Spent Coffee Grounds-Derived Micro-Nano-Architectured Air Filtration Membranes for Multiwavelength Light-Triggered Bioaerosol Inactivation

Single-step co-electrospinning of MOF-177/polyacrylonitrile nanofibres with electrosprayed ZIF-8-decorated spent coffee ground microparticles. At 4.5x the NIOSH respirator test face velocity the membrane captured 99.7% of $PM_{2.5}$ at about half the pressure drop of a MOF-free control and removed 83.4% of ammonia; photothermal heating >80 C under simulated sunlight or NIR restored microbial suppression after three days of office air. Limitation: one challenge aerosol and short loading; no size-resolved efficiency at the MPPS, no long-term loading curve, and MOF stability under humidity is not shown. Relevance: low - a filtration material, not a measurement; at most a candidate inlet pre-filter for bioaerosol-sensitive sensor housings.

doi: 10.1002/smll.76528

Burden, policy & mitigation — 11 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (11 OF 11 RECORDS IN THIS SUBTOPIC)

BEST-COST all-cause vs disease-specific exposure-response functions burden $PM_{2.5}$ NO_2 2026 Eur J Public Health (conference abstract)

Effect of all-cause versus disease-specific exposure-response functions on air pollution burden

BEST-COST burden estimates (healthiar R package) in five countries, comparing WHO all-cause mortality ERFs with disease-specific ERFs. Disease-specific YLL were 2-4x lower (ratio 0.26-0.52; median 0.42 for $PM_{2.5}$, 0.31 for NO_2). Country rankings were stable (Spearman rho >=0.90), but absolute agreement was moderate for $PM_{2.5}$ (ICC 0.46) and poor for NO_2 (0.15). Limitation: five countries is too few for inference (Wilcoxon p=0.0625); the gap partly reflects causes of death excluded from the disease-specific set, which is a substantive question, not a method artefact. Conference abstract. Relevance: moderate - ERF choice, not exposure error, dominates burden uncertainty.

doi: 10.1093/eurpub/ckag169.810

Grzeszczyk-Nojszewska Poland air pollution interventions public attitudes survey 2026 Int J Occup Med Environ Health

Public attitudes toward personal and population interventions to reduce air pollution exposure: a 2025 nationwide cross-sectional survey in Poland

CAWI survey, May 2025, quota sample of 1,092 adults. 76.4% saw air pollution as requiring public action (women aOR 1.47, 1.11-1.96; age >=60 aOR 2.49, 1.59-3.91). Commonest protective behaviours: closing windows during episodes (44.2%) and avoiding outdoor time (36.5%). Fines for burning prohibited materials in home furnaces had 70.2% support. Limitation: quota online panels over-represent the engaged; no link to local concentrations, and stated behaviours are unverified. Relevance: medium - window closing is the dominant action, so indoor/outdoor paired sensing is what tells people whether it works.

doi: 10.13075/ijomeh.1896.02831

Imman caregivers $PM_{2.5}$ prevention preschool children northern Thailand qualitative 2026 Child Health Nurs Res

Caregivers' experiences of $PM_{2.5}$ prevention for preschool children in Thailand: a qualitative study with focus group interviews

Focus groups plus in-depth interviews with 20 primary caregivers (14 women) of preschool children in a rural northern Thai community subject to recurrent haze; content analysis (Graneheim and Lundman). Themes: barriers of awareness and responsibility against pervasive local sources; individual prevention driven by visible symptoms rather than air-quality information; and a shift toward collective action and calls for structural change (waste-burning control). Limitation: one community, 20 participants, no link to measured exposure or behaviour change. Relevance: medium - caregivers act on symptoms, not data; local sensor displays only help if they reach this audience in an actionable form.

doi: 10.4094/chnr.2025.047

K-HEALTHinAIR indoor $PM_{2.5}$ COPD attributable burden Spanish households 2026 Eur J Public Health (conference abstract)

Attributable Burden of COPD from Indoor $PM_{2.5}$ Exposure in Spanish Households, 2025

Comparative risk assessment combining measured household $PM_{2.5}$ (K-HEALTHinAIR), GBD 2023 Spanish rates and published exposure-response functions. Mean attributable COPD burden ~116 thousand DALYs; by household type, homes with smokers 327 thousand, air-conditioning for heating 183 thousand, no climate control 192 thousand, gas cooking 119 thousand. Limitation: applying ambient-derived ERFs to indoor $PM_{2.5}$ of different composition (tobacco, cooking) and scaling household-type estimates to national totals as if every home were that type; the subgroup figures are not additive and no uncertainty is given. Conference abstract. Relevance: moderate - burden estimates driven by low-cost indoor measurements.

doi: 10.1093/eurpub/ckag169.1053

Lee type 2 diabetes burden Asia GBD 2023 particulate matter leading risk 2026 Diabet Med

The evolving burden of type 2 diabetes mellitus in Asia, 1990-2023: A Global Burden of Disease Study 2023

GBD 2023 secondary analysis, 34 Asian countries in five regions, 1990-2023. Age-standardised T2DM prevalence and DALY rates rose almost everywhere, most in South Asia (prevalence +66.5%, DALYs +75.6%); High-income Asia Pacific DALYs fell 2.4% (UI crosses zero). Particulate matter pollution was the leading attributable risk factor in every Asian region, though low-fruit diet led in South Asia (157 DALYs/100,000). Limitation: attribution inherits the GBD $PM_{2.5}$-diabetes exposure-response curve and satellite-model exposure surfaces; it is not new evidence of causality, and the abstract gives no PM-attributable numbers. Relevance: low - burden framing; the exposure layer is exactly the satellite-fusion product low-cost networks are used to correct in Asia.

doi: 10.1111/dme.70421

Lu kitchen air purifier health-economic outcomes 338 Chinese cities 2026 Environ Int

Heterogeneous health-economic outcomes of kitchen air purifier interventions: A 338-city study in urban China

METADATA ONLY. Title and authors only (Lu, Yao, Jiang, Fu et al.). No abstract was retrievable at harvest. The title reports a 338-city assessment of heterogeneous health-economic outcomes of kitchen air-purifier interventions in urban China. Exposure model, health endpoints, cost basis and effect sizes are not available; nothing is inferred. Relevance: moderate - cooking-generated PM is the largest indoor source a low-cost sensor sees, and any intervention-scale cost-effectiveness estimate depends on indoor concentration inputs such sensors could supply. Retry abstract first.

doi: 10.1016/j.envint.2026.110587

Mohamadi Isfahan $PM_{2.5}$ AirQ+ WRF-SMOKE-CMAQ health damage 2026 J Environ Health Sci Eng

Assessing the health impacts of $PM_{2.5}$ from anthropogenic and natural sources in Isfahan, Iran

Station annual $PM_{2.5}$ fed AirQ+; at a 2020 GDP of USD 2,283 per capita, annual health damage was USD 441 M (253-628 M). A year-long WRF-SMOKE-CMAQ run attributed 7.9% of damage to iron and steel, 2.2% to cement and 1.7% to brick kilns; 'environmental' (natural) particles accounted for USD 115.3 M (26%). Limitation: attributing damage linearly by modelled mass share assumes equal toxicity, and the share of dust (natural) depends on how the inventory handles anthropogenic fugitive dust. Few stations in a large city. Relevance: low - a standard AirQ+ exercise; denser monitoring would mainly tighten the exposure input.

doi: 10.1007/s40201-026-01011-0

Ruengdit active-flow respirator fit children Chiang Mai 2026 Front Public Health

Active-flow systems enhance respirator fit success in children: a comparative study with standard disposable models

82 children aged 5-10, modified OSHA quantitative fit test. A standard triple-fold KN95-equivalent FFR passed in 3.7%; adding a portable HEPA unit supplying 66.5 L/min positive pressure raised the pass rate to 23.2% (OR 17.0) and median fit factor from 9.5 to 39.5. Wider bizygomatic breadth (aOR 1.31) and longer menton-sellion (1.11) favoured fit; head nodding was worst. Limitation: no CI given for the OR; crossover order and fit-test learning effects are not described, and 77% still failed with the active system. Relevance: high - a median fit factor of 9.5 means adult masks give children roughly 90% leakage-limited protection at best, which matters for exposure-reduction advice issued from LCS haze alerts.

doi: 10.3389/fpubh.2026.1851331

Ruwali Lary US infant life expectancy environmental determinants machine learning 2026 Sensors (Basel)

Estimating Life Expectancy in the U.S.: Identifying Environmental Determinants and Disparities (2003 to 2019)

County-level IHME life expectancy for the under-1 age group, contiguous US 2003-2019, regressed on 33 ECMWF environmental variables with ML. Marked north-south disparity, southern counties lowest. With only five variables the model reached R2 0.81 and RMSE 1.06 years on an independent test set. Beyond PM, O_3 and CO, lower formaldehyde, sulphate and dust aerosol, more vegetation and lower wet-bulb temperature ranked as predictors of higher life expectancy. Limitation: no socioeconomic, race or healthcare covariates, so the south-north environmental gradient absorbs deprivation; feature importance is predictive, not causal, and IHME county estimates are themselves smoothed models. Relevance: moderate - a reanalysis-driven baseline that dense LCS data could sharpen.

doi: 10.3390/s26206478

Wang carbon footprint air-pollutant emissions Beijing 2022 Winter Olympics 2026 Atmosphere

Linking Carbon Footprints to Air-Pollutant Emissions from Mega Sporting Events: Evidence from the Beijing 2022 Winter Olympics

Screening-level inventory converting the verified 713,788 t CO_2e footprint of the Beijing 2022 Winter Olympics (2016-2022) into co-emitted pollutants via Chinese pollutant-to-CO_2 ratios over eight emission carriers, with Shapley attribution and Monte Carlo/Sobol uncertainty. Construction produced 63% of CO_2e, ~70% of 1,335 t NOx and 76% of 279 t $PM_{2.5}$. Green electricity avoided 31% of no-measure emissions; offsets reduced no air pollutant. Limitation: emission-side only - ratio-based $PM_{2.5}$ with no dispersion, concentration or exposure step, so there is no health quantity, and national average ratios may misstate construction-site fugitive dust. Relevance: low - construction dust is the source class that site-perimeter low-cost PM monitoring is typically deployed to police.

doi: 10.3390/atmos17111072

Wang climate-adaptive city pilot $PM_{2.5}$ health dividend China 2026 Front Public Health

Does climate adaptation policy deliver a health dividend? Exposure, modeled health burden, and defensive capacity in China's climate-adaptive city pilots

Difference-in-differences on a 2011-2022 panel of 264 prefecture cities, 21 treated by China's 2017 climate-adaptive city pilot. The pilot is associated with $PM_{2.5}$ lower by 3.87 $\mu g\,m^{-3}$ (95% CI -6.49 to -1.25; wild-bootstrap p=0.0067); GEMM translates this into an ~8.1% lower modelled burden (-11.32 to -4.76%) - an accounting result, not observed mortality. Defensive-capacity responses (physicians, beds, insurance) are imprecise. Limitation: the estimate shrinks and loses precision in the matched sample and after pollution-convergence adjustment, so selection of already-improving cities is not excluded; 21 treated units is a small base. Relevance: low-medium - policy evaluation at city scale rests on gridded $PM_{2.5}$ whose own error is not propagated.

doi: 10.3389/fpubh.2026.1987084

Other clinical endpoints — 8 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (8 OF 8 RECORDS IN THIS SUBTOPIC)

Deng ultrafine particles kidney disease ED visits New York 2026 Environ Res

A mysterious danger for kidney disease exacerbations: short-term impacts of ultrafine particles on hospital visits

New York State discharge data 2013-2018 (95% of hospitals), 269,945 kidney-disease ED visits, conditional logistic regression per IQR of modelled UFP. Associations appeared at lag 1 and on every cumulative lag 0-1 to 0-6, peaking at lag 0-3: excess risk 0.74% (0.28-1.20) per IQR. Larger in adults >65, non-African American patients, hot and dry days, and fall/winter. Limitation: UFP is a CTM product, not a measurement; its spatial error is large and likely correlated with $PM_{2.5}$ and NO_2, and the abstract reports no co-pollutant adjustment, so 'UFP-specific' toxicity is not isolated. The effect per IQR is small. Relevance: high - a state-scale UFP health signal built entirely on modelled number concentration is the case for CPC or OPC-extended nodes to validate it.

doi: 10.1016/j.envres.2026.126022

Duan landscape fire-sourced $PM_{2.5}$ kidney function eGFR children China 2026 Environ Res

Landscape fire-sourced $PM_{2.5}$ and kidney function in children: evidence from a national cross-sectional study in China

China National Nutrition and Health Surveillance of Children and Lactating Women 2016-2018, n=12,102 aged 5-18. Per IQR (0.505 $\mu g\,m^{-3}$) of fire-sourced $PM_{2.5}$: eGFR -0.83 mL/min/1.73 m2 (-1.03 to -0.62); middle-school students -1.46 (-1.83 to -1.09) vs primary -0.59 (-0.84 to -0.34). Associations were stronger with low NDVI and low vitamin or carbohydrate intake. Limitation: an IQR of 0.5 $\mu g\,m^{-3}$ is a tiny contrast for an effect this size, which points to residual spatial confounding (fire-PM tracks rural, agricultural regions); creatinine-based eGFR in children depends on muscle mass and hydration, which also vary by region. Cross-sectional. Relevance: low - source-specific PM from a chemical-transport model, no sensing.

doi: 10.1016/j.envres.2026.125978

Hoang $PM_{2.5}$ minimal residual disease childhood ALL REDIAL 2026 JNCI Cancer Spectr

Fine particulate matter and minimal residual disease in childhood acute lymphoblastic leukemia from the REDIAL consortium

2,125 children with ALL; mean daily residential $PM_{2.5}$ during induction in quartiles; logistic regression. In B-ALL, positive end-of-induction MRD: Q2 aOR 1.56 (1.08-2.24), Q3 1.51 (1.05-2.17), Q4 1.19 (n.s.). Restricted to NCI high-risk B-ALL, aORs rose to 1.73-2.28 across Q2-Q4. Limitation: a non-monotone quartile pattern with the top quartile null is not a dose-response; census-tract $PM_{2.5}$ over a ~4-week induction is tightly tied to treatment centre and ethnicity, which REDIAL was built to study and which also predicts MRD. Relevance: low-medium - induction-window exposure is short enough that a home monitor would be the right instrument.

doi: 10.1093/jncics/pkag116

Hu air pollution built environment lung cancer recurrence after surgery China 2026 Research Square (preprint)

Ambient Air Pollution, Residential Built Environment, and Lung Cancer Recurrence After Curative-Intent Surgery: A Retrospective Cohort Study in Three Chinese Cities

542 adults after curative resection at three tertiary hospitals (2019-2026); 60 recurrences, median follow-up 27.3 months. Site-stratified meta-analysis, per within-site SD of first-year post-diagnosis exposure: $PM_{2.5}$ HR 1.41 (1.08-1.84), PM_{10} 1.38 (1.10-1.73), NO_2 1.67, CO 1.51, SO_2 1.59; O_3 null. A co-exposure component gave HR 1.51 (1.20-1.89); green-space diversity HR 0.59/SD. Limitation: 60 events across six pollutants, mixtures and built-environment terms is badly underpowered for adjustment; within-site SD contrasts are small and residence tracks stage, histology and access to follow-up imaging. Preprint. Relevance: low.

doi: 10.21203/rs.3.rs-11311605/v1

Li ambient air pollution gastric cancer incidence 204 countries ecological 2026 Curr Med Sci

Ambient Air Pollution and Gastric Cancer Incidence: A Multinational Ecological Analysis (1990-2020)

Negative-binomial fixed-effects panel over 204 countries, 1990-2020, from GBD 2021. Per unit $PM_{2.5}$: +0.3% (0.1-0.5%) in age-standardised gastric cancer incidence; NO_2 +1.8% (1.2-2.4%), O_3 +0.6% (0.3-0.9%). Splines show nonlinearity; $PM_{2.5}$ lag window 3-4 years; male and high-middle-SDI strata most affected. Limitation: both exposure and outcome are GBD modelled quantities that share covariates (SDI), and a 3-4 year lag for a cancer with a decades-long Helicobacter-driven course is biologically implausible; country fixed effects do not remove the H. pylori decline that tracks development. The authors concede ecological status and call for individual-level validation. Relevance: low.

doi: 10.1007/s11596-026-00255-z

Slovakia $PM_{2.5}$ lung cancer mortality gender lag ecological 2026 Eur J Public Health (conference abstract)

$PM_{2.5}$ exposure and lung cancer mortality in Slovakia: gender differences and delayed effects

National time-series ecology, 2010-2023: population-weighted $PM_{2.5}$ fell from 24.4 to 15.2 $\mu g\,m^{-3}$ while lung-cancer mortality fell from 30.7 to 25.3 per 100,000 (AAPC -1.5%), driven by men (56.8 to 42.6; AAPC -2.2%); female rates were flat (12.6). Correlation with $PM_{2.5}$ r=0.95 overall and 0.97 in men, none in women; lags suggested emerging female associations. Limitation: 14 annual points of two monotonic trends correlate trivially; male smoking decline over the same period explains the male-only pattern, and the outcome is a GBD model output. Conference abstract. Relevance: low.

doi: 10.1093/eurpub/ckag169.1260

Wang $PM_{2.5}$ lung cancer nonsmokers narrative review IL-1beta EGFR screening 2026 J Toxicol Environ Health B Crit Rev

Environmental $PM_{2.5}$ exposure associated with lung cancer in nonsmokers: a narrative review of high-risk groups, mechanisms, and implications for screening

Narrative review of cohort, meta-analytic and toxicological evidence on $PM_{2.5}$ and lung cancer in never-smokers (<100 lifetime cigarettes), weighted to East Asian women. Emphasises macrophage IL-1beta signalling promoting EGFR-mutant adenocarcinoma, and lists high-exposure microenvironments: incense burning, cooking and solid fuel, subway/bus/motorbike commuting and professional driving, firefighting, construction, informal e-waste recycling. Argues for cumulative $PM_{2.5}$ in risk-prediction models and screening of exposed never-smokers. Limitation: non-systematic; no new quantitative synthesis, and the microenvironment list is not dose-ranked. Relevance: medium - cumulative personal $PM_{2.5}$ as a screening criterion would need exactly the personal and microenvironment data wearable sensors produce.

doi: 10.1080/10937404.2026.2747089

Zhong air pollution years of life lost cancer Pudong Shanghai time series 2026 PLoS One

The short-term impact of ambient air pollution on years of life lost from cancer in Pudong New Area, Shanghai: A time series study

Daily time series of cancer deaths and YLL in Pudong, 2013-2021, with DLNM (linear for YLL, quasi-Poisson for deaths). Per 10 $\mu g\,m^{-3}$ at lag 0-1: $PM_{2.5}$ +2.22 YLL (0.71-3.74), PM_{10} +1.04 (0.17-1.91); SO_2, NO_2 and O_3 also positive. Effects were marginally larger in men, younger and more educated people. Limitation: short-term harvesting of cancer deaths is mostly mortality displacement in terminally ill patients, and single-pollutant models leave the $PM_{2.5}$ estimate inseparable from co-pollutants; YLL per day is driven by a few young deaths, so the scale is unstable. Relevance: low - single-district monitor-average exposure.

doi: 10.1371/journal.pone.0360049

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Al-Atram early-life air pollution child neurodevelopment critical review	Early Life Ambient Air Pollution and Child Neurodevelopment: Critical Review of Cohort Evidence	*J Xenobiot*	Critical review	Prenatal and early-childhood $PM_{2.5}$, PM_{10}, NO2	Global / multi-region
Holm CHAPS Fresno neurobehavior BDNF children air pollution	Neurobehavioral assessments and biomarkers in school children who are highly exposed to air pollution	*Front Public Health*	Prospective birth cohort + DLNM	Daily prenatal and first-year $PM_{2.5}$, PM_{10}, O3, NO2, NOx, PAH456 and EC	USA (Fresno, California)
Li built environment sleep duration air pollution mediation Yinzhou cohort	Association between built environment and sleep duration mediated by air pollution: a cohort study in China	*BMC Public Health*	Prospective cohort	Residential PM_{10}, $PM_{2.5}$, NO2 (modelled annual) as mediators	China (Yinzhou, Ningbo)
Papatheodorou prenatal wildfire $PM_{2.5}$ ASD ADHD Medicaid	Prenatal Exposure to Wildfire $PM_{2.5}$ and Risk of ASD and ADHD in Children Enrolled in Medicaid	*JAMA Pediatr*	Retrospective cohort + DLNM	Weekly wildfire-smoke-attributable $PM_{2.5}$ (1 km), aggregated to residential ZIP code across pregnancy	USA (nationwide Medicaid, 2007-2014)
Santoso air pollution noise greenness salivary cortisol BAMSE	Air pollution, road traffic noise, and greenness in relation to salivary cortisol in young adults	*J Expo Sci Environ Epidemiol*	Cross-sectional (birth cohort)	Address-level $PM_{2.5}$, PM_{10}, black carbon, NOx (dispersion models), prior year	Sweden (Stockholm, BAMSE)
Wang air pollution adolescent anxiety depression Sichuan cross-sectional	Air pollution and mental health in adolescents: a cross-sectional study in Sichuan, China	*BMC Public Health*	School-based cross-sectional	PM_{10}, $PM_{2.5}$, O3, CO, SO2, NO2 assigned via IP-geolocated survey location; winter periods	China (Sichuan)
Yang heat anomaly wildfire smoke $PM_{2.5}$ incident depression All of Us	Heat anomaly, wildfire smoke, social-material vulnerability, and incident depression in a national EHR-linked cohort: A preregistered secondary analysis	*J Affect Disord*	Prospective cohort (registry linkage)	30-day mean wildfire-smoke $PM_{2.5}$ (Stanford ECHO model) at ZIP3	USA (national, All of Us)
Zhou short-term indoor $PM_{2.5}$ cognition HRV randomized crossover	Short-term indoor fine particulate matter exposure impedes cognitive performance and autonomic nervous system in adults	*iScience*	Randomised crossover trial	Candle-generated indoor $PM_{2.5}$, 1.70 vs 31.22 $\mu g/m^3$ (purifier vs sham filter)	UK (laboratory)
Cardiovascular & metabolic					
Andreotti active fire detections circulatory hospital admissions Brazil ecological panel	Active Fire Detections and Hospital Admissions for Circulatory Diseases in Brazil: An Ecological Panel Study	*Cureus*	Ecological panel, fixed effects	Annual satellite active-fire detection density (INPE) as smoke proxy	Brazil (27 federative units)
Cao $PM_{2.5}$ chemical components heart disease longitudinal China 123 cities	Association between $PM_{2.5}$ and its chemical components exposure and heart disease risk: a longitudinal study from China	*BMC Cardiovasc Disord*	Prospective cohort	$PM_{2.5}$ and components (SO4, NO3, NH4, OM, BC), gridded	China (123 cities)
Cerin air pollution road environment cardiometabolic walking AusDiab	Associations of Ambient Air Pollution and Road-Related Environmental Characteristics with Cardiometabolic Health: The Role of Walking	*Toxics*	Cross-sectional (cohort baseline)	Residential $PM_{2.5}$ and NO2 (modelled); road density, distance to nearest busy road	Australia (AusDiab wave 3)
Musa outdoor air pollution cardiovascular health school children Dar es Salaam	Effects of outdoor air pollution on the cardiovascular health of school-aged children residing near polluted areas and the efficacy of mitigation measures in Dar es Salaam, Tanzania	*BMC Public Health*	Cross-sectional survey	None measured; residence near industry, major roads or waste burning used as proxy	Tanzania (Dar es Salaam)
Zhang short-term air pollution stroke incidence Guiyang karst city	Time-series analysis of the impact of short-term exposure to atmospheric pollutants on stroke risk among residents in Guiyang City, 2016-2021	*Front Public Health*	Time-series (generalized additive model)	Daily fixed-site $PM_{2.5}$ and PM_{10} (city mean), with SO2, NO2, O3, CO	China (Guiyang)

First author	Focus	Journal	Design	Metric	Region
Zhao interpretable machine learning multi-window $PM_{2.5}$ hypertension CHNS Beijing	An Interpretable Machine Learning Framework Integrating Multi-Window Environmental Exposure for Hypertension Risk Assessment	*Toxics*	Gradient-boosting model + SHAP interpretability	Cumulative $PM_{2.5}$ over 0, 7, 15, 30 and 60 days	China (Beijing, CHNS subsample)
Reproductive & developmental					
Ai air pollution IVF outcomes Chengdu retrospective cohort	Association between Ambient Air Pollution Exposure and Reproductive Outcomes following In Vitro Fertilization: A Retrospective Cohort Study	*Int J Fertil Steril*	Retrospective cohort	CHAP 1 km $PM_{2.5}$, PM_{10}, O3, NO2 at residence by IVF stage	China (Chengdu)
Gaylord DNA methylation signature prenatal air pollution ECHO cohort	A methodology to identify a DNA methylation-based signature of prenatal air pollution exposure in the ECHO Cohort	*Environ Epigenet*	Cross-sectional (birth cohort)	Prenatal $PM_{2.5}$, NO2, 8-h max O3 from geocoded ML exposure models	USA (multi-site ECHO cohorts)
ICMR-SPIC stillbirth risk factors biomass fuel India pooled cohort	Prevalence and risk factors for stillbirths in India: secondary analysis of ICMR-Stillbirth Pooled India Cohort	*Lancet Reg Health Southeast Asia*	Prospective cohort	Household biomass fuel use (questionnaire); no PM measurement	India (ten pooled cohorts)
Lin prenatal air pollution transient tachypnea newborn Taiwan	Prenatal air pollution exposure and the risk of transient tachypnea of the newborn	*Front Pediatr*	Case-cohort study	40-week prenatal $PM_{2.5}$, CO, NOx from nearest of 71 monitoring stations (postal code)	Taiwan (national)
Scotland prenatal $PM_{2.5}$ infant amoxicillin birthweight mediation	Prenatal $PM_{2.5}$ and Infant Antibiotics: Birth-Weight Mediation via Causal Analysis	*Eur J Public Health (conference abstract)*	Prospective cohort (registry linkage)	Annual postcode $PM_{2.5}$, pregnancy-month weighted (mean 7 $\mu g/m^3$)	United Kingdom (Scotland, 2009-2018)
Valuvunina $PM_{2.5}$ PurpleAir quarry low birth weight Kinshasa DRC	Atmospheric Pollution by Fine Particulate Matter ($PM_{2.5}$) and Low Birth Weight: A Cross-Sectional Study in Binza Ozone Health Zone, Democratic Republic of the Congo, 2022	*J Toxicol*	Cross-sectional survey	Calibrated PurpleAir PA-II-SD $PM_{2.5}$ (48-72 h per health area; 0.52 x raw + 3.2 vs BAM)	Democratic Republic of the Congo (Kinshasa)
Zhu prenatal PM1 $PM_{2.5}$ PM_{10} childhood hemoglobin anemia Fujian birth cohort	Association of prenatal PM1, $PM_{2.5}$, and PM_{10} exposure with early childhood hemoglobin levels and anemia: Identification of susceptibility windows	*Ecotoxicol Environ Saf*	Prospective birth cohort	Trimester-specific PM1, $PM_{2.5}$, PM_{10} (ChinaHighAirPollutants)	China (Fujian)
Respiratory & allergic					
Dublin air pollution fungal spores respiratory admissions time series	Short-term effects of air pollution and aeroallergens on respiratory admissions	*Eur J Public Health (conference abstract)*	Time-series	Daily NO2, $PM_{2.5}$, PM1 (EPA Ireland); fungal spore counts	Ireland (Dublin)
ISVOLCAN Tajogaite volcano La Palma lung function obstructive pattern	Lung function after Tajogaite volcano eruption in La Palma: ISVOLCAN study	*Eur J Public Health (conference abstract)*	Cross-sectional (cohort baseline)	Residence area during the 2021 eruption; SO2 and PM exposure categories	Spain (La Palma, Canary Islands)
Kobza respiratory ambulatory care air quality Upper Silesia	The assessment of respiratory diseases in ambulatory care attributable to air quality in the Upper Silesian agglomeration	*Int J Occup Med Environ Health*	Ecological	Daily $PM_{2.5}$, PM_{10}, SO2, NOx, Pb, benzene from 50 regional stations	Poland (Upper Silesia)
Kusumaningtiar school children $PM_{2.5}$ PM_{10} health risk systematic review	Environmental Health Risks in School Children Due to Air Pollution Exposure: Systematic Literature Review	*Research Square (preprint)*	Systematic review (PRISMA)	Indoor and outdoor school $PM_{2.5}$ and PM_{10}	Global / multi-region
Li spatiotemporal causal machine learning $PM_{2.5}$ asthma Colorado counties	A Spatiotemporal Causal Machine Learning Framework for Quantifying the Impact of $PM_{2.5}$ on Asthma at the Community Level	*GeoHealth*	Causal ML on monitoring records	County-month $PM_{2.5}$ (exposure surface not stated in abstract)	USA (Colorado)
Lovric ragweed sensitisation PM_{10} pollen model comparison Croatia ATOPICA	Outdoor Environmental Risk Factors for Paediatric Ragweed Sensitisation: A Formal Model Comparison	*Children (Basel)*	School-based cross-sectional	Area-level PM_{10} and NO2; regional ragweed pollen; CORINE land cover (5 km)	Croatia (ATOPICA / EDIAQI)
Luglio air pollution pediatric asthma hospitalizations Louisiana vulnerable groups	Associations between air pollution and pediatric asthma hospitalizations across vulnerable groups in Louisiana	*Environ Epidemiol*	Time-stratified case-crossover	Daily $PM_{2.5}$, O3 and NO2 (NO2 2010-2016 only)	USA (Louisiana)
Oliveira air pollution HRV lung function schoolchildren Cubatao	Effect of air pollution on heart rate variability and pulmonary function in schoolchildren: a cross-sectional study, Cubatao, Brazil, 2017.	*Epidemiol Serv Saude*	School-based cross-sectional	Regulatory monitor concentrations (CETESB), daily and 2-3 day averages	Brazil (Cubatao, Sao Paulo State)

First author	Focus	Journal	Design	Metric	Region
Pedersen early-life air pollution lung function COPSAC	Impact of air pollution exposure in early life on lung function development in the COPSAC cohorts	*BMJ Open Respir Res*	Prospective birth cohort	Modelled multi-source $PM_{2.5}$ and NO2 at home address (DEHM/UBM/AirGIS-type chain)	Denmark (COPSAC2000 and COPSAC2010)
Perez-Garcia nasal EWAS respiratory biomarkers Mexican American CHAMACOS	Nasal Epigenome-Wide Association Study of Respiratory and Allergic Biomarkers in Mexican American Young Adults	*medRxiv*	Cross-sectional (cohort baseline)	Prenatal and past-year $PM_{2.5}$ and O3 (residential)	USA (Salinas Valley, California, CHAMACOS; Project Viva replication)
Poland air pollution COPD asthma emergency admissions DLNM heating season	Air pollution and respiratory admissions in Poland: a nationwide DLNM analysis of COPD and asthma	*Eur J Public Health (conference abstract)*	Distributed-lag time-series	PM_{10}, $PM_{2.5}$, NO2, SO2, PAHs and metals in PM_{10} (national monitoring)	Poland (nationwide, 2012-2019)
Shi air pollution pulmonary tuberculosis incidence China GTWR	Analysis of influencing factors of atmospheric pollution on pulmonary tuberculosis incidence in China from 2014 to 2020	*Front Public Health*	Ecological / spatial	Provincial annual $PM_{2.5}$, SO2, NO2, CO, O3	China (31 provinces)
Wang air pollutants sleep apnea polysomnography architecture Shenzhen	Associations of multiple ambient air pollutants and sleep apnea, sleep architecture and nocturnal oxygenation in adult patients with or without obstructive sleep apnea	*Environ Res*	Retrospective cohort	$PM_{2.5}$, PM_{10}, $PM_{2.5}$ inorganic components, NO2, SO2, CO, O3; lag 0-180 d	China (Shenzhen)
Wang climate air pollutants influenza 48 countries convergent cross mapping	Delayed and nonlinear associations and predictive causal signals between climate factors, air pollutants, and influenza epidemics in 48 countries	*Glob Health Res Policy*	Bayesian spatiotemporal model	Weekly cumulative SO2, organic carbon and dust $PM_{2.5}$ (gridded; source not stated in abstract)	Global (48 countries)
Wilner public safety power shutoff wildfire $PM_{2.5}$ healthcare California	Co-exposure to planned power outages and wildfire $PM_{2.5}$ and acute healthcare utilization in California: A time-stratified case-crossover study	*PLoS Med*	Time-stratified case-crossover	ZIP-level daily wildfire $PM_{2.5}$ (per 10 $\mu g/m^3$) jointly with public safety power shutoffs	USA (California, 2013-2019)
Yuksel COPD cigarette biomass combined exposure exacerbations Turkiye	Clinical outcomes of COPD associated with cigarette smoke, biomass smoke, and combined exposure: A retrospective observational study	*Tob Induc Dis*	Retrospective clinical series	Household biomass smoke (fuel type and duration from records); no PM measurement	Turkiye (multicentre)
Zhou preoperative short-term $PM_{2.5}$ postoperative pulmonary complications lung cancer surgery	Preoperative short-term $PM_{2.5}$ Exposure and Postoperative Pulmonary Complications After Lung Cancer Surgery	*Am J Respir Crit Care Med*	Retrospective cohort	Short-term preoperative $PM_{2.5}$ (metadata only)	China
Mechanistic toxicology					
Awasthi air pollution oxidative stress inflammation molecular mechanisms review	Molecular Mechanisms Linking Air Pollution-Induced Oxidative Stress to Inflammation and Human Disease	*J Xenobiot*	Mechanistic review	PM, NOx, O3, SO2, VOCs (generic)	Global / multi-region
Chen $PM_{2.5}$ mitochondria-associated membranes mouse liver	Exposure to airborne $PM_{2.5}$ suppresses mitochondrial associated membranes and MAM-associated metabolic regulators in mouse liver	*Am J Physiol Cell Physiol*	Animal exposure study + in vitro receptor overexpression	Concentrated ambient $PM_{2.5}$ inhalation (mouse); $PM_{2.5}$ particles on hepatocytes/macrophages	USA (laboratory)
Chen $PM_{2.5}$ mixtures Parkinson disease INPP5D SHIP1 PI3K-AKT multi-omics	Neurotoxicity of $PM_{2.5}$-Bound Multi-Pollutant Mixtures in Parkinson's Disease: Exploring the INPP5D/SHIP1-PI3K-AKT Axis via an Integrative Multi-Omics Study	*J Xenobiot*	Mendelian randomisation + TWAS	GWAS-instrumented $PM_{2.5}$; NIST SRM 1648a urban PM in SH-SY5Y cells	Not stated (public GWAS; laboratory)
Chen pirfenidone fine particulate matter EMT mitochondrial fission autophagy	Pirfenidone reduces fine particulate matter-induced epithelial to mesenchymal transition in lung epithelial cells through regulating mitochondrial fission and autophagy	*Respir Res*	Animal + in vitro	Fine PM (standard reference material type not stated), 100 ug/mL in vitro; intratracheal in mice	Laboratory (no field site)
Do $PM_{2.5}$ superior cervical ganglion calcium channels satellite glia	$PM_{2.5}$ Exposure Disrupts Sympathetic Neurotransmission by Impairing Voltage-Gated Ca2+ Channels and Remodeling Satellite Glia in the Superior Cervical Ganglion	*Int J Mol Sci*	Animal + in vitro	Collected $PM_{2.5}$, intracerebroventricular and ex vivo direct exposure	Korea (laboratory)

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Donohoe ultrafine amorphous silica epithelial stress pro-fibrotic	Ultrafine amorphous silica induces chronic epithelial cell stress and pro-fibrotic reprogramming in a particle size- and exposure duration-dependent manner	*Part Fibre Toxicol*	Metadata only (no abstract)	Ultrafine amorphous silica particles	Not stated in record
Ganse diesel exhaust particles cardiac metabolomics glycolytic shift rats	Diesel Exhaust Particles Exposure Disrupts Cardiac Bioenergetics and Induces a Shift Towards Glycolytic Metabolism in Rats	*Biomolecules*	Animal model	Diesel exhaust particles, intratracheal instillation 3x/week for 1 or 3 weeks	Spain (laboratory)
Gautam diesel exhaust miR-133b-3p CTGF Hippo ECM remodeling mouse lung	Diesel Exhaust-Induced miR-133b-3p Downregulation and CTGF-Linked Hippo Pathway Inactivation Promote ECM Remodeling and Inflammation in Mouse Lungs	*Environ Mol Mutagen*	Animal exposure study	Whole diesel exhaust, 30 min/day, 5 d/week, 8 weeks vs filtered air	India (inferred from author group; laboratory)
He FOXM1 lncRNA COPD-R biomass $PM_{2.5}$ pulmonary inflammation rat	FOXM1/LncRNA COPD-R regulates pulmonary inflammation in chronic obstructive pulmonary disease induced by biomass-related $PM_{2.5}$	*Ecotoxicol Environ Saf*	Animal + in vitro	Biomass-fuel smoke (rats, 7 months) and biomass-related $PM_{2.5}$ (L2 cells)	China (inferred from author group; laboratory)
Liu Streptococcus D19T fermentation supernatant asthmatic mice $PM_{2.5}$	Protective effect and mechanism of Streptococcus D19T fermentation supernatant in asthmatic mice exposed to $PM_{2.5}$	*Environ Int*	Metadata only (no abstract)	$PM_{2.5}$ exposure in an asthmatic mouse model	Laboratory (not stated)
Liu cerium dioxide scavenging environmentally persistent free radicals $PM_{2.5}$ filtration review	Cerium Dioxide-Based Scavenging of Environmentally Persistent Free Radicals in Fine Particulate Matter: A Prospective Review	*Int J Mol Sci*	Narrative review	EPFRs on $PM_{2.5}$ surfaces; CeO2-functionalised filter media (proposed)	Not stated (review)
Nored lithium-ion battery fire soot PM pulmonary toxicity mice fluoride	Pulmonary Toxicity of Particulate-Phase Combustion Byproducts from Lithium-Ion Battery Fires	*Environ Res*	Animal exposure study	LIB thermal-runaway soot and smoke PM (<10 um), chemical composition	USA (fire test facility; mouse exposures)
Zakari PM1 microbial profiles lung gut microbiome mice	Associations of particulate matter-associated microbial profiles with lung and intestinal microbiomes in healthy mice	*J Hazard Mater*	Murine inhalation exposure	Ambient PM1 (ions, metals, carbon, oxidative potential, 16S) vs HEPA-filtered and clean air	Not stated in record (ambient-air exposure chamber)
Zhao coal dust macrophage extracellular vesicles thyroid cancer miR-205-5p	Coal dust-stimulated macrophage-derived extracellular vesicles promote thyroid cancer cell phenotypes involving miR-205-5p, CAMK1G, and NF-kB-associated signaling	*Front Immunol*	Experimental / toxicology	Characterised coal dust (macrophage exposure; murine exposure)	China (laboratory)
Occupational & indoor					
Arif Agra footwear cluster ventilation IAQ tiers	Ventilation, indoor air quality and thermal comfort in India's leather footwear manufacturing cluster: a tier-resolved field study	*Research Square (preprint)*	Cross-sectional (occupational)	Breathing-zone $PM_{2.5}$ with CO2, TVOC, formaldehyde, air velocity	India (Agra)
Blanco-Alegre ironing indoor ultrafine particles ventilation eBC	Ironing as a Source of Indoor Aerosol Particles: Effects of Ventilation and Ironing Conditions	*Atmosphere*	Measurement campaign	Particle number concentration and size distribution, eBC	Not stated in record
Fan wildfire smoke indoor $PM_{2.5}$ absenteeism Oklahoma Head Start OK-AIR	Wildfire smoke, indoor and outdoor $PM_{2.5}$, and sick-related absenteeism in Oklahoma Head Start early care and education centers	*J Expo Sci Environ Epidemiol*	Cohort (secondary trial analysis)	Continuous classroom $PM_{2.5}$ paired with hourly EPA outdoor $PM_{2.5}$; HMS smoke days; FIRMS fire detections	USA (Oklahoma)
Geng grain dust simulator manikin respirator workplace protection factor	Using a Grain Dust Simulator with a Manikin System to Evaluate Performances of Commonly Used Respirators	*J Agric Saf Health*	Laboratory + field measurement	Respirable grain dust, 5-20 mg/m3; real-time on-farm grain-bin dust	USA (laboratory + on-farm)
Hunter e-cigarette aerosol early post-emission particle decay controlled conditions	Early post-emission particle decay dynamics of E-cigarette aerosols under controlled conditions	*Sci Total Environ*	Metadata only (no abstract)	E-cigarette aerosol particle size/number decay	Not stated in record
K-HEALTHinAIR multicountry indoor air homes schools low-cost sensors	Multicountry evaluation of indoor air in homes and schools: results from K-HEALTHinAIR project	*Eur J Public Health (conference abstract)*	Long-term indoor monitoring campaign	Low-cost sensor CO2, PM_{10}, $PM_{2.5}$ (annual 24 h means, 2025)	Europe (Spain, Austria, the Netherlands, Poland)

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Kulmala determination deposition rates indoor particle dynamics	Determination of deposition rates from indoor particle dynamics	*Aerosol Sci Technol*	Metadata only (no abstract)	Indoor particle concentration decay (size-resolved, presumed)	Finland (inferred from author affiliation; not stated in record)
Martin simplified method portable air cleaner CADR low-cost sensors	A simplified method for evaluating fine particulate matter removal performance of portable air cleaners	*MethodsX*	Field co-location + chamber	$PM_{2.5}$ decay measured with low-cost sensors vs research-grade instruments	USA (residence and laboratory)
Nahian Siegel in-situ testing in-duct air cleaners residential	In-situ testing of in-duct air cleaners in residential buildings	*Build Environ*	Metadata only (no abstract)	Residential in-duct air cleaner performance (particle removal presumed)	Not stated (record metadata only)
Ng particulate matter respiratory deposition inhalation dose biomass burning homes indigenous communities	Particulate Matter Exposure, Respiratory Deposition and Inhalation Dosage Assessment in Biomass Burning Homes in Rural Indigenous Communities	*Atmos Environ*	Metadata only (no abstract)	Household PM from biomass burning; respiratory deposition / inhaled dose	India (inferred from author affiliation; not stated in record)
Patras primary school classroom PM airborne viruses ventilation	Dual burden of PM and airborne viruses in primary schools: A pilot study in Patras, Greece	*Eur J Public Health* (*conference abstract*)	School-based cross-sectional	Classroom $PM_{2.5}/PM_{10}$ (triplicate) plus airborne virus RT-qPCR	Greece (Patras)
Saborit gas cooking indoor air quality households 7 European countries	Characterization of the impact of gas cooking on indoor air quality in households across 7 European countries	*Indoor Environments*	Long-term indoor monitoring campaign	Household indoor air during gas cooking (metadata only)	Europe (7 countries)
Stefaniak recycled plastic bag HDPE filament 3-D printing emissions	Recycling Plastic Bags into Filament: Consideration of Emissions during Melting, Extrusion, and 3-D Printing.	*J Chem Health Saf*	Chamber emission characterisation	Particle number size distribution (1.2-560 nm), particle yields per g plastic; acetone and formaldehyde	United States (laboratory; South African and US HDPE bags)
Zargarian spent coffee grounds MOF filtration membrane $PM_{2.5}$ bioaerosol photothermal	Spent Coffee Grounds-Derived Micro-Nano-Architectured Air Filtration Membranes for Multiwavelength Light-Triggered Bioaerosol Inactivation	*Small*	Bench / device experiment	$PM_{2.5}$ filtration efficiency (laboratory aerosol)	Laboratory (no field site)
Sensing, forecasting & instrumentation					
Ai optical trapping Raman single carbonaceous aerosol smoke	In-Situ Single-Particle Sensing of Carbonaceous Aerosols in Freshly Generated Smoke Using Optical Trapping-Raman Spectroscopy	*Sensors* (*Basel*)	Laboratory instrument development	Single airborne smoke particles (several um, 10 pg), Raman spectra	Not stated (laboratory)
Bei UPLC-MS/MS 74 antibiotics in $PM_{2.5}$ Beijing	Optimization and application of a sensitive UPLC-MS/MS method for the quantitative determination of antibiotics in $PM_{2.5}$	*J Chromatogr A*	Analytical method validation	Filter-collected 24 h $PM_{2.5}$, 74 antibiotics by SPE + UPLC-MS/MS	China (Beijing)
Kara Pythagorean fuzzy $PM_{2.5}$ monitoring station placement Trabzon	A Pythagorean Fuzzy decision making framework for $PM_{2.5}$ monitoring station placement in urban areas: A case study of Trabzon, Turkiye	*J Air Waste Manag Assoc*	Delphi consensus	$PM_{2.5}$ monitoring network siting (criteria-weighted)	Turkiye (Trabzon)
Manchanda hyperlocal BC sensing Bayesian inverse modeling West Oakland	Hyperlocal sensing and inverse modeling reveal community impacts of urban air pollutant emissions	*Sci Adv*	Method / algorithm	Black carbon from dense fixed-site network (6.7 sites/km2) plus mobile mapping	USA (West Oakland, California)
Mercado-Velazquez low-cost sensors three-tier London calibration transferability	How Accurate Are Low-Cost Air Quality Sensors? A 10-Month Study of Three-Tier Comparison with Three Calibration Methods and Cross-Site Transferability	*Sensors* (*Basel*)	Calibration model comparison	Low-cost optical $PM_{2.5}$ plus MOS/electrochemical NO2, CO, O3 vs reference	United Kingdom (London, two sites 1.6 km apart)
Ouimette PMS5003 PurpleAir internal heating hygroscopic mass reduction	Air Flow, Heat Transfer, and Hygroscopic Aerosol Mass Reduction in the Plantower PMS5003 Sensor and PurpleAir Monitor	*Preprints.org* (*preprint*)	Laboratory + field measurement	PMS5003 / PurpleAir $PM_{2.5}$ vs Teledyne T640x; detection-zone T and RH	USA (Santa Rosa, California; Winston-Salem, North Carolina)
Schmidt AOD gap filling iterative singular spectrum analysis AERONET Cuiaba	Gap filling in long-term AOD time series using Iterative Singular Spectrum Analysis at a tropical AERONET site	*Environ Monit Assess*	Method / algorithm	AERONET daily aerosol optical depth	Brazil (Cuiaba-Miranda, Mato Grosso)
Spinosa volcanic Saharan dust biomass burning aerosol advections Naples	Multiple advections of volcanic, Saharan dust and biomass burning aerosols over Naples	*Atmos Res*	Metadata only (no abstract)	Columnar/profiled aerosol type (likely lidar-based)	Italy (Naples)

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Trivanovic periodic technical inspection particle number devices gasoline exhaust	Assessment of existing PN-PTI devices to measure gasoline exhaust	*J Aerosol Sci*	Metadata only (no abstract)	Solid particle number (PN) at tailpipe, PTI-grade counters	Switzerland (inferred from author group; not stated in record)
Vannini biochar passive biomonitor potentially toxic elements lichens	Wood-derived biochar as a potential sustainable alternative to lichens for monitoring airborne potentially toxic elements	*Atmos Environ*	Metadata only (no abstract)	Airborne potentially toxic elements accumulated on exposed biochar	Not stated in record
Waghmare machine learning explainable AI $PM_{2.5}$ prediction Nagpur	Machine learning models with explainable AI for $PM_{2.5}$ prediction for Urban air quality assessment	*MethodsX*	Gradient-boosting model + SHAP interpretability	Station $PM_{2.5}$ with co-pollutants and meteorology	India (Nagpur, Maharashtra)
Wang $PM_{2.5}$-bound PAH association rules machine learning forecasting Nagoya	Characterizing statistical association patterns and local forecasting of $PM_{2.5}$-bound polycyclic aromatic hydrocarbon concentrations in Nagoya, Japan	*Environ Monit Assess*	Machine-learning model	Hourly $PM_{2.5}$-bound PAHs, BC, NOx, oxidants, meteorology	Japan (Nagoya)
Washeem machine learning condensation bioaerosol sampler collection volume	Machine-Learning-Guided Prediction and Temperature Optimization for Collection Volume Control in a Condensation-Based Bioaerosol Sampler	*J Aerosol Sci*	Metadata only (no abstract)	Bioaerosol (condensation-growth liquid collection)	Not stated in record
Wu source-dependent organic aerosol hygroscopicity eastern China biomass burning	Source-dependent organic aerosol hygroscopicity in eastern China: observational evidence for conditional enhancement under biomass-burning influence	*Atmos Chem Phys*	Field campaign	Humidified light scattering f(RH) with OA composition; ZSR closure for kappa_OA	China (eastern, rural site)
Zhao spin-trapping EPR radicals atmospheric fine particulate matter cavity quality factor	Optimized spin-trapping method for radicals detection induced by atmospheric fine particulate matter: Critical roles of temporal evolution and cavity quality factor	*Aerosol Sci Technol*	Metadata only (no abstract)	$PM_{2.5}$-induced radicals (EPR spin trapping)	Laboratory (not stated)
zum Berge multicopter vibration optical particle counter accuracy	Influence of multicopter-induced vibrations on the measurement accuracy of optical particle counter	*Aerosol Sci Technol*	Metadata only (no abstract)	Optical particle counter size-resolved counts on a multicopter UAV	Not stated in record
Exposure assessment & modelling					
Accra $PM_{2.5}$ elemental composition PMF source shift e-waste burning	Shifting Dominant Air Pollution Sources in Sub-Saharan Africa's Changing Urban Environment: Analysis of Elemental Composition and Source Contributions of Ambient Fine Particulate Matter ($PM_{2.5}$) Pollution in Accra, Ghana	*Environ Health Perspect*	Intensive field campaign, PMF source apportionment	Weekly gravimetric $PM_{2.5}$ filters, ED-XRF elements, PMF	Ghana (Accra, 146 sites)
Aktypis nucleation-mode particles marine atmosphere downwind urban	Frequent Appearance of Nucleation-Mode Particles in the Marine Atmosphere Downwind of an Urban Area	*Atmos Environ*	Metadata only (no abstract)	Nucleation-mode particle number size distributions	Not stated in record
Aloraini negative control PSCF CWT trajectory dust attribution Riyadh	Transport Pathways Versus Dust Sources: A Negative-Control Test of Trajectory-Statistical Attribution in Riyadh, Saudi Arabia	*Atmosphere*	Monitoring-record analysis	METAR present-weather dust codes (locally lifted vs suspension), 2016-2025	Saudi Arabia (Riyadh, Arabian Peninsula)
Alvi PMF trace elements PM_{10} Faisalabad health risk	PMF Resolved Source Identification & Health Risk Assessment of Trace-elements in Airborne Particulate Matter (PM_{10}) from South-Asian Megacity	*J Hazard Mater*	Source apportionment	Filter PM_{10} with ICP-OES trace elements; PMF	Pakistan (Faisalabad, Indo-Gangetic Plain)
Anjum Idaho $PM_{2.5}$ 2005-2024 wildfire vs wood-smoke regimes	Long-Term Spatiotemporal Variability of Fine Particulate Matter ($PM_{2.5}$) Across Idaho, USA, Using Statistical and Geospatial Tools	*Research Square (preprint)*	Monitoring-record analysis	EPA AQS daily $PM_{2.5}$, 32 sites (4,260 site-months)	USA (Idaho)
Baliaka Eaton urban firestorm airborne toxic metals lead Los Angeles	Fire at the boundary: Airborne toxic metals from the 2025 Los Angeles urban firestorm	*Sci Adv*	Field measurement	Real-time airborne trace elements (XRF) in PM during the Eaton fire	USA (Los Angeles, California)
Bei air moisture particulate matter winter Guanzhong Basin	Contributions of air moisture to particulate matter pollution during wintertime in the Guanzhong Basin, China	*Atmos Pollut Res*	Metadata only (no abstract)	Wintertime particulate matter (fraction not stated in record)	China (Guanzhong Basin)
Chakraborty fog intensity droplet size aqueous SOA Kanpur	Fog Intensity and Droplet Size-Resolved Aqueous Secondary Organic Aerosol (aqSOA) Formation in Polluted Winter Fog: New Insights from a Combined Observational and Modeling Study	*Environ Sci Technol*	Field campaign	Size-resolved fogwater, offline HR-AMS with PMF; ambient OA	India (Kanpur)

First author	Focus	Journal	Design	Metric	Region
Dhital on-road black carbon PM exposure Nepal route travel mode	On-road exposure to black carbon and particulate matter in Nepal: Effects of route attributes and travel modes	*J Air Waste Manag Assoc*	Mobile monitoring campaign	Equivalent BC (aethalometer), PM1, $PM_{2.5}$, PM_{10} (portable optical)	Nepal (Kathmandu Valley and intercity highways)
Fang end of uniformity spatial climatic toxicological redefinitions air quality	The End of Uniformity: Spatial, Climatic, and Toxicological Redefinitions of Air Quality	*Environ Sci Technol*	Metadata only (no abstract)	Not stated (perspective on air-quality metrics)	Not stated in record
Jabbari-Hichri on-road emissions light-duty vehicles particulate gaseous pollutants	On-road emissions of light-duty vehicles: detailed analysis of particulate and gaseous pollutants	*Atmos Pollut Res*	Metadata only (no abstract)	Light-duty vehicle on-road particulate and gaseous emissions	France (inferred from author affiliation; not stated in record)
Julaha vertical aerosol chemical composition 230 m 4 m ACSM rural background	Meteorological Effects on the Vertical Variability of Aerosol Chemical Composition at a Rural Background Site	*Environ Pollut*	Field measurement	Submicron Org, SO4, NO3, NH4 (ToF-AMS at 230 m, ToF-ACSM at 4 m), eBC, OC/EC	Czech Republic (rural background tower site)
Kalkavouras ammonia sulfuric acid BVOC urban new particle formation Athens	Assessing the combined roles of ammonia, sulfuric acid, and biogenic volatile organic compounds in urban new particle formation	*Environ Pollut*	Long-term field measurement campaign	Particle number size distributions with NH3, BVOCs and proxy H2SO4	Greece (Athens, Thissio)
Kwon seasonal $PM_{2.5}$ secondary inorganic aerosol livestock rural Ulsan	Seasonal Composition of $PM_{2.5}$ and Secondary Inorganic Aerosols at a Rural Site near Livestock Facilities in Ulsan, South Korea	*Atmosphere*	Filter sampling + composition	Weekly 24 h $PM_{2.5}$ filters, eight water-soluble ions	South Korea (Ulsan, rural)
Lian firework barium aerosols single-particle SPAMS Guangzhou	Evolution of firework-related barium aerosols: insights from single-particle analysis and mass concentration monitoring	*Atmos Chem Phys*	Field measurement	Online Ba mass concentration plus single-particle aerosol mass spectrometry (SPAMS)	China (Guangzhou, four sites)
Liang deep learning low-visibility events fog-haze eastern China	Classification and quantification of low-visibility events using deep learning over eastern China	*Atmos Chem Phys*	Deep-learning model	Surface $PM_{2.5}$ with geostationary satellite and meteorology	China (eastern)
Lu heterogeneous ammonia uptake SOA WRF-Chem North China Plain	Impacts of the heterogeneous ammonia uptake on air quality in the North China Plain	*Atmos Chem Phys*	Chemical transport modelling	WRF-Chem $PM_{2.5}$, inorganic aerosol and nitrogen-containing organics (NOCs)	China (North China Plain)
Mandujano change-point detection trend estimation long-term PM records	Assessing the Effect of Change-Point Detection Techniques on Trend Estimation in Long-Term Particulate Matter Records	*Environ Sci Atmos*	Metadata only (no abstract)	Long-term PM monitoring records	Not stated in record
Netto metal distribution $PM_{2.5}$ smelting Rio de Janeiro solubility risk	Metal distribution in atmospheric particulate matter ($PM_{2.5}$) from urban areas with smelting activities in the Rio de Janeiro state, Brazil: Solubility, correlation and risk assessment	*Atmos Pollut Res*	Metadata only (no abstract)	$PM_{2.5}$ metals, soluble fractions	Brazil (Rio de Janeiro state)
Pani Southeast Asian biomass burning smoke aerosol light extinction apportionment high-altitude western North Pacific	Impact of peninsular Southeast Asian biomass-burning smoke on chemical apportionment of aerosol light extinction at a high-altitude remote environment in the western North Pacific	*Atmos Res*	Metadata only (no abstract)	Aerosol chemical composition and light extinction (IMPROVE-type apportionment)	Taiwan (inferred from author affiliation; not stated in record)
Park $PM_{2.5}$ composition sources COVID-19 social distancing Seoul	Restructuring of $PM_{2.5}$ Composition and Source Contributions Across COVID-19 Social Distancing Periods: A Case Study at an Urban Monitoring Site in Seoul, South Korea	*Toxics*	Source apportionment	$PM_{2.5}$ mass and speciation, PMF (7 factors), GAM meteorological adjustment	South Korea (Seoul, Gwangjin-gu)
Qu $PM_{2.5}$-bound heavy metals source apportionment Laibin winter	Source Apportionment and Health Risk Assessment of $PM_{2.5}$-Bound Heavy Metals in Wintertime Laibin, Guangxi, China	*Toxics*	Source apportionment	$PM_{2.5}$ mass and nine metals (Zn, Cr, Ni, Pb, V, Cu, Sb, Cd, Co), day/night filters	China (Laibin, Guangxi)
Rosu multi-instrument Saharan dust event South-Eastern Europe	A multi-instrument analysis of a Saharan dust event in South-Eastern Europe	*Atmos Res*	Field campaign / remote sensing	Saharan dust event (metadata only)	Europe (South-Eastern)
Somsunun $PM_{2.5}$-bound metals health risk haze season Chiang Mai	Health Risk Assessment of Ambient $PM_{2.5}$-Bound Metals During Haze Seasons in Urban and Rural Chiang Mai, Thailand	*Toxics*	Filter sampling + composition	24 h $PM_{2.5}$ filters (n=101) and metals, three land-use sites	Thailand (Chiang Mai)
Szalata waste fire toxic emissions Gaussian plume environmental risk	Environmental Risk Assessment of Toxic Emissions into the Air Resulting from Waste Fires	*Toxics*	Physical exposure model	Modelled PM_{10}, $PM_{2.5}$ and gases from a waste-storage fire (Gaussian plume)	Poland

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Szczepanik Warsaw wintertime air quality improvement 2016-2025	Improvement in wintertime air quality in urban and suburban areas of Warsaw within 2016-2025	*Aerosol Air Qual Res*	Metadata only (no abstract)	Wintertime ambient air quality (pollutants not stated)	Poland (Warsaw)
UK Air Quality Cell PM_{10} plume back trajectory tyre fire Mexborough vulnerability	Back trajectory modelled PM_{10} plumes: population vulnerability profile from an uncontrolled fire	*Eur J Public Health (conference abstract)*	Event-based monitoring	Incident PM_{10} monitoring + ADMS-STAR back-trajectory plume (22 hourly plumes)	United Kingdom (Mexborough, England)
Upaul Khulna $PM_{2.5}$ meteorology HYSPLIT random forest STL	Meteorological and HYSPLIT-Derived Transport Influences on $PM_{2.5}$ Exposure in a Khulna City, Bangladesh	*Research Square (preprint)*	Monitoring-record analysis	Hourly $PM_{2.5}$ from the CAMS-8 station (2024); NASA POWER meteorology; HYSPLIT	Bangladesh (Khulna)
Vikas wavelet background $PM_{2.5}$ India MERRA-2	Estimating background $PM_{2.5}$ concentrations using a wavelet-based convolution framework	*Environ Monit Assess*	Algorithm development	MERRA-2 reanalysis $PM_{2.5}$; CPCB monitors for evaluation	India
Wang urban morphology street-scale $PM_{2.5}$ PM_{10} change points	Non-linear associations, model-supported change points, and seasonal variations of urban morphology with street-scale $PM_{2.5}$ and PM_{10} concentrations	*Atmos Pollut Res*	Metadata only (no abstract)	Street-scale $PM_{2.5}$ and PM_{10}	Not stated (metadata only)
Wei $PM_{2.5}$ ozone nonlinear coupling Puyang oilfield city GAM 2015-2025	Long-Term Trends and Nonlinear $PM_{2.5}$-O3 Coupling in a Typical Oilfield City of the "2 + 26" Region	*Atmosphere*	Statistical model	Daily $PM_{2.5}$ and MDA8 O3 (fixed monitors), 2015-2025	China (Puyang, Henan)
Yingkar $PM_{2.5}$-bound PAHs Central Asia urban trends sources risk	Environment-to-Health Linkages of $PM_{2.5}$-Bound Polycyclic Aromatic Hydrocarbons in Urban Cities of Central Asia: Trends, Sources, and Risk Assessment	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$-bound PAHs	Kazakhstan (Central Asian cities; inferred from title and author group)
Yousaf nanoscale curvature hygroscopic growth respiratory deposition ultrafine	Nanoscale Curvature Controls Hygroscopic Growth and Respiratory Deposition of Atmospheric Nanoparticles	*Environ Pollut*	Laboratory measurement	15-100 nm NaCl particles, HTDMA growth factors fed to MPPD dosimetry	Laboratory (no field site)
Yu Paris urban plume aerosol composition optical properties airborne ground	Urban-to-downwind variations in aerosol composition and optical properties in the Paris urban plume from coordinated airborne and ground-based observations	*Atmos Chem Phys*	Aircraft measurement campaign	PM1 composition (AMS), particle number 80-200 nm, SSA, refractive index, AAE	France (Paris and downwind region)
Burden, policy & mitigation					
BEST-COST all-cause vs disease-specific exposure-response functions burden $PM_{2.5}$ NO2	Effect of all-cause versus disease-specific exposure-response functions on air pollution burden	*Eur J Public Health (conference abstract)*	Health impact model	EEA $PM_{2.5}$ and NO2 exposure; WHO-recommended ERFs	Europe (Belgium, Estonia, France, Norway, Portugal)
Grzesczyk-Nojszewska Poland air pollution interventions public attitudes survey	Public attitudes toward personal and population interventions to reduce air pollution exposure: a 2025 nationwide cross-sectional survey in Poland	*Int J Occup Med Environ Health*	Cross-sectional population survey	Air pollution (perceived); no measurement	Poland (nationwide)
Imman caregivers $PM_{2.5}$ prevention preschool children northern Thailand qualitative	Caregivers' experiences of $PM_{2.5}$ prevention for preschool children in Thailand: a qualitative study with focus group interviews	*Child Health Nurs Res*	Qualitative / ethnographic study	Seasonal haze $PM_{2.5}$ (perceived); no measurement	Thailand (rural north)
K-HEALTHinAIR indoor $PM_{2.5}$ COPD attributable burden Spanish households	Attributable Burden of COPD from Indoor $PM_{2.5}$ Exposure in Spanish Households, 2025	*Eur J Public Health (conference abstract)*	Burden estimation	Measured indoor $PM_{2.5}$ in Spanish homes (K-HEALTHinAIR)	Spain
Lee type 2 diabetes burden Asia GBD 2023 particulate matter leading risk	The evolving burden of type 2 diabetes mellitus in Asia, 1990-2023: A Global Burden of Disease Study 2023	*Diabet Med*	Burden estimation (GBD)	GBD particulate matter pollution risk factor (ambient + household)	Asia (34 countries)
Lu kitchen air purifier health-economic outcomes 338 Chinese cities	Heterogeneous health-economic outcomes of kitchen air purifier interventions: A 338-city study in urban China	*Environ Int*	Metadata only (no abstract)	Kitchen (cooking) particulate exposure - not stated in record	China (338 cities)
Mohamadi Isfahan $PM_{2.5}$ AirQ+ WRF-SMOKE-CMAQ health damage	Assessing the health impacts of $PM_{2.5}$ from anthropogenic and natural sources in Isfahan, Iran	*J Environ Health Sci Eng*	Integrated emission-air quality-health modelling	Monitoring-station annual $PM_{2.5}$; WRF-SMOKE-CMAQ source contributions	Iran (Isfahan)

First author	Focus	Journal	Design	Metric	Region
Ruengdit active-flow respirator fit children Chiang Mai	Active-flow systems enhance respirator fit success in children: a comparative study with standard disposable models	*Front Public Health*	Laboratory metrology	Quantitative fit factor (TSI PortaCount, ambient particle counting)	Thailand (Chiang Mai)
Ruwali Lary US infant life expectancy environmental determinants machine learning	Estimating Life Expectancy in the U.S.: Identifying Environmental Determinants and Disparities	*Sensors (Basel)*	Machine-learning model	33 ECMWF reanalysis environmental variables incl. PM, sulphate and dust aerosol, O3, CO, HCHO	USA (contiguous, county level)
Wang carbon footprint air-pollutant emissions Beijing 2022 Winter Olympics	Linking Carbon Footprints to Air-Pollutant Emissions from Mega Sporting Events: Evidence from the Beijing 2022 Winter Olympics	*Atmosphere*	Emission inventory	Co-emitted $PM_{2.5}$, NOx, SO2, CO, NMVOC from pollutant-to-CO2 ratios	China (Beijing)
Wang climate-adaptive city pilot $PM_{2.5}$ health dividend China	Does climate adaptation policy deliver a health dividend? Exposure, modeled health burden, and defensive capacity in China's climate-adaptive city pilots	*Front Public Health*	Difference-in-differences evaluation	City-level annual $PM_{2.5}$ (gridded product), GEMM burden	China (264 cities)
Other clinical endpoints					
Deng ultrafine particles kidney disease ED visits New York	A mysterious danger for kidney disease exacerbations: short-term impacts of ultrafine particles on hospital visits	*Environ Res*	Time-stratified case-crossover	Daily UFP number concentration from a validated chemical transport model (street-geocoded residence)	USA (New York State)
Duan landscape fire-sourced $PM_{2.5}$ kidney function eGFR children China	Landscape fire-sourced $PM_{2.5}$ and kidney function in children: evidence from a national cross-sectional study in China	*Environ Res*	Cross-sectional survey	Three-year mean landscape fire-sourced $PM_{2.5}$ at school address (modelled)	China (national surveillance)
Hoang $PM_{2.5}$ minimal residual disease childhood ALL REDIAL	Fine particulate matter and minimal residual disease in childhood acute lymphoblastic leukemia from the REDIAL consortium	*JNCI Cancer Spectr*	Retrospective cohort	CDC daily census-tract $PM_{2.5}$ during induction (residence; centre then residence)	USA (REDIAL consortium, Texas-led)
Hu air pollution built environment lung cancer recurrence after surgery China	Ambient Air Pollution, Residential Built Environment, and Lung Cancer Recurrence After Curative-Intent Surgery: A Retrospective Cohort Study in Three Chinese Cities	*Research Square (preprint)*	Retrospective cohort	Satellite-derived 1 km $PM_{2.5}$, PM_{10}, NO2, CO, SO2, O3 (post-diagnosis first year)	China (three cities)
Li ambient air pollution gastric cancer incidence 204 countries ecological	Ambient Air Pollution and Gastric Cancer Incidence: A Multinational Ecological Analysis	*Curr Med Sci*	Ecological panel, fixed effects	GBD 2021 national population-weighted $PM_{2.5}$, NO2, O3	Global (204 countries)
Slovakia $PM_{2.5}$ lung cancer mortality gender lag ecological	$PM_{2.5}$ exposure and lung cancer mortality in Slovakia: gender differences and delayed effects	*Eur J Public Health (conference abstract)*	Ecological	National population-weighted $PM_{2.5}$ (Indicator of Population Exposure), 2010-2023	Slovakia
Wang $PM_{2.5}$ lung cancer nonsmokers narrative review IL-1beta EGFR screening	Environmental $PM_{2.5}$ exposure associated with lung cancer in nonsmokers: a narrative review of high-risk groups, mechanisms, and implications for screening	*J Toxicol Environ Health B Crit Rev*	Narrative review	Ambient and microenvironmental $PM_{2.5}$ (incense, cooking, transit, occupational)	Global/multi-region
Zhong air pollution years of life lost cancer Pudong Shanghai time series	The short-term impact of ambient air pollution on years of life lost from cancer in Pudong New Area, Shanghai: A time series study	*PLoS One*	Distributed-lag time-series	Daily $PM_{2.5}$, PM_{10}, SO2, NO2, O3 (monitoring network)	China (Shanghai, Pudong New Area, 2013-2021)

Provenance and method

Period **25 – 31 October 2026**, Sunday to Saturday, **seven daily issues**, **135 records**, **135 unique DOIs**, no record without a DOI. This rollup is **aggregation only**: it re-reads `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over the period and **queries no literature API**; every record was gated by `check_dois.py` at daily-issue time. Tier mix **16 A, 54 B, 65 C**. The 25–29 October dailies were built on the **30 October recovery run**; the 30 and 31 October dailies and this rollup on the **1 November run**; no record is held. Effect estimates are transcribed verbatim from the daily issues on heterogeneous contrasts and are **not** meta-analysed. Abstracts remain the copyright of their respective publishers.